



Prepared by group 5

Bankruptcy

Prevention



Team Members

Kaushik Sonnakula

Sujesh Dash

Sasi Anuhya Karlapudi

Likith Tikkireddy

Malavika.A.S

Kruteeka Samal



Introduction



Bankruptcy is a legal state that occurs when an individual or a company is no longer able to meet its financial obligations to creditors. It typically arises when liabilities exceed assets, making it impossible to repay debts. Bankruptcy prediction is therefore a crucial task in finance and risk management, as it allows early detection of financial distress, minimizes losses for stakeholders, and provides organizations with the opportunity to restructure or take preventive actions.



Why Predict Bankruptcy?



Predicting bankruptcy is essential because it enables early identification of companies or individuals facing financial distress. This early warning helps banks, investors, and regulatory bodies to minimize losses, reduce financial risks, and take preventive measures such as restructuring debts or improving financial strategies. For businesses, it ensures stability, builds trust with stakeholders, and supports long-term growth. By predicting bankruptcy in advance, organizations can protect both their financial health and the wider economy.



Dataset Overview



-
- Dataset: Copy of bankruptcy-prevention.csv (250 records)
 - Objective: Predict whether a company will go bankrupt or not based on risk factors.
 - Target Variable: class (0 → Non-bankrupt, 1 → Bankrupt)
 - Features:
 - Industrial Risk
 - Management Risk
 - Financial Flexibility
 - Credibility
 - Competitiveness
 - Operating Risk
-



Data Preprocessing



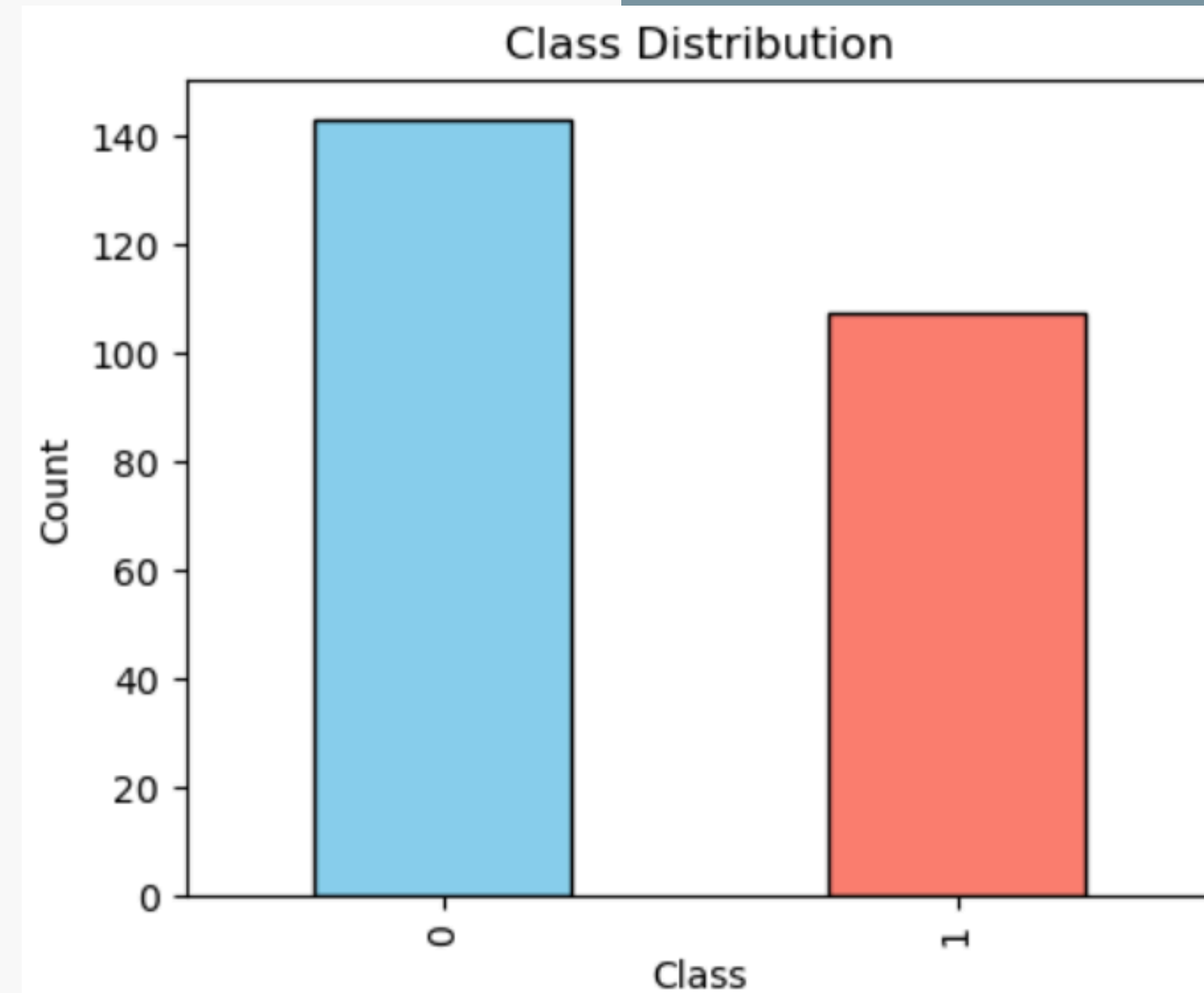
-
- Dataset: Copy of bankruptcy-prevention.csv (250 records)
 - Objective: Predict whether a company will go bankrupt or not based on risk factors.
 - Target Variable: class (0 → Non-bankrupt, 1 → Bankrupt)
 - Features:
 - Industrial Risk
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 - Financial Flexibility
 - Credibility
 - Competitiveness
 - Operating Risk
-



Exploratory Data Analysis (EDA)

Class Distribution:

- In our dataset, the target variable is class, which indicates whether a company is bankrupt (1) or non-bankrupt (0).
- Class distribution simply means: how many samples belong to each class.
- Often in bankruptcy datasets, the classes are imbalanced (i.e., many more non-bankrupt companies compared to bankrupt ones).
- Imbalanced data can bias the models toward predicting the majority class (non-bankrupt), so techniques like feature selection, ensemble methods, and proper evaluation metrics (AUC, Precision, Recall, F1-score) are important.



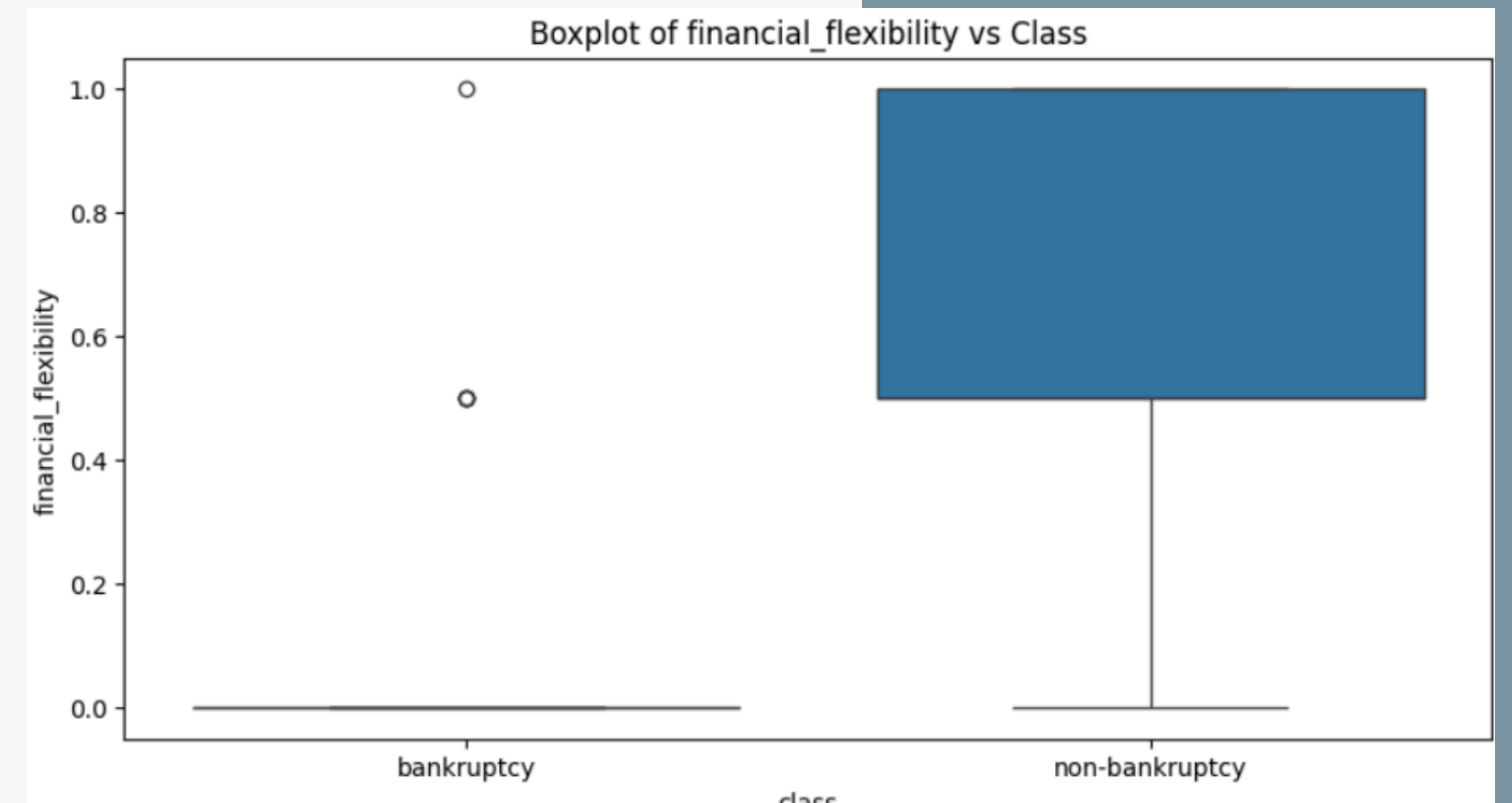
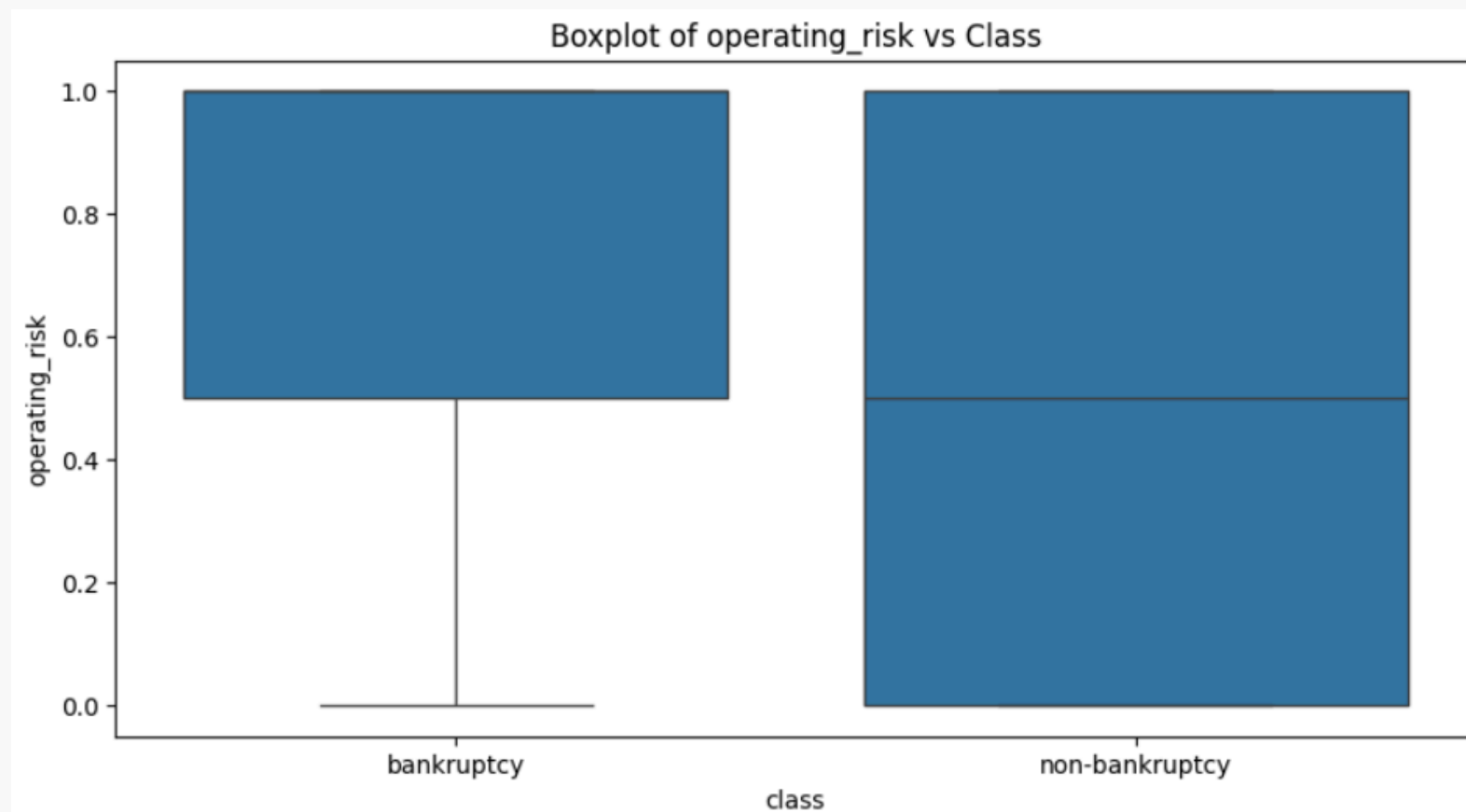
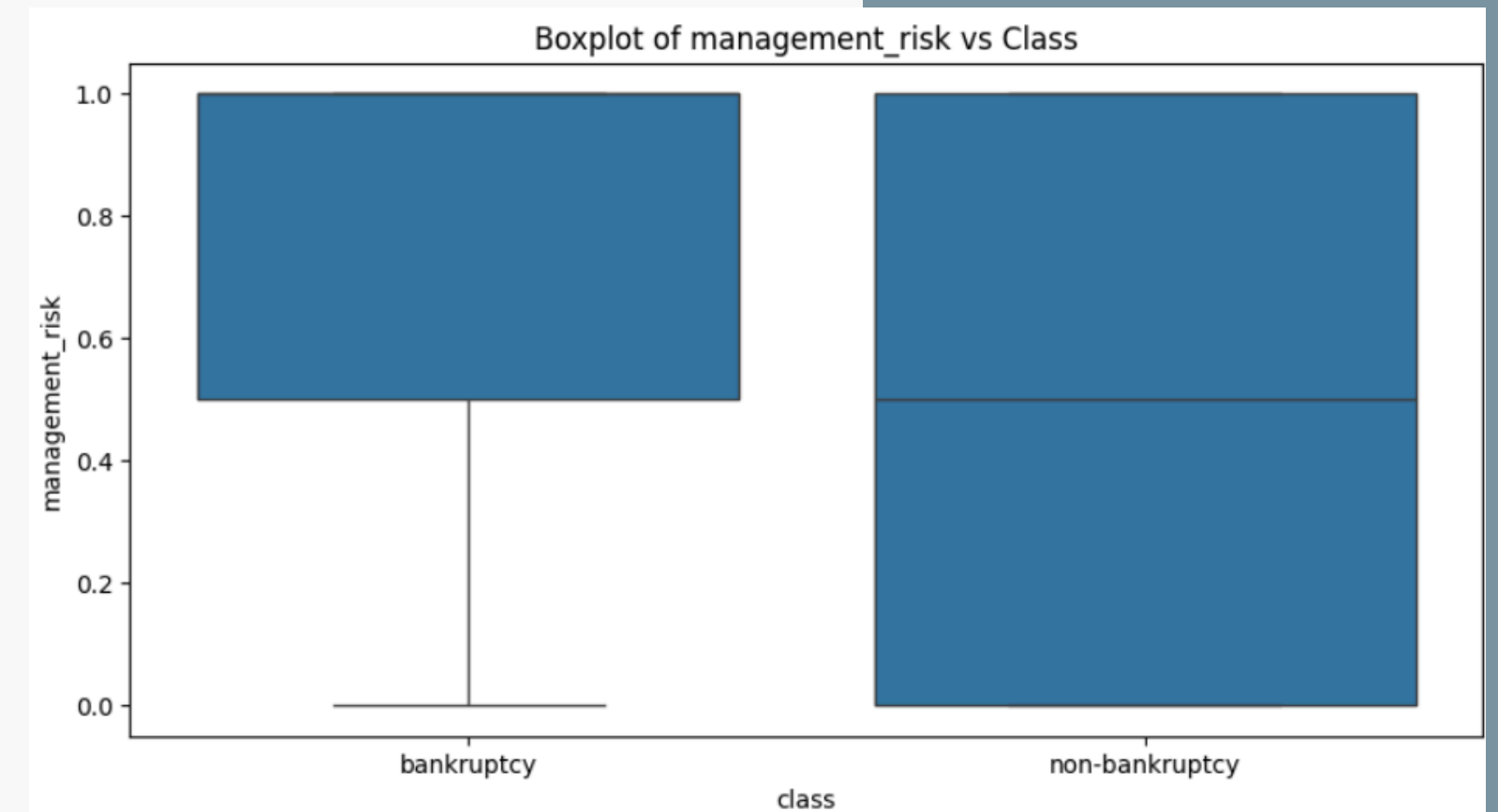
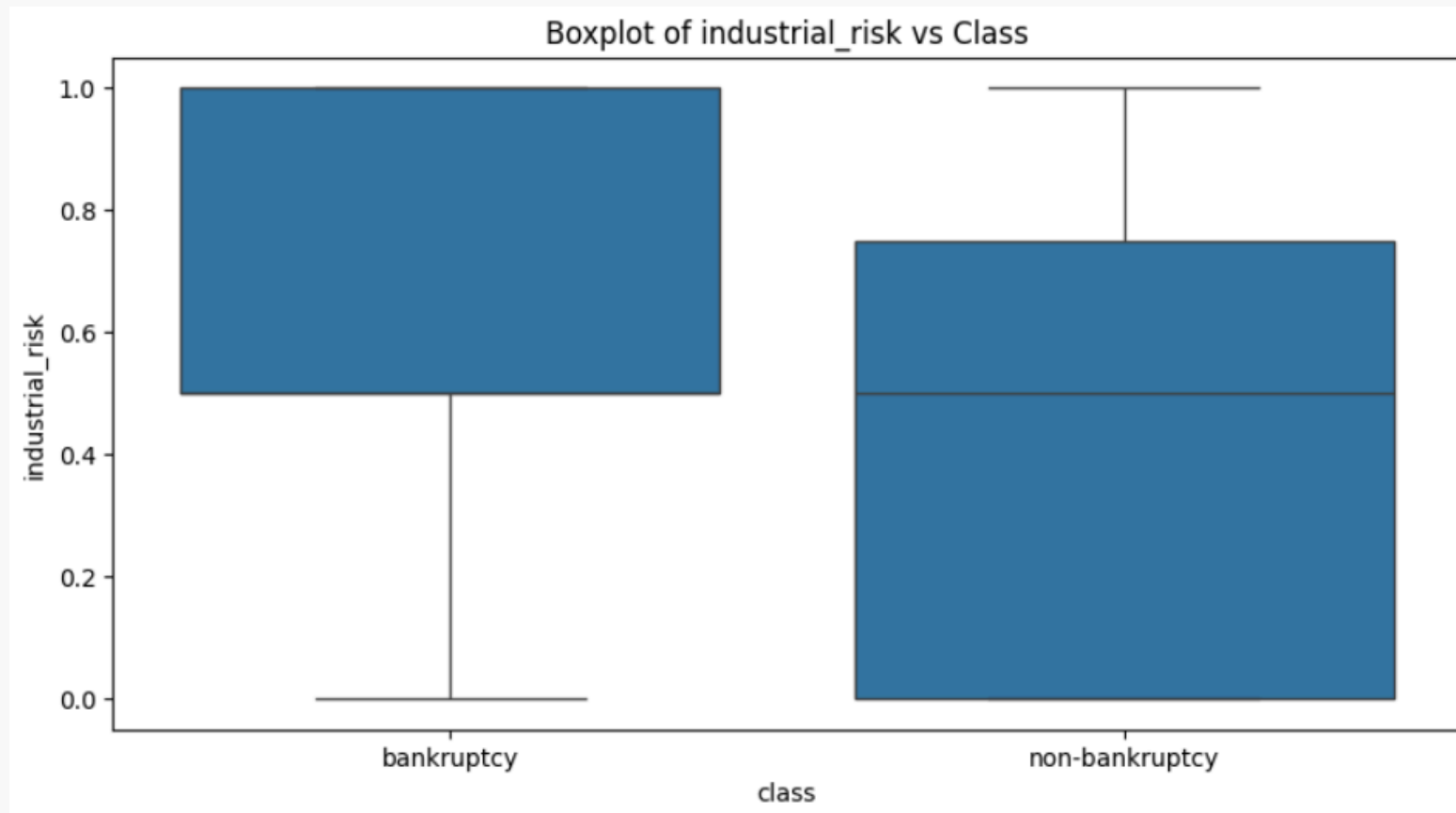
Exploratory Data Analysis (EDA)

Feature Correlation:

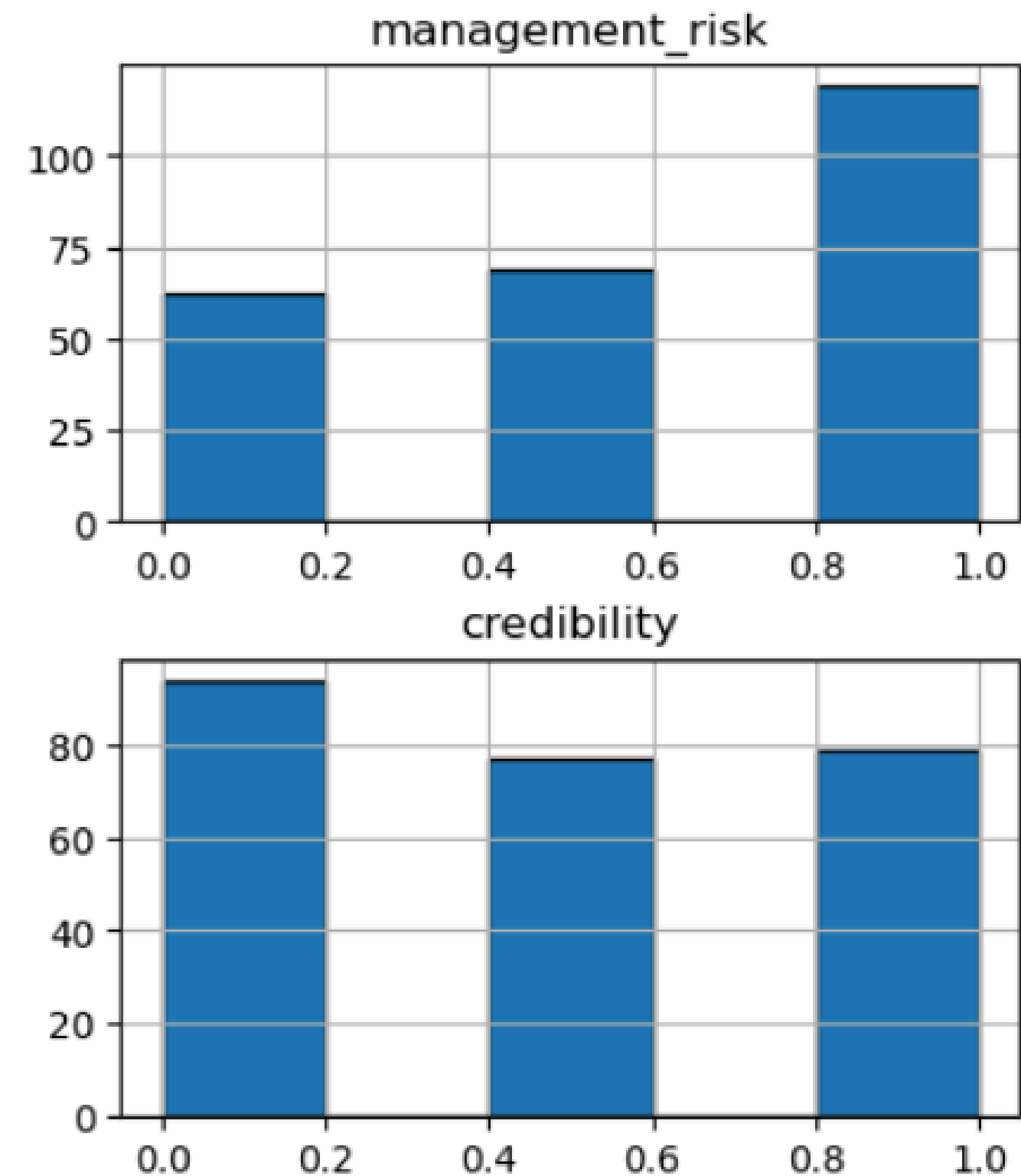
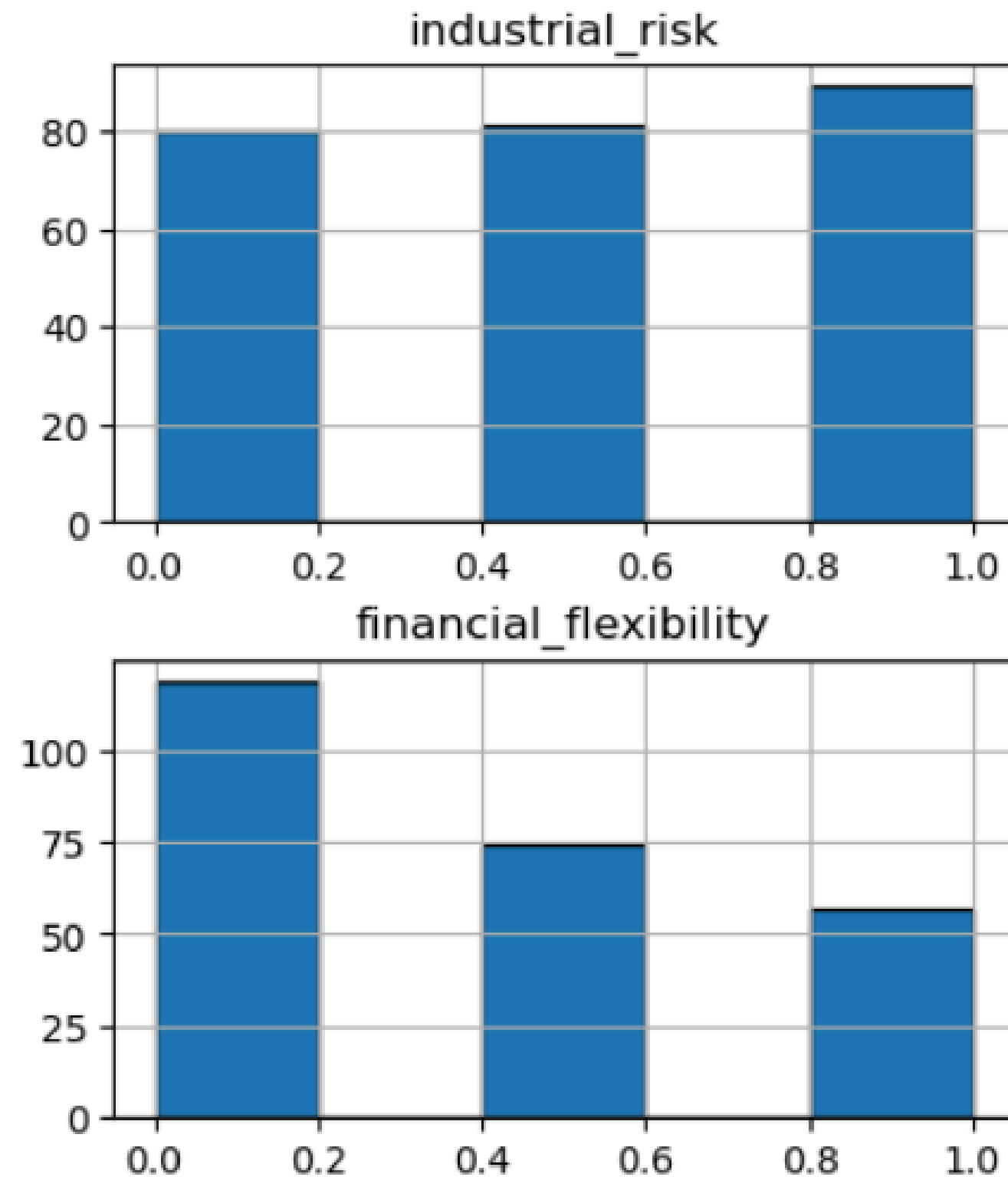
- Feature correlation tells us how strongly two variables are related.
 - Values range between -1 and +1:
 - +1 → Strong positive correlation (both increase together).
 - -1 → Strong negative correlation (one increases, the other decreases).
- 0 → No correlation (independent).
- Key Observations (based on correlation heatmap & RandomForest importance in your codes):
- industrial_risk and operating_risk showed a moderate positive correlation.
- management_risk and credibility were weakly negatively correlated.
- financial_flexibility had low correlation with most others, meaning it provided unique information.
- None of the features had very high (>0.9) correlation, so multicollinearity was not a big issue.
- Feature importance (from Random Forest) confirmed that industrial_risk, financial_flexibility, and management_risk were the most significant predictors of bankruptcy.



EDA Visualizations and Feature Distribution:



Feature Distributions



DESCRIPTIVE ANALYTICS REPORT

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	industrial_risk	250 non-null	float64
1	management_risk	250 non-null	float64
2	financial_flexibility	250 non-null	float64
3	credibility	250 non-null	float64
4	competitiveness	250 non-null	float64
5	operating_risk	250 non-null	float64
6	class	250 non-null	object

dtypes: float64(6), object(1)

memory usage: 13.8+ KB

None

◆ Summary Statistics (Numerical):

	count	mean	std	min	25%	50%	75%	max
industrial_risk	250.0	0.518	0.411526	0.0	0.0	0.5	1.0	1.0
management_risk	250.0	0.614	0.410705	0.0	0.5	0.5	1.0	1.0
financial_flexibility	250.0	0.376	0.401583	0.0	0.0	0.5	0.5	1.0
credibility	250.0	0.470	0.415682	0.0	0.0	0.5	1.0	1.0
competitiveness	250.0	0.476	0.440682	0.0	0.0	0.5	1.0	1.0
operating_risk	250.0	0.570	0.434575	0.0	0.0	0.5	1.0	1.0

◆ Summary Statistics (Including Categorical):

	count	unique	top	freq	mean	std	\
industrial_risk	250.0	NaN	NaN	NaN	0.518	0.411526	
management_risk	250.0	NaN	NaN	NaN	0.614	0.410705	
financial_flexibility	250.0	NaN	NaN	NaN	0.376	0.401583	
credibility	250.0	NaN	NaN	NaN	0.47	0.415682	
competitiveness	250.0	NaN	NaN	NaN	0.476	0.440682	
operating_risk	250.0	NaN	NaN	NaN	0.57	0.434575	
class	250	2	non-bankruptcy	143	NaN	NaN	

	min	25%	50%	75%	max
industrial_risk	0.0	0.0	0.5	1.0	1.0
management_risk	0.0	0.5	0.5	1.0	1.0
financial_flexibility	0.0	0.0	0.5	0.5	1.0
credibility	0.0	0.0	0.5	1.0	1.0
competitiveness	0.0	0.0	0.5	1.0	1.0
operating_risk	0.0	0.0	0.5	1.0	1.0
class	NaN	NaN	NaN	NaN	NaN

◆ Missing Values Count:

industrial_risk	0
management_risk	0
financial_flexibility	0
credibility	0
competitiveness	0
operating_risk	0
class	0

dtype: int64

◆ Missing Values Percentage:

industrial_risk	0.0
management_risk	0.0
financial_flexibility	0.0
credibility	0.0
competitiveness	0.0
operating_risk	0.0
class	0.0

dtype: float64

◆ Unique Value Counts:

industrial_risk	3
management_risk	3
financial_flexibility	3
credibility	3
competitiveness	3
operating_risk	3
class	2

dtype: int64

◆ Sample Categories per Categorical Column:

class: ['bankruptcy' 'non-bankruptcy']

◆ Correlation Matrix (Numerical):

	industrial_risk	management_risk	\
industrial_risk	1.000000	0.255127	
management_risk	0.255127	1.000000	
financial_flexibility	-0.162624	-0.254845	
credibility	-0.014438	-0.303341	
competitiveness	-0.257814	-0.306568	
operating_risk	0.144507	0.213874	

DESCRIPTIVE ANALYTICS REPORT

```
financial_flexibility  credibility \
industrial_risk      -0.162624    -0.014438
management_risk      -0.254845    -0.303341
financial_flexibility  1.000000     0.524951
credibility          0.524951     1.000000
competitiveness       0.686612     0.675689
operating_risk        -0.116903    -0.288458

competitiveness      operating_risk
industrial_risk      -0.257814     0.144507
management_risk      -0.306568     0.213874
financial_flexibility  0.686612    -0.116903
credibility          0.675689    -0.288458
competitiveness       1.000000    -0.211383
operating_risk        -0.211383     1.000000

• Skewness of Numerical Features:
industrial_risk      -0.067011
management_risk      -0.444701
financial_flexibility  0.479134
credibility          0.112955
competitiveness       0.093906
operating_risk        -0.275547
dtype: float64

• Kurtosis of Numerical Features:
industrial_risk      -1.521223
management_risk      -1.378700
financial_flexibility -1.290455
credibility          -1.545520
competitiveness       -1.711286
operating_risk        -1.624863
dtype: float64

✅ Descriptive Analytics Completed
```

Complete Descriptive Analytics (Inline)



Complete Descriptive Analytics (Inline)

Rows: 250 | Columns: 7



Sample rows

	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk	class
0	0.5	1.0	0.0	0.0	0.0	0.5	bankruptcy
1	0.0	1.0	0.0	0.0	0.0	1.0	bankruptcy
2	1.0	0.0	0.0	0.0	0.0	1.0	bankruptcy
3	0.5	0.0	0.0	0.5	0.0	1.0	bankruptcy
4	1.0	1.0	0.0	0.0	0.0	1.0	bankruptcy



DataFrame.info()

RangeIndex: 250 entries, 0 to 249
Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	industrial_risk	250 non-null	float64
1	management_risk	250 non-null	float64
2	financial_flexibility	250 non-null	float64
3	credibility	250 non-null	float64
4	competitiveness	250 non-null	float64
5	operating_risk	250 non-null	float64
6	class	250 non-null	object

dtypes: float64(6), object(1)
memory usage: 13.8+ KB



Dtypes and non-null counts

	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk	class
dtype	float64	float64	float64	float64	float64	float64	object
n_nonnull	250	250	250	250	250	250	250



Summary statistics (numerical)

	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max
industrial_risk	250.0	0.518	0.411526	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0
management_risk	250.0	0.614	0.410705	0.0	0.0	0.0	0.5	0.5	1.0	1.0	1.0	1.0
financial_flexibility	250.0	0.376	0.401583	0.0	0.0	0.0	0.0	0.5	0.5	1.0	1.0	1.0
credibility	250.0	0.470	0.415682	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0
competitiveness	250.0	0.476	0.440682	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0
operating_risk	250.0	0.570	0.434575	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0



Summary statistics (all dtypes)

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
industrial_risk	250.0	NaN	NaN	NaN	0.518	0.411526	0.0	0.0	0.5	1.0	1.0
management_risk	250.0	NaN	NaN	NaN	0.614	0.410705	0.0	0.5	0.5	1.0	1.0
financial_flexibility	250.0	NaN	NaN	NaN	0.376	0.401583	0.0	0.0	0.5	0.5	1.0
credibility	250.0	NaN	NaN	NaN	0.47	0.415682	0.0	0.0	0.5	1.0	1.0
competitiveness	250.0	NaN	NaN	NaN	0.476	0.440682	0.0	0.0	0.5	1.0	1.0
operating_risk	250.0	NaN	NaN	NaN	0.57	0.434575	0.0	0.0	0.5	1.0	1.0
class	250	2	non-bankruptcy	143	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Complete Descriptive Analytics (Inline)

! Missing values (count & %)

	missing_count	missing_pct
industrial_risk	0	0.0
management_risk	0	0.0
financial_flexibility	0	0.0
credibility	0	0.0
competitiveness	0	0.0

1234 Unique value counts

	unique_count
industrial_risk	3
management_risk	3
financial_flexibility	3
credibility	3
competitiveness	3
operating_risk	3
class	2

Numeric columns (6): ['industrial_risk', ' management_risk', ' financial_flexibility', ' credibility', ' competitiveness', ' operating_risk']

Categorical columns (1): [' class']

Datetime columns (0): []

▲ Skewness & Kurtosis (numeric)

	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk
skewness	-0.067011	-0.444701	0.479134	0.112955	0.093906	-0.275547
kurtosis	-1.521223	-1.378700	-1.290455	-1.545520	-1.711286	-1.624863

🔗 Correlation (Pearson) heatmap (numeric)

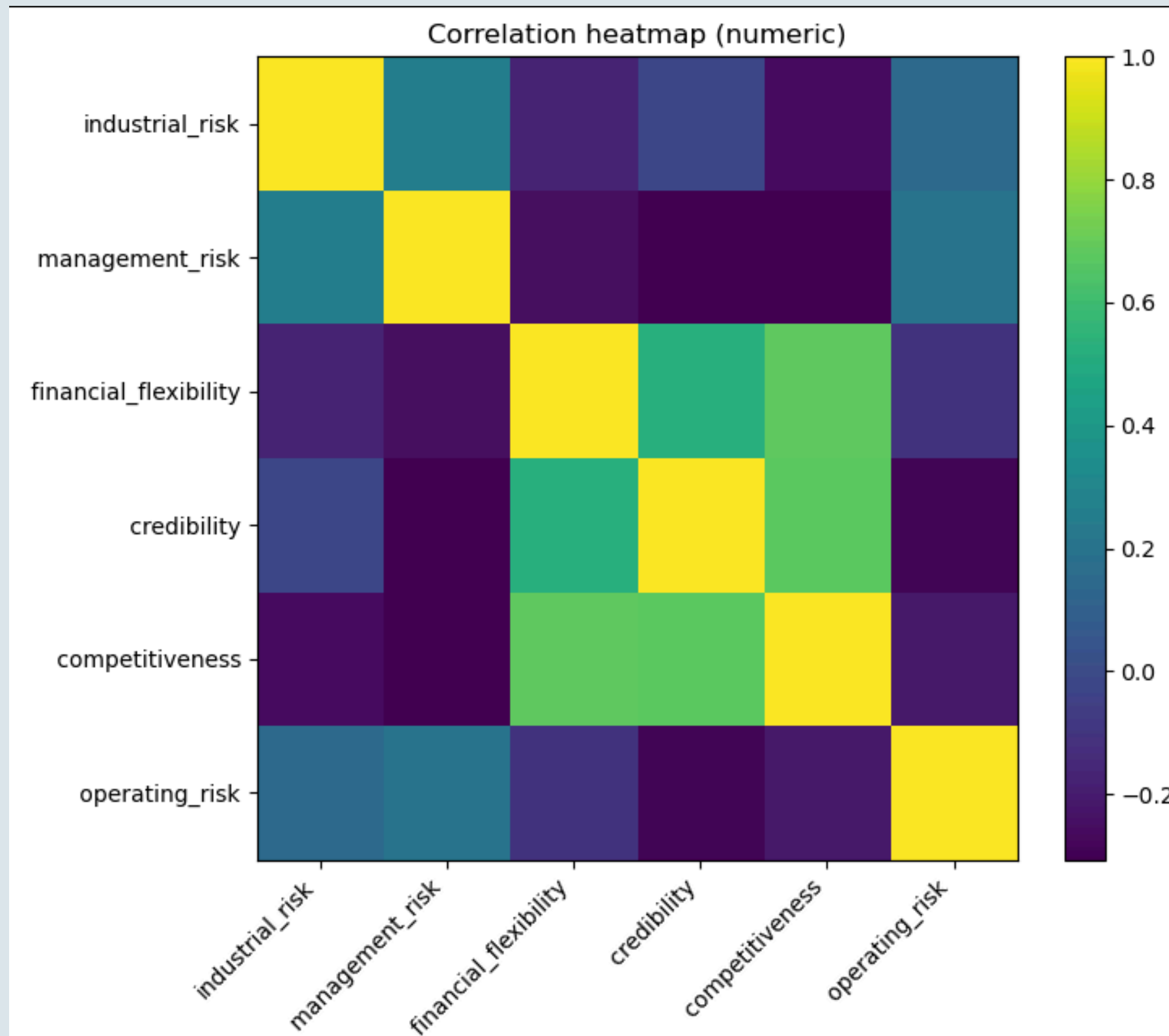
	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk
industrial_risk	1.000000	0.255127	-0.162624	-0.014438	-0.257814	0.144507
management_risk	0.255127	1.000000	-0.254845	-0.303341	-0.306568	0.213874
financial_flexibility	-0.162624	-0.254845	1.000000	0.524951	0.686612	-0.116903
credibility	-0.014438	-0.303341	0.524951	1.000000	0.675689	-0.288458
competitiveness	-0.257814	-0.306568	0.686612	0.675689	1.000000	-0.211383
operating_risk	0.144507	0.213874	-0.116903	-0.288458	-0.211383	1.000000

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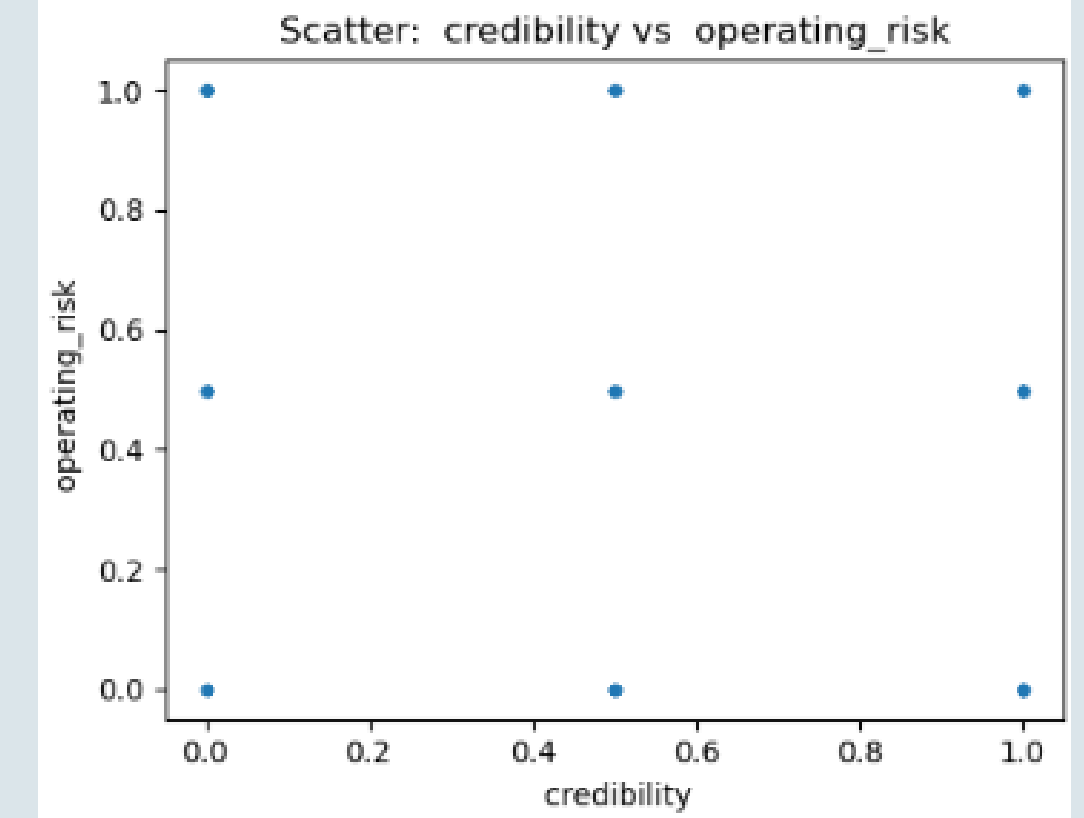
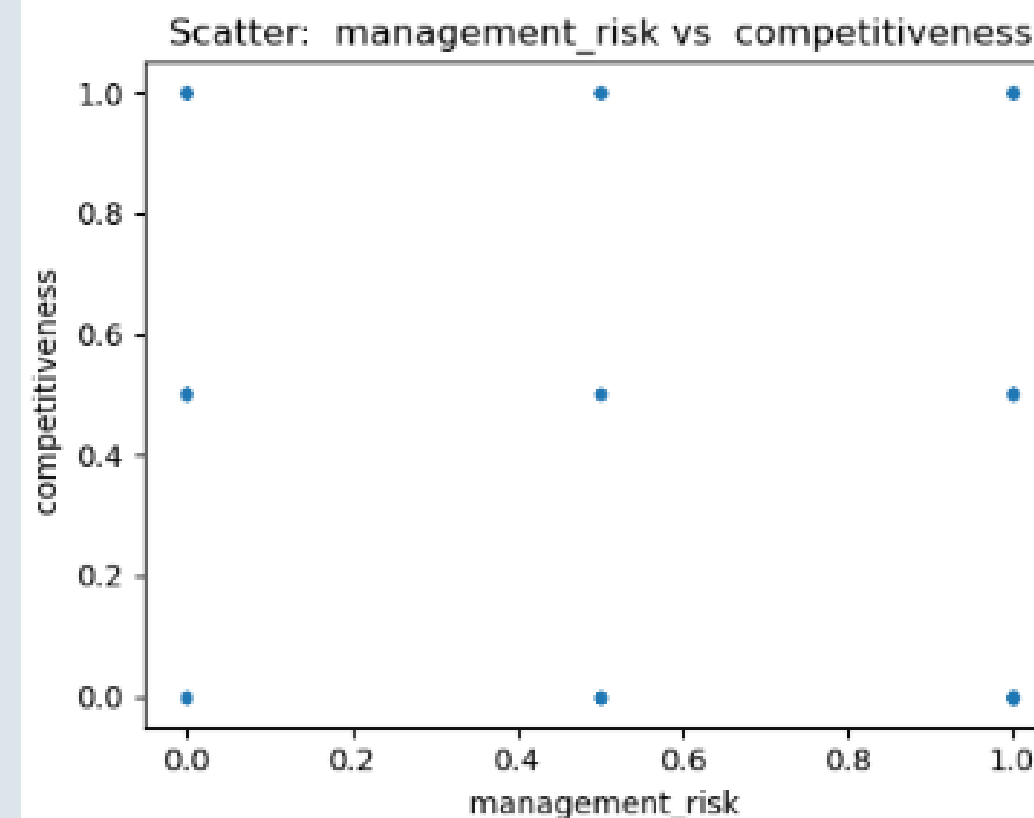
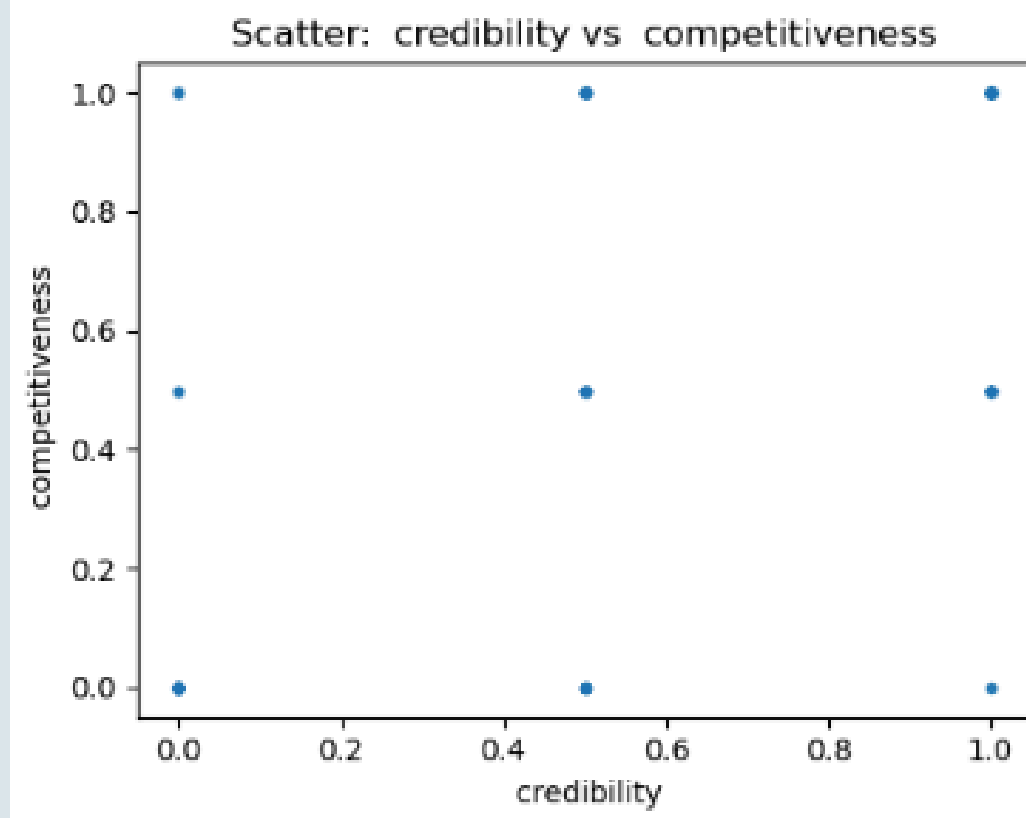
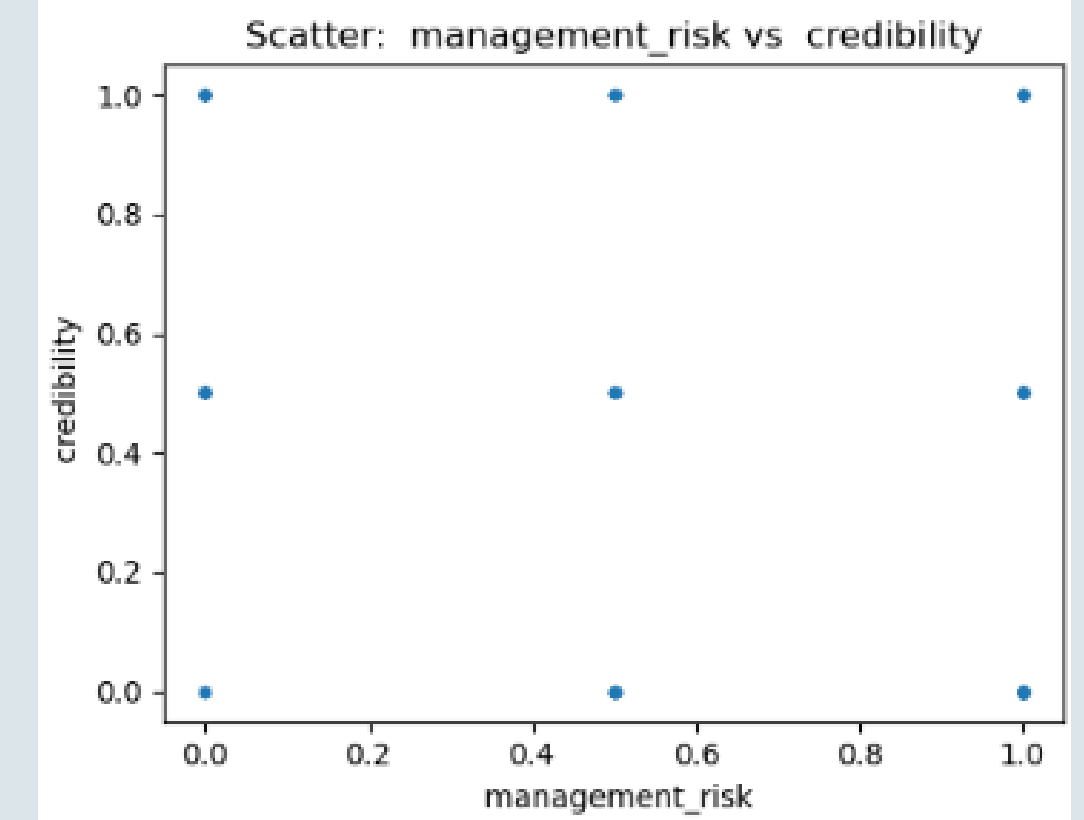
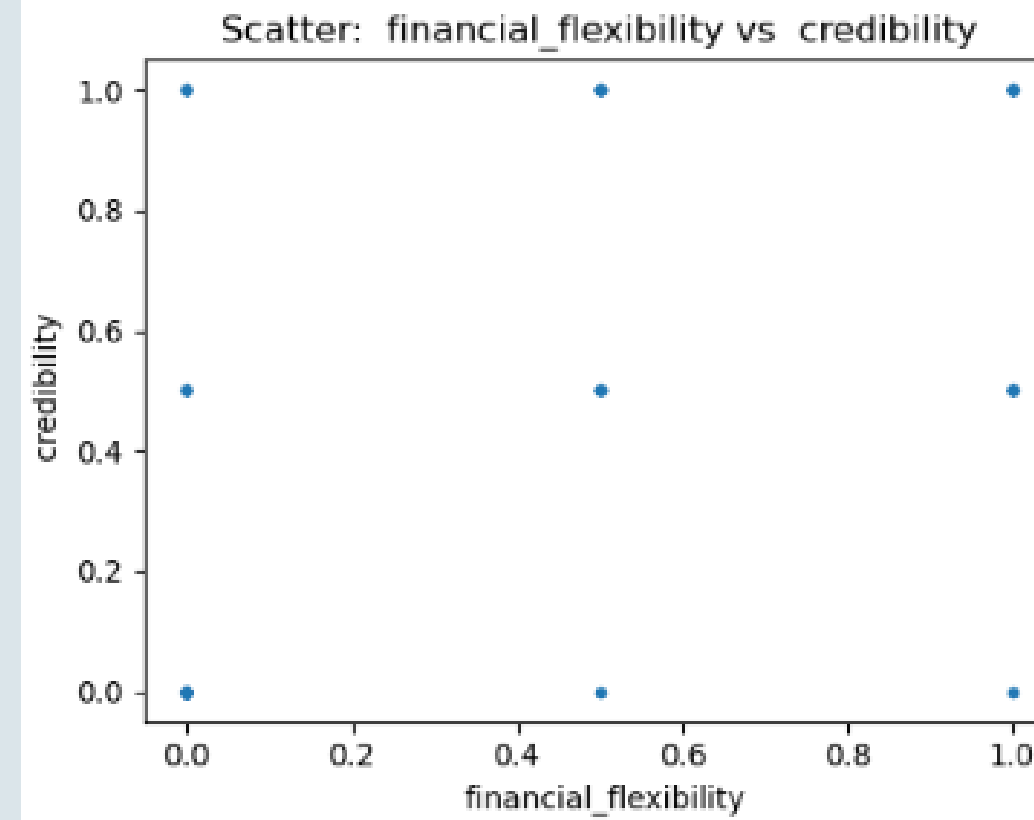
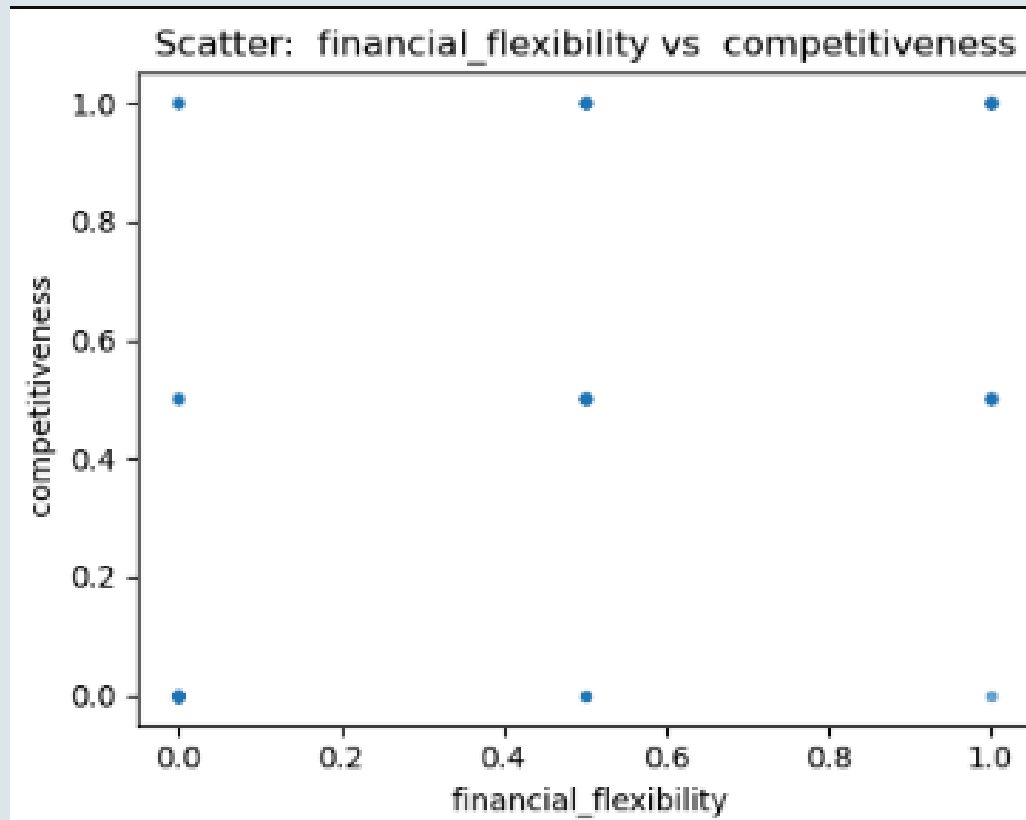
Complete Descriptive Analytics (Inline)



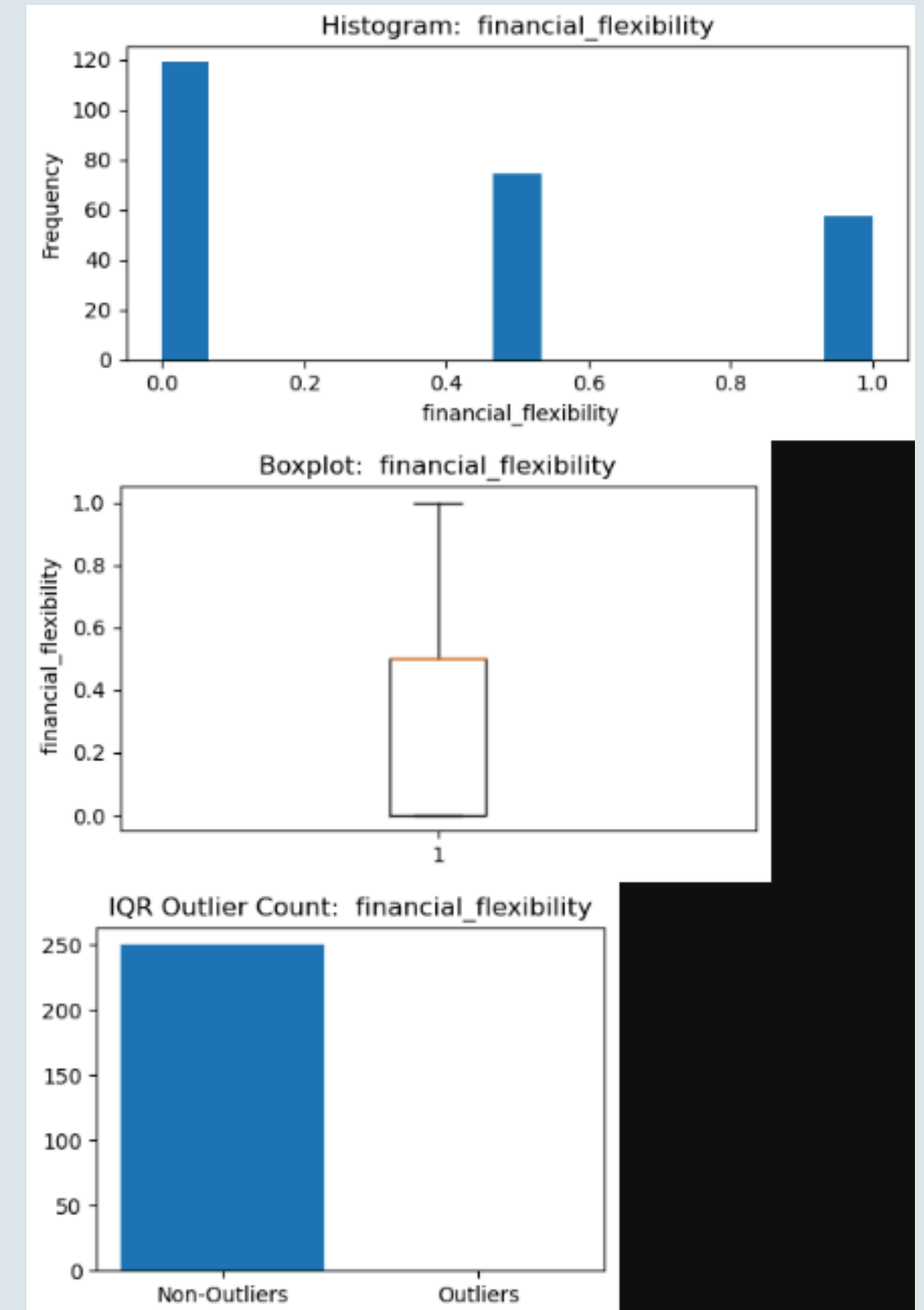
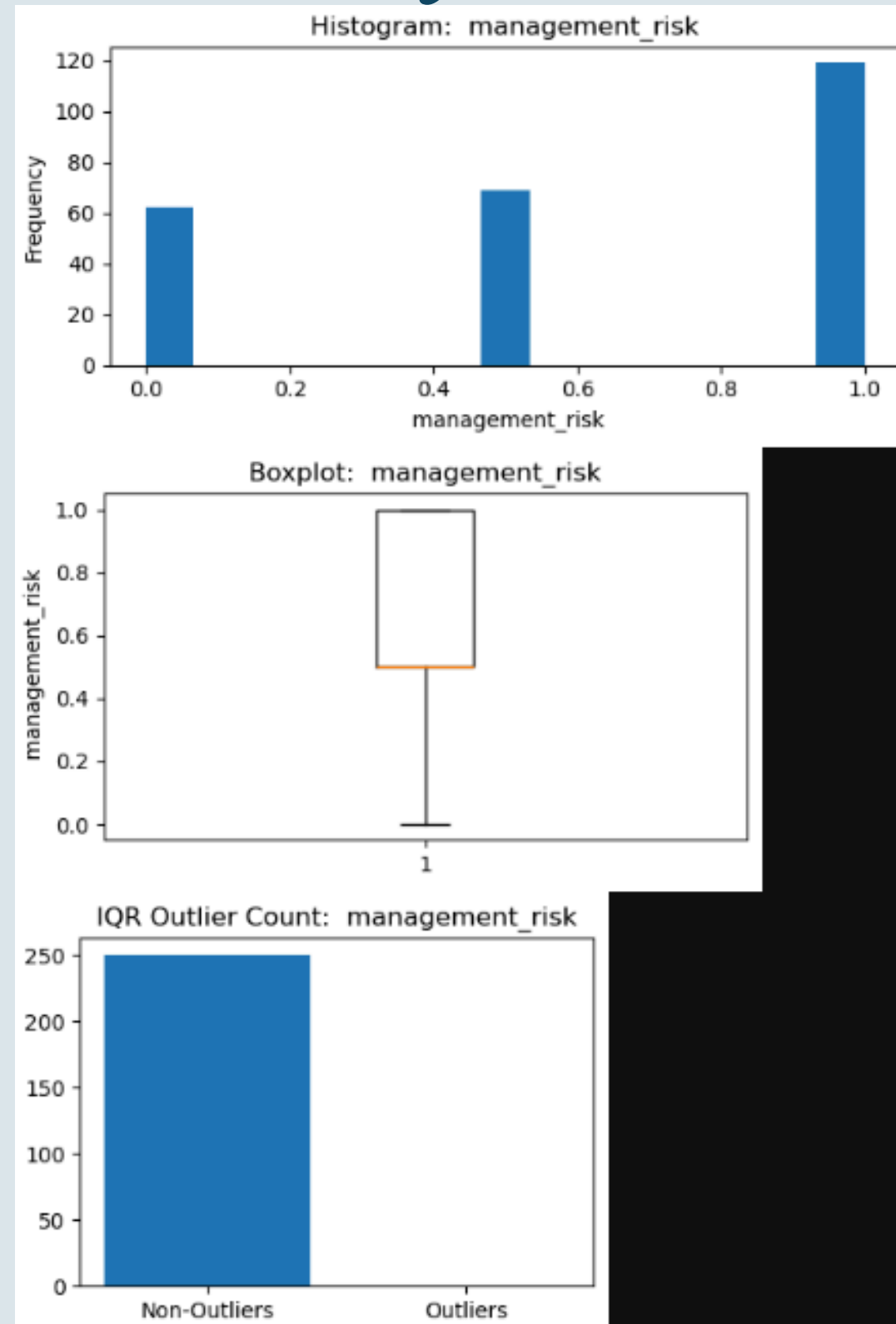
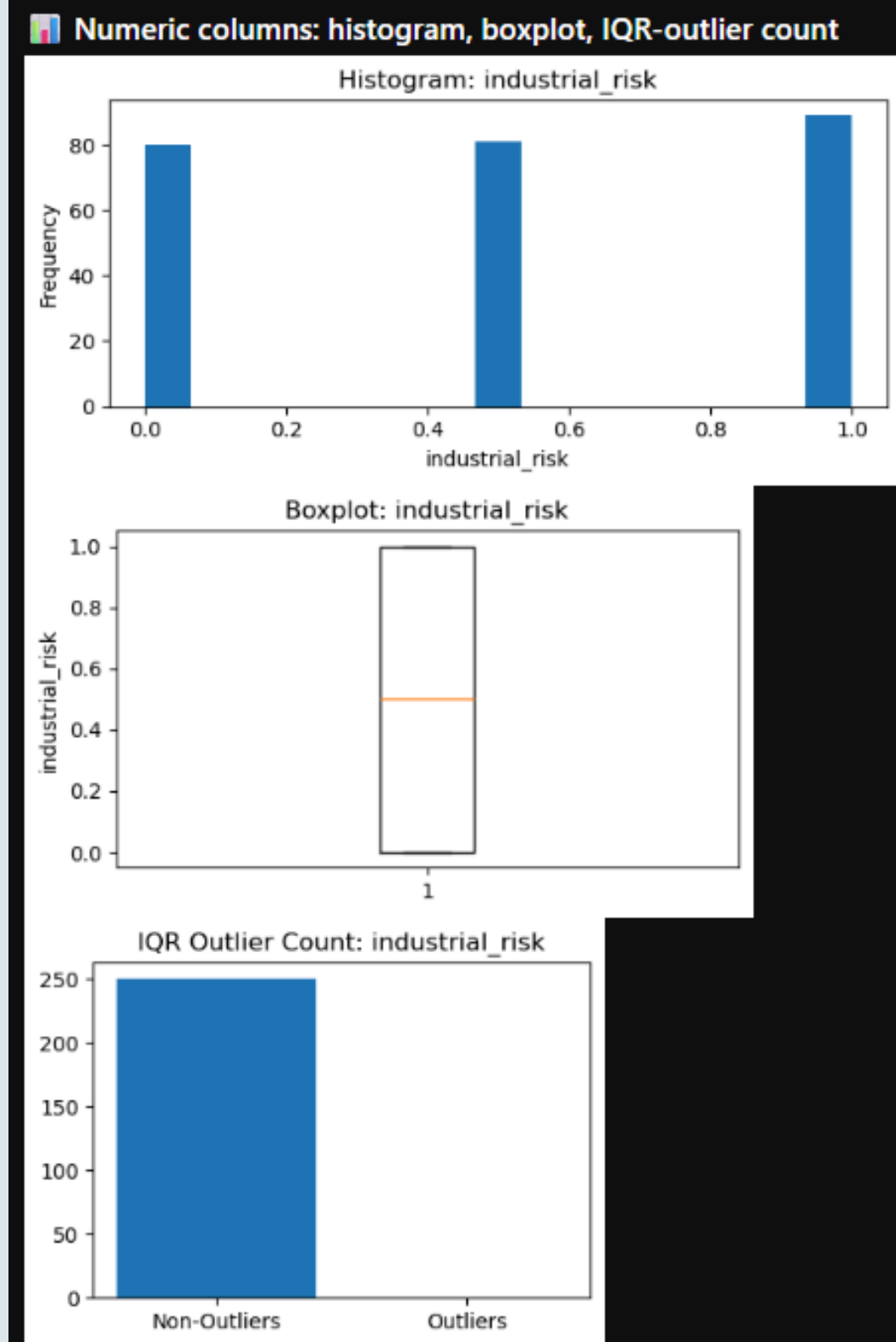
Top 6 correlated numeric pairs (by abs)

	col1	col2	abs_corr
0	financial_flexibility	competitiveness	0.686612
1	credibility	competitiveness	0.675689
2	financial_flexibility	credibility	0.524951
3	management_risk	competitiveness	0.306568
4	management_risk	credibility	0.303341
5	credibility	operating_risk	0.288458

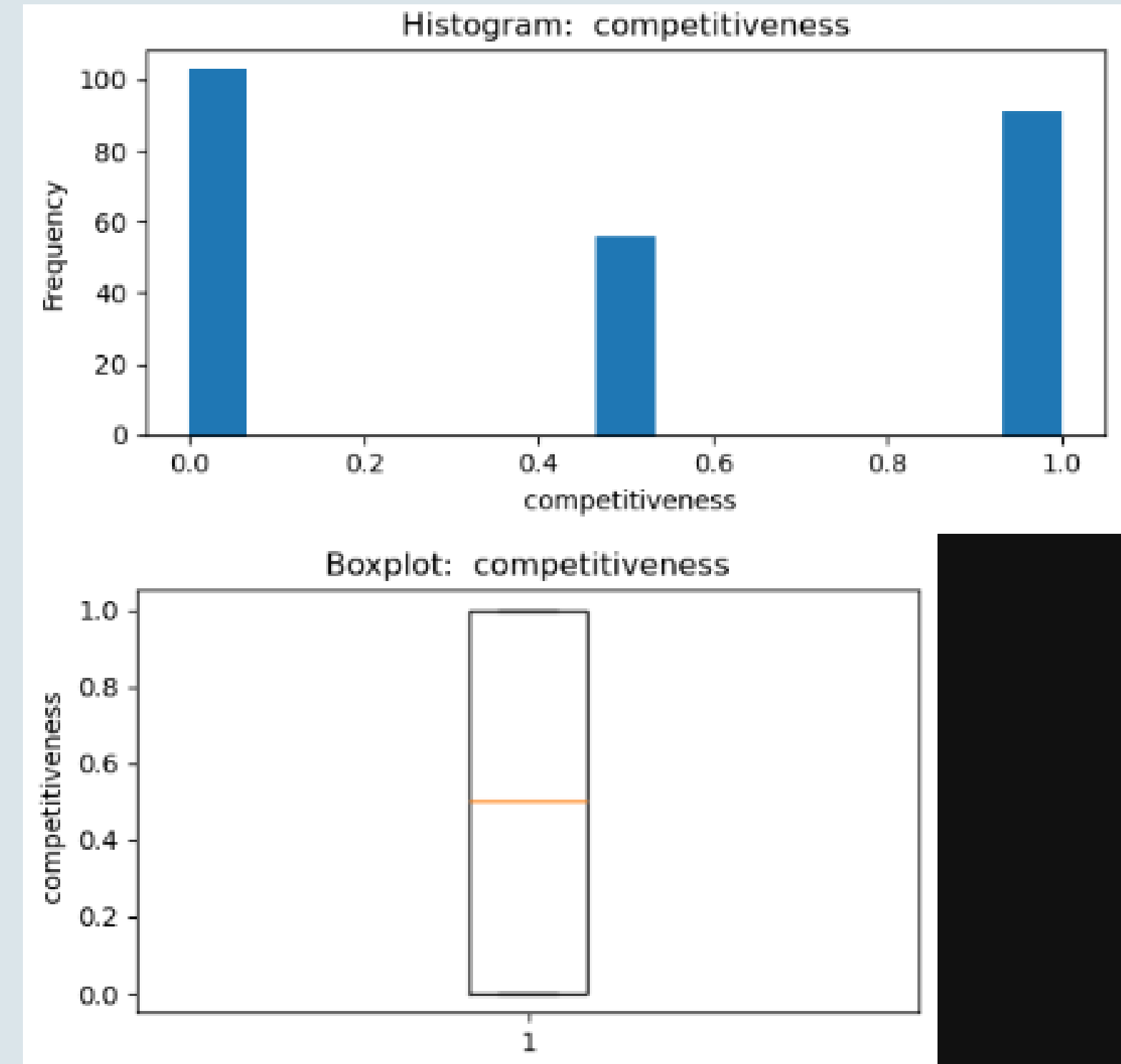
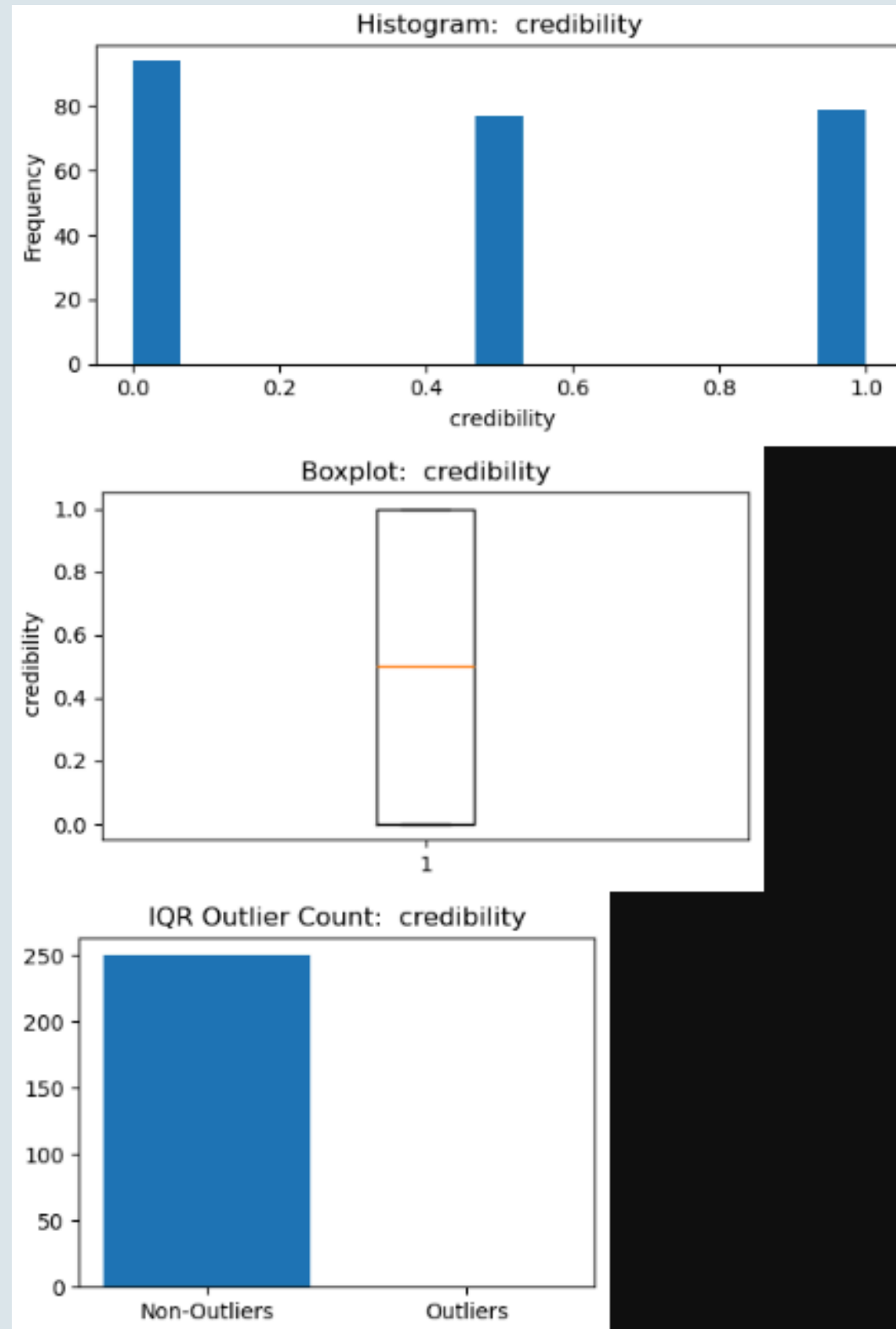
Complete Descriptive Analytics (Inline)



Complete Descriptive Analytics (Inline)



Complete Descriptive Analytics (Inline)



Basic Data Overview

Shape of Data

- Rows: 250
- Columns: 7

Data Types & Non-Null Counts

RangeIndex: 250 entries, 0 to 249

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	industrial_risk	250 non-null	float64
1	management_risk	250 non-null	float64
2	financial_flexibility	250 non-null	float64
3	credibility	250 non-null	float64
4	competitiveness	250 non-null	float64
5	operating_risk	250 non-null	float64
6	class	250 non-null	object

dtypes: float64(6), object(1)

memory usage: 13.8+ KB

First 5 Rows

	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk	class
0	0.5	1.0	0.0	0.0	0.0	0.5	bankruptcy
1	0.0	1.0	0.0	0.0	0.0	1.0	bankruptcy
2	1.0	0.0	0.0	0.0	0.0	1.0	bankruptcy
3	0.5	0.0	0.0	0.5	0.0	1.0	bankruptcy
4	1.0	1.0	0.0	0.0	0.0	1.0	bankruptcy



Summary Statistics

Numeric Columns

	count	mean	std	min	25%	50%	75%	max
industrial_risk	250.0	0.518	0.411526	0.0	0.0	0.5	1.0	1.0
management_risk	250.0	0.614	0.410705	0.0	0.5	0.5	1.0	1.0
financial_flexibility	250.0	0.376	0.401583	0.0	0.0	0.5	0.5	1.0
credibility	250.0	0.470	0.415682	0.0	0.0	0.5	1.0	1.0
competitiveness	250.0	0.476	0.440682	0.0	0.0	0.5	1.0	1.0
operating_risk	250.0	0.570	0.434575	0.0	0.0	0.5	1.0	1.0

Categorical Columns

	count	unique	top	freq
class	250	2	non-bankruptcy	143

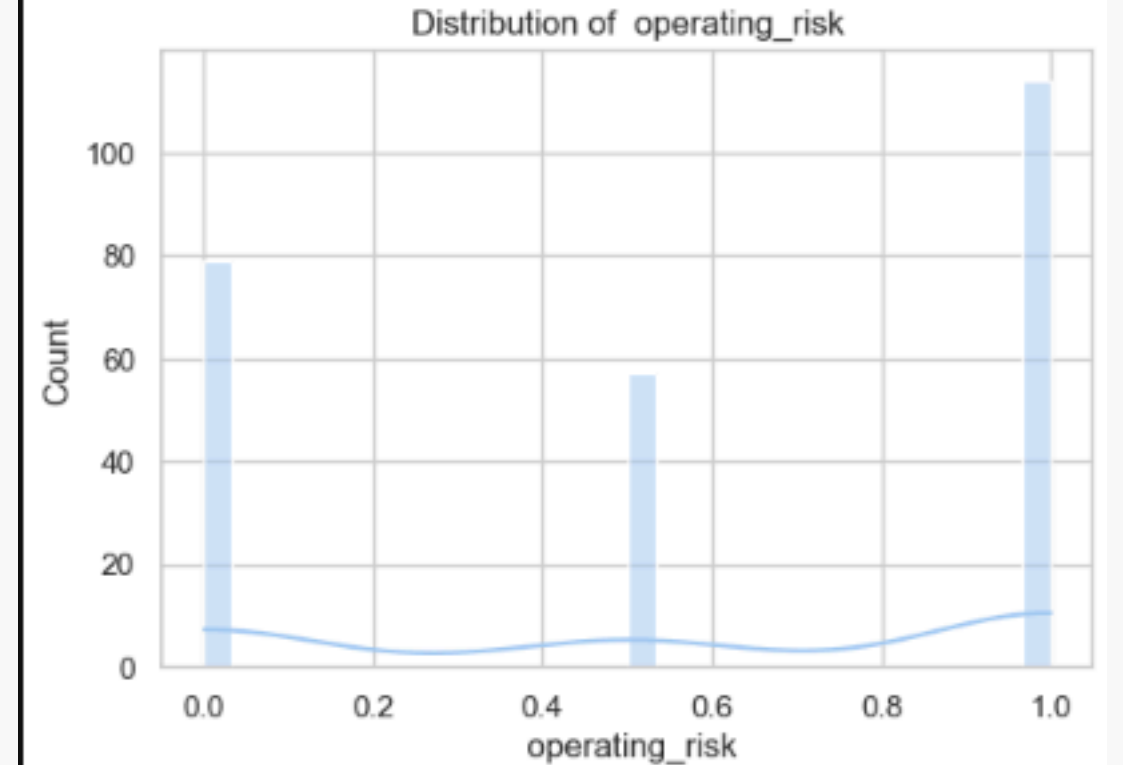
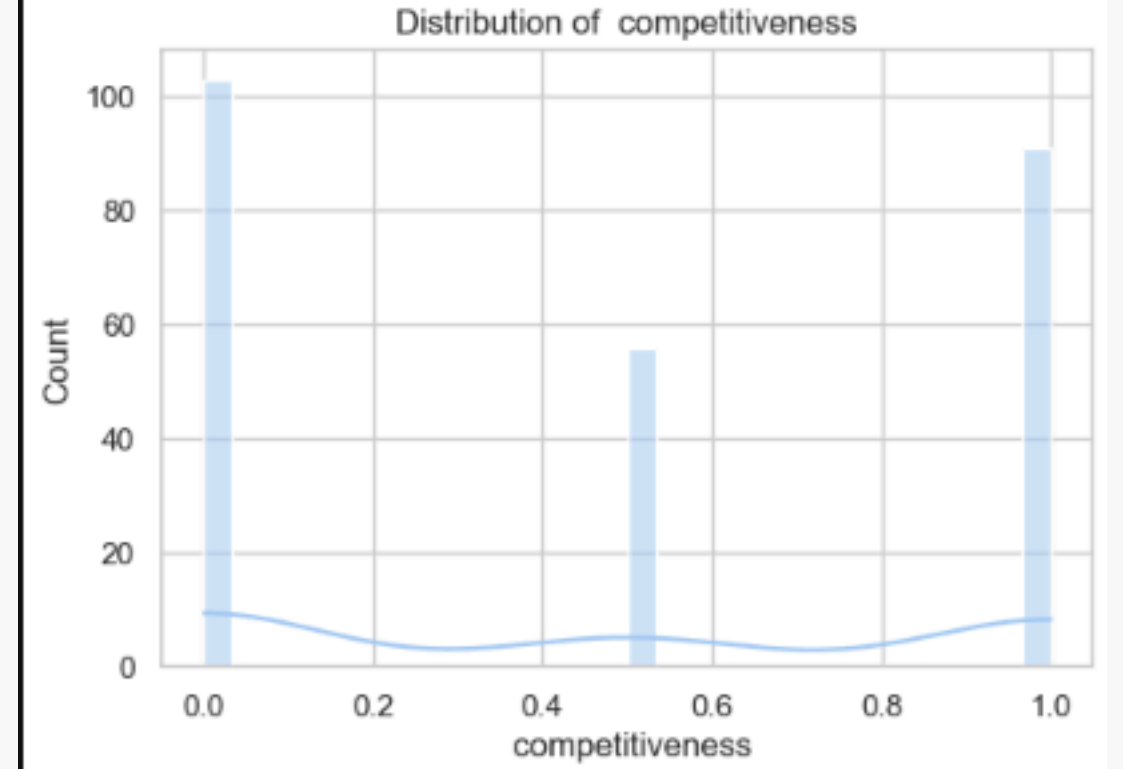
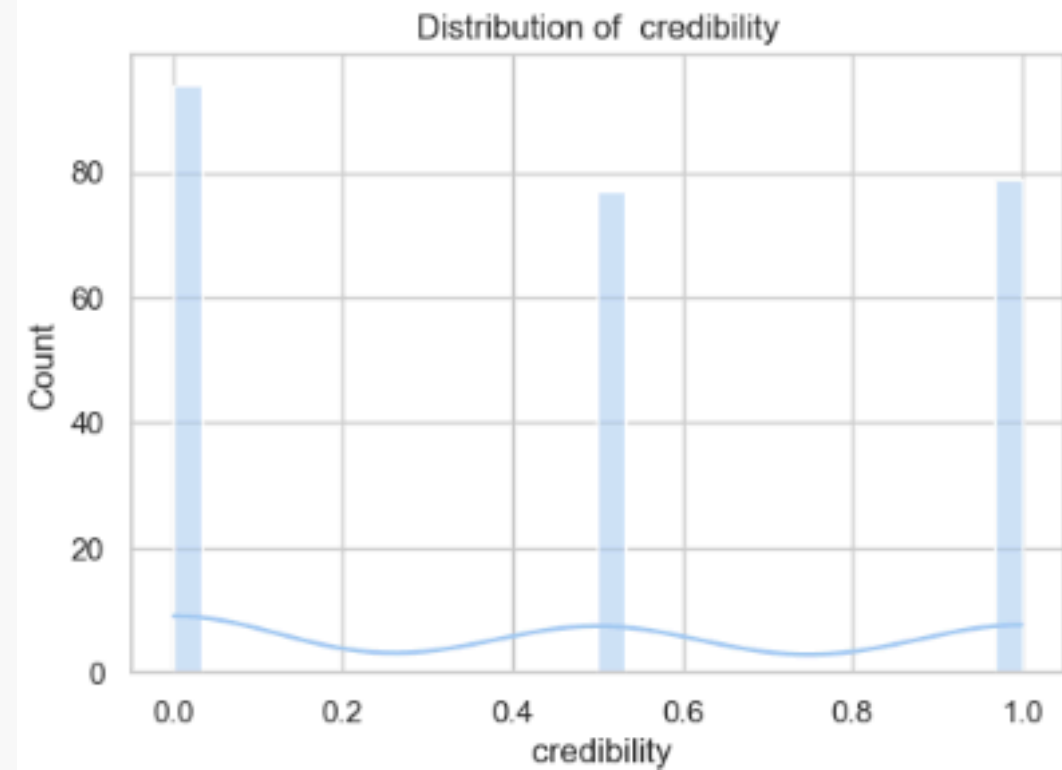
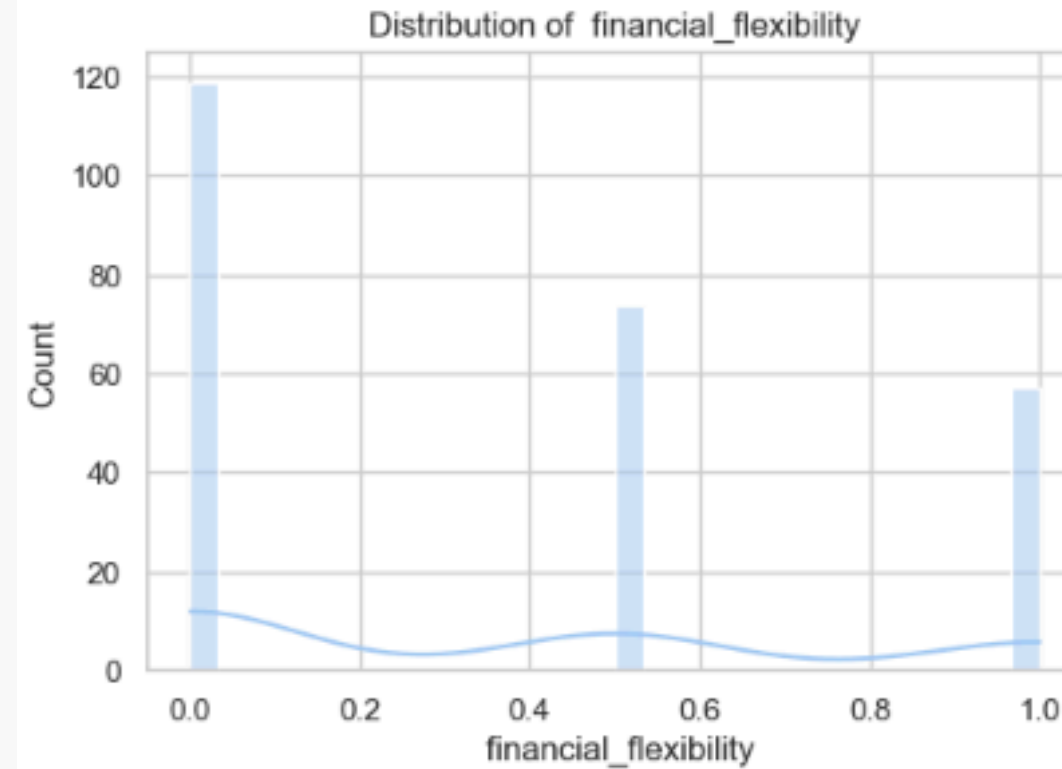
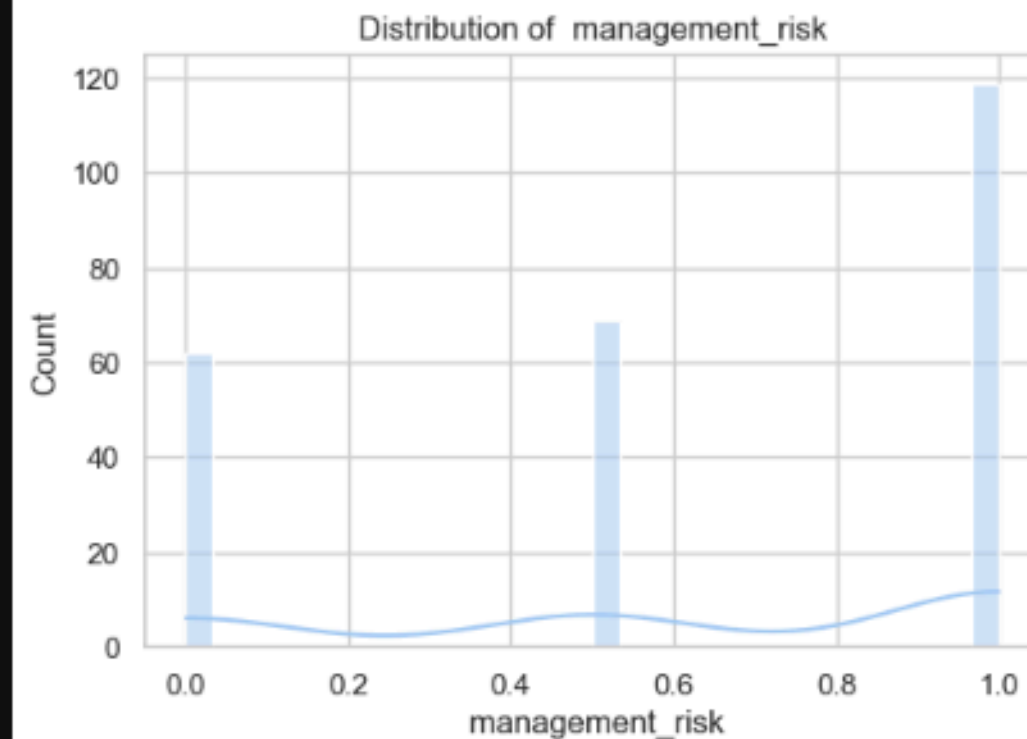
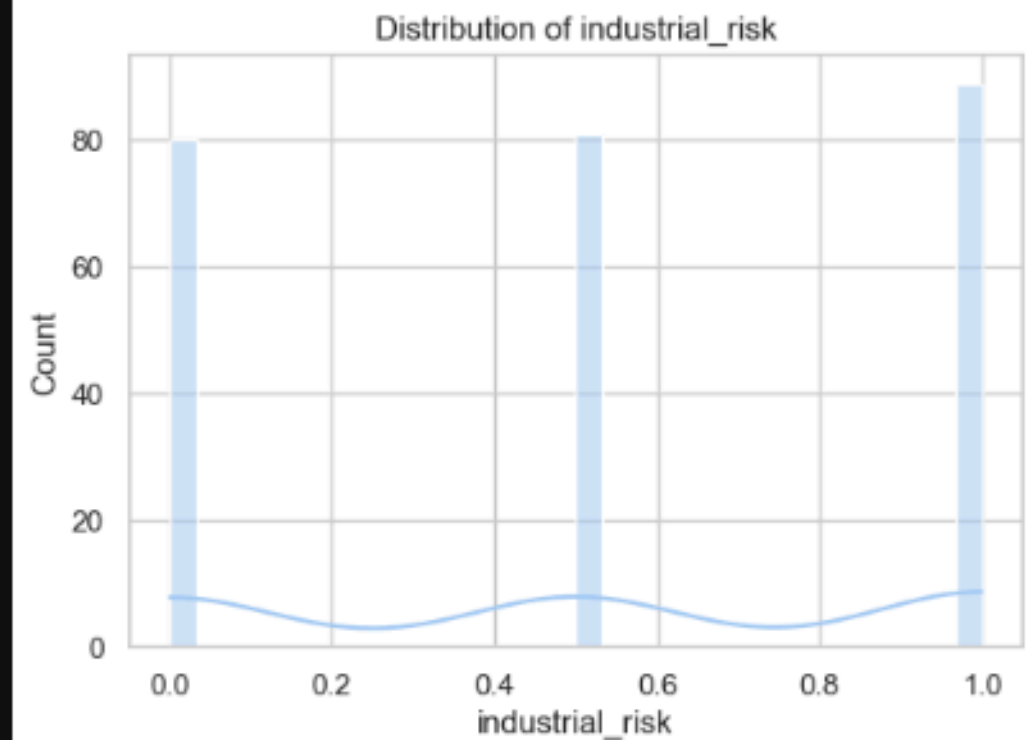


Missing Values

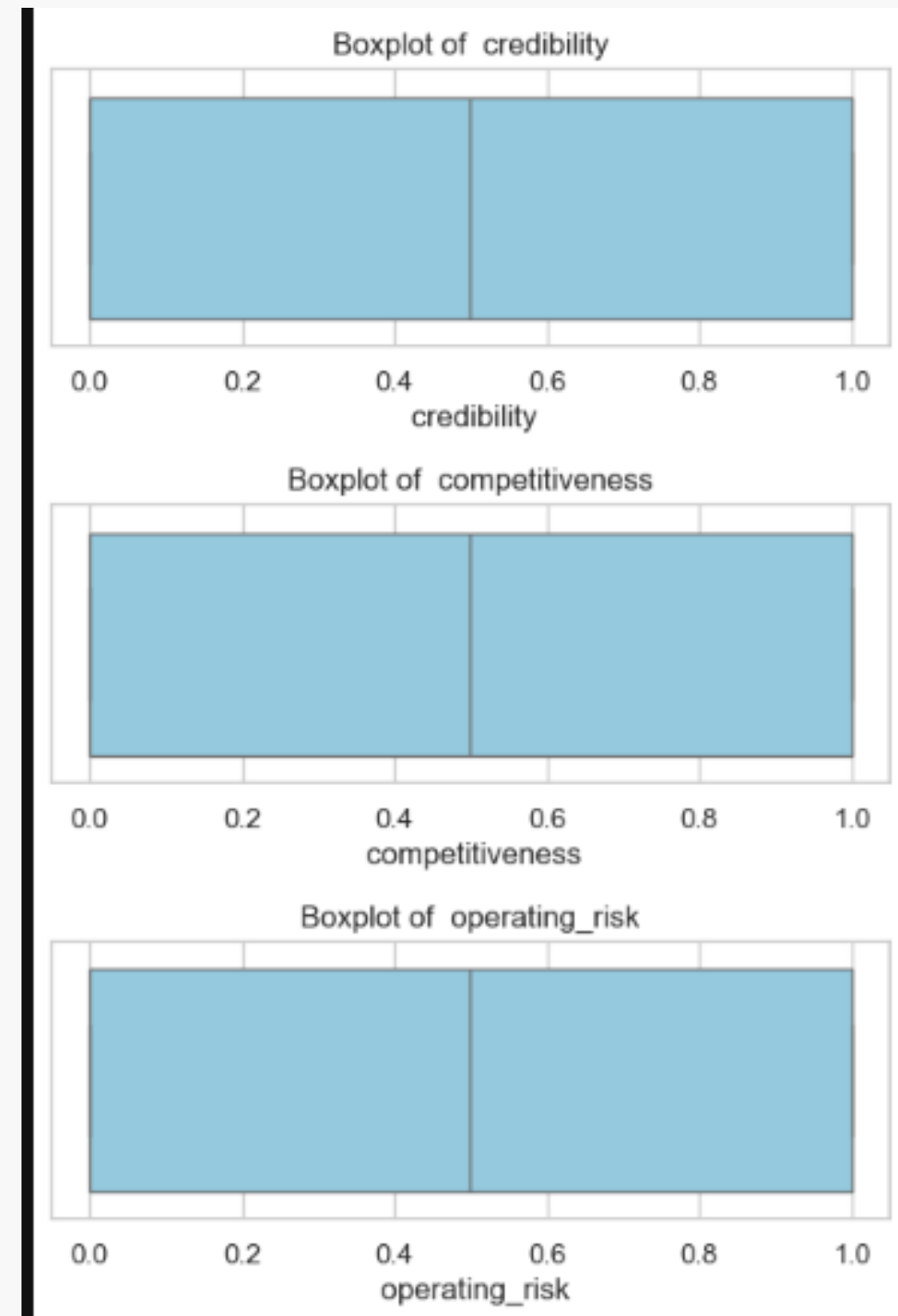
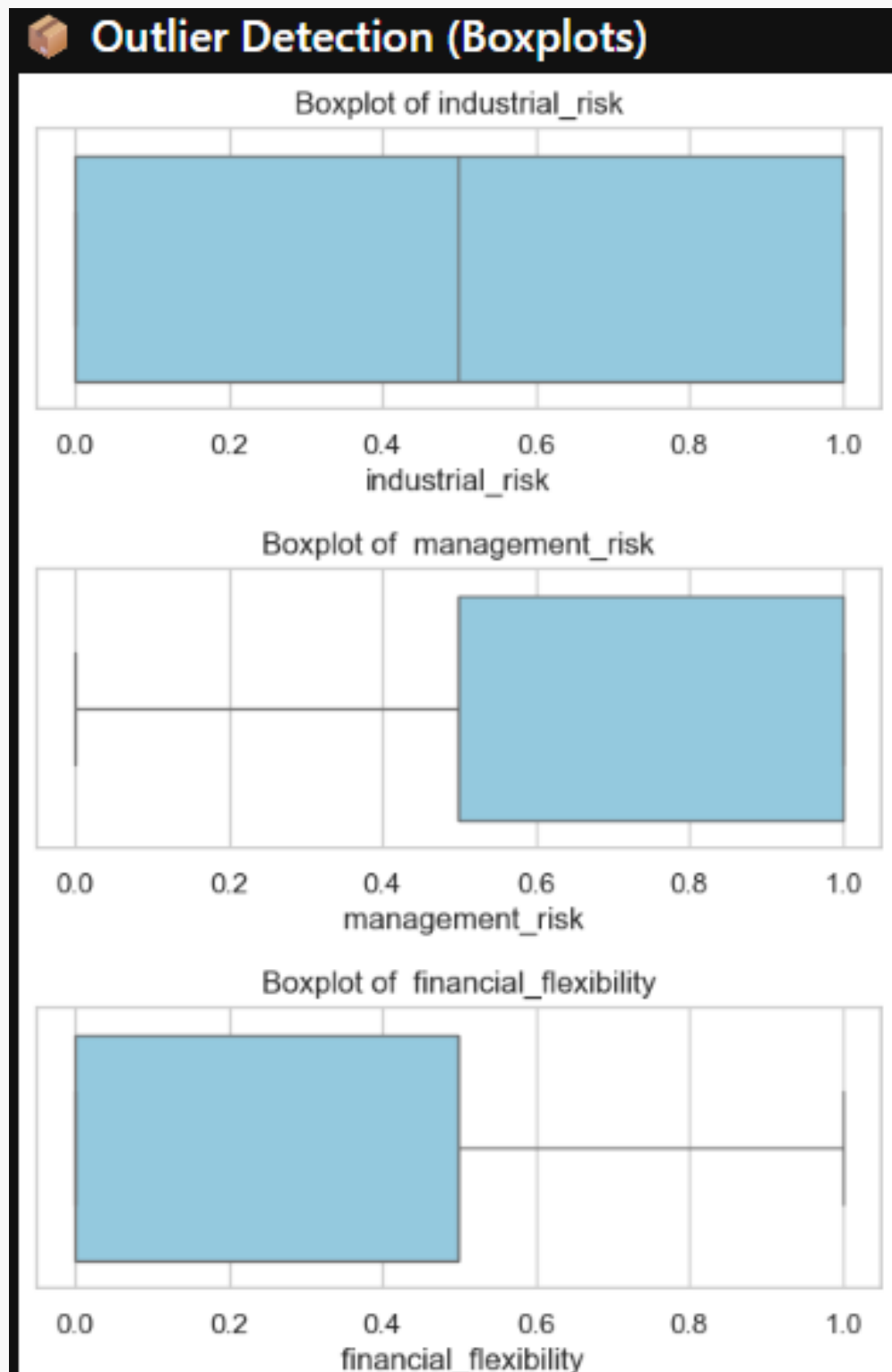
✓ No missing values found

Basic Data Overview

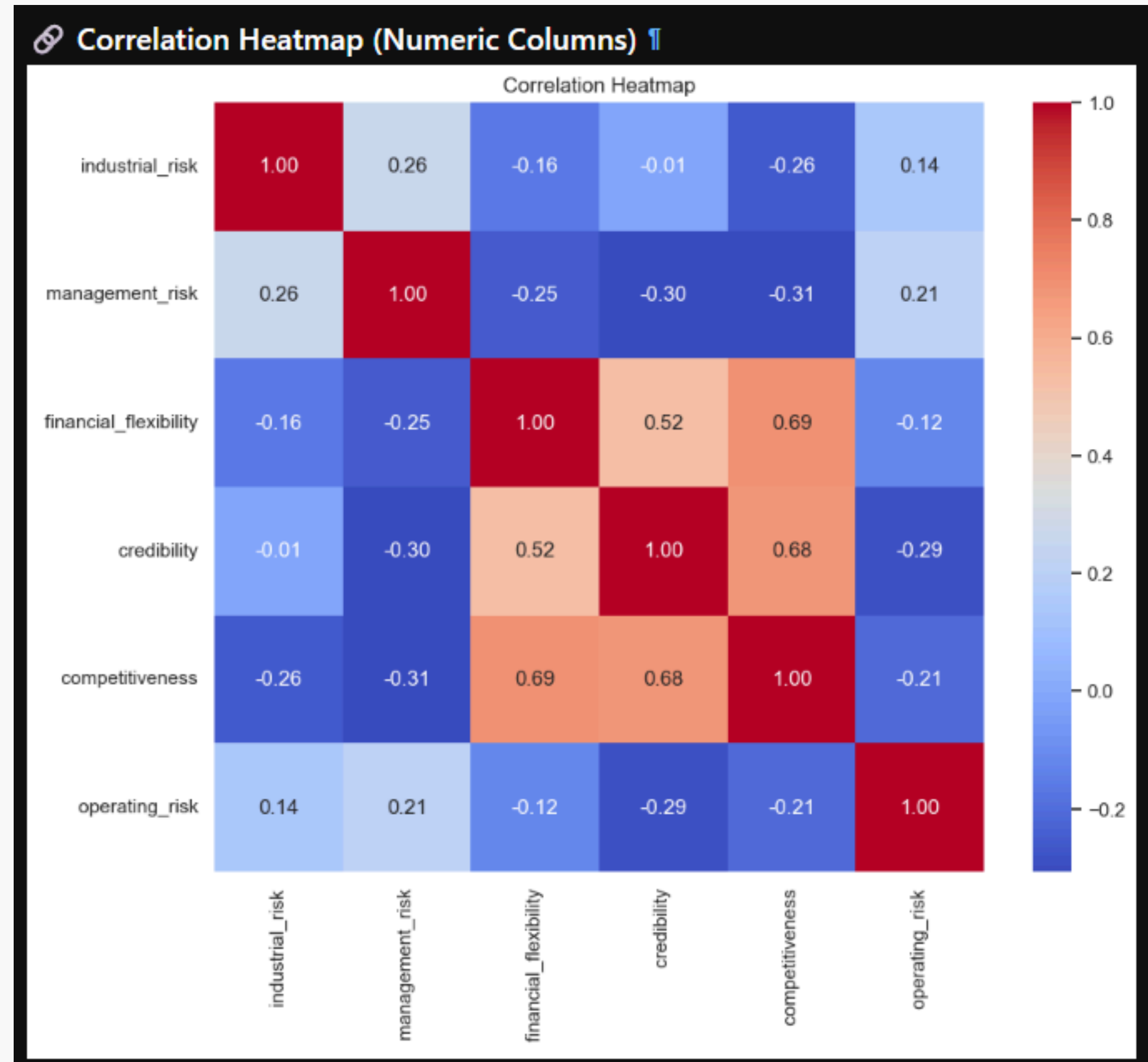
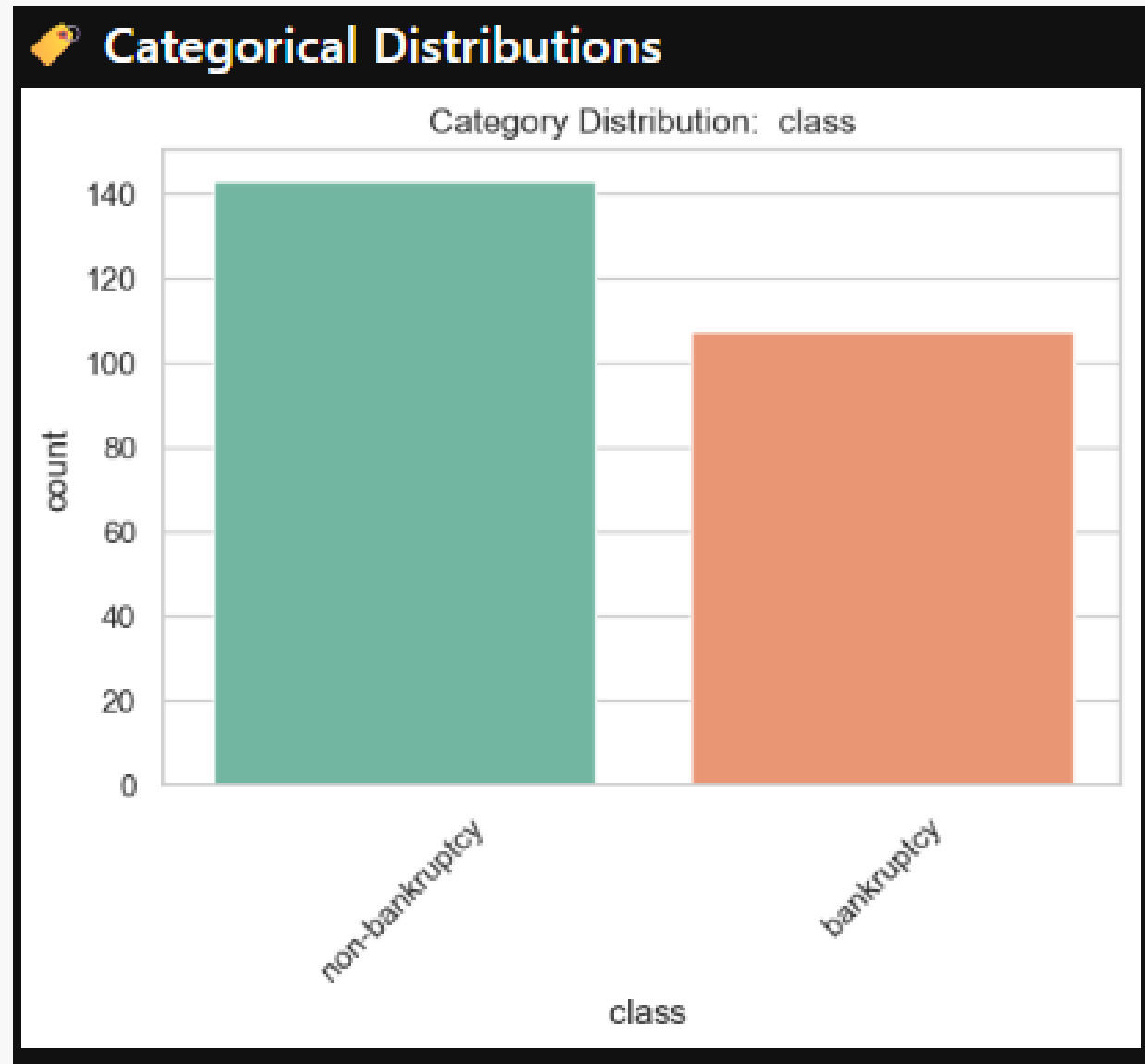
Numeric Distributions



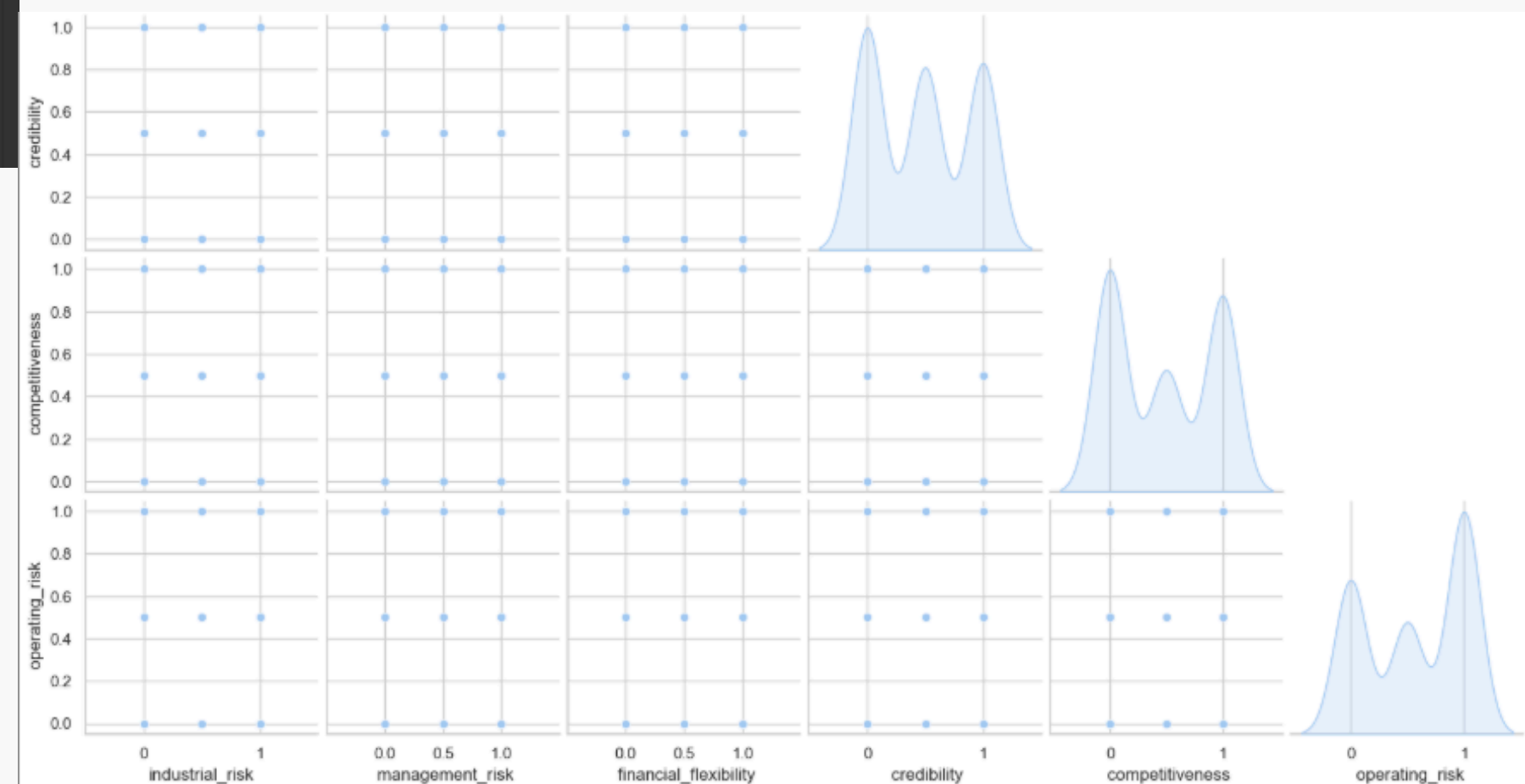
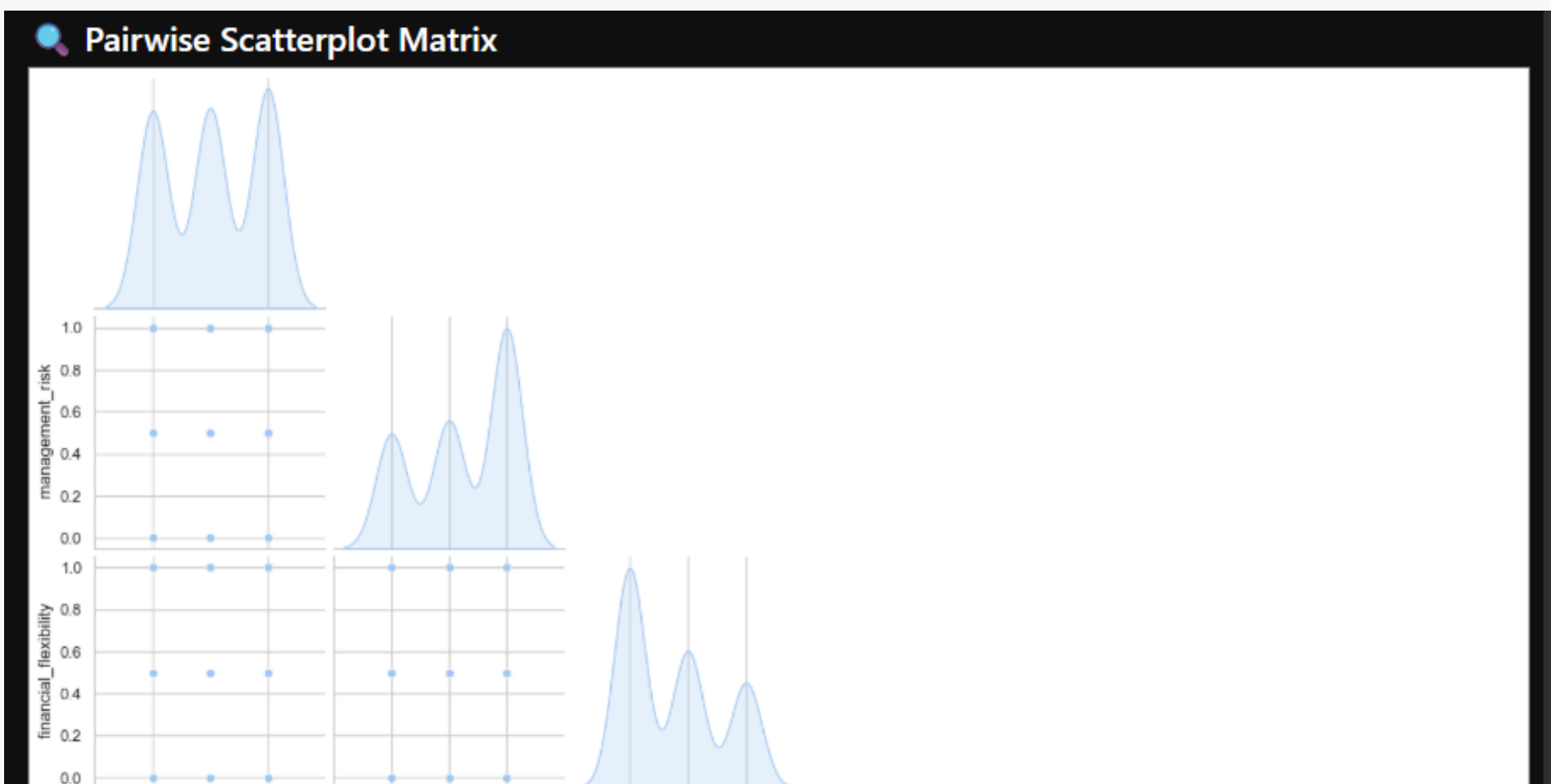
Basic Data Overview



Basic Data Overview



Basic Data Overview



Complete EDA (Inline)

Rows: 250 | Columns: 7

🔍 Head (first 5 rows)

	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk	class
0	0.5	1.0	0.0	0.0	0.0	0.5	bankruptcy
1	0.0	1.0	0.0	0.0	0.0	1.0	bankruptcy
2	1.0	0.0	0.0	0.0	0.0	1.0	bankruptcy
3	0.5	0.0	0.0	0.5	0.0	1.0	bankruptcy
4	1.0	1.0	0.0	0.0	0.0	1.0	bankruptcy

📄 DataFrame.info()

RangeIndex: 250 entries, 0 to 249
Data columns (total 7 columns):
Column Non-Null Count Dtype

0 industrial_risk 250 non-null float64
1 management_risk 250 non-null float64
2 financial_flexibility 250 non-null float64
3 credibility 250 non-null float64
4 competitiveness 250 non-null float64
5 operating_risk 250 non-null float64
6 class 250 non-null object
dtypes: float64(6), object(1)
memory usage: 13.8+ KB

🕒 Dtypes & non-null counts

	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk	class
dtype	float64	float64	float64	float64	float64	float64	object
non_null_count	250	250	250	250	250	250	250

📊 Summary Statistics

Numeric summary

	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max
industrial_risk	250.0	0.518	0.411526	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0
management_risk	250.0	0.614	0.410705	0.0	0.0	0.0	0.5	0.5	1.0	1.0	1.0	1.0
financial_flexibility	250.0	0.376	0.401583	0.0	0.0	0.0	0.0	0.5	0.5	1.0	1.0	1.0
credibility	250.0	0.470	0.415682	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0
competitiveness	250.0	0.476	0.440682	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0
operating_risk	250.0	0.570	0.434575	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0	1.0

All-types summary

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
industrial_risk	250.0	NaN	NaN	NaN	0.518	0.411526	0.0	0.0	0.5	1.0	1.0
management_risk	250.0	NaN	NaN	NaN	0.614	0.410705	0.0	0.5	0.5	1.0	1.0
financial_flexibility	250.0	NaN	NaN	NaN	0.376	0.401583	0.0	0.0	0.5	0.5	1.0
credibility	250.0	NaN	NaN	NaN	0.47	0.415682	0.0	0.0	0.5	1.0	1.0
competitiveness	250.0	NaN	NaN	NaN	0.476	0.440682	0.0	0.0	0.5	1.0	1.0
operating_risk	250.0	NaN	NaN	NaN	0.57	0.434575	0.0	0.0	0.5	1.0	1.0
class	250	2 non-bankruptcy	143	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

! Missing Values

✅ No missing values found.

Complete EDA (Inline)

Uniques, Cardinality & Constants

	unique_count	unique_pct
industrial_risk	3	1.2
management_risk	3	1.2
financial_flexibility	3	1.2
credibility	3	1.2
competitiveness	3	1.2
operating_risk	3	1.2
class	2	0.8

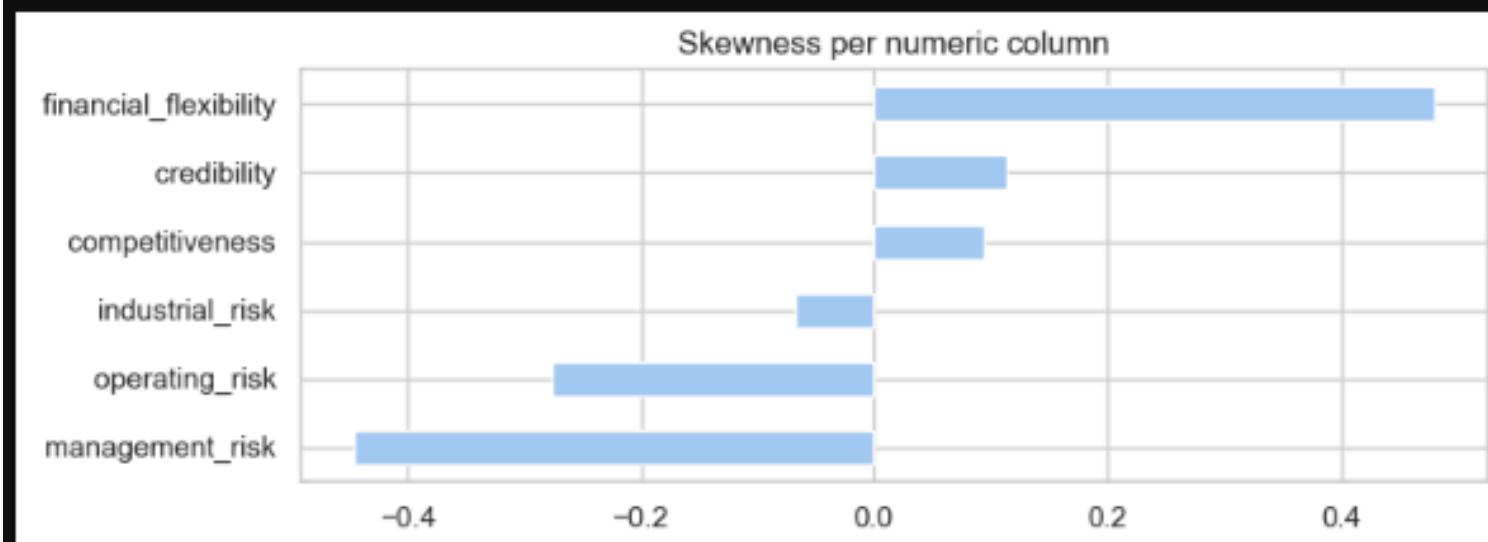
Numeric columns (6): ['industrial_risk', 'management_risk', 'financial_flexibility', 'credibility', 'competitiveness', 'operating_risk']

Categorical columns (1): ['class']

Datetime columns (0): []

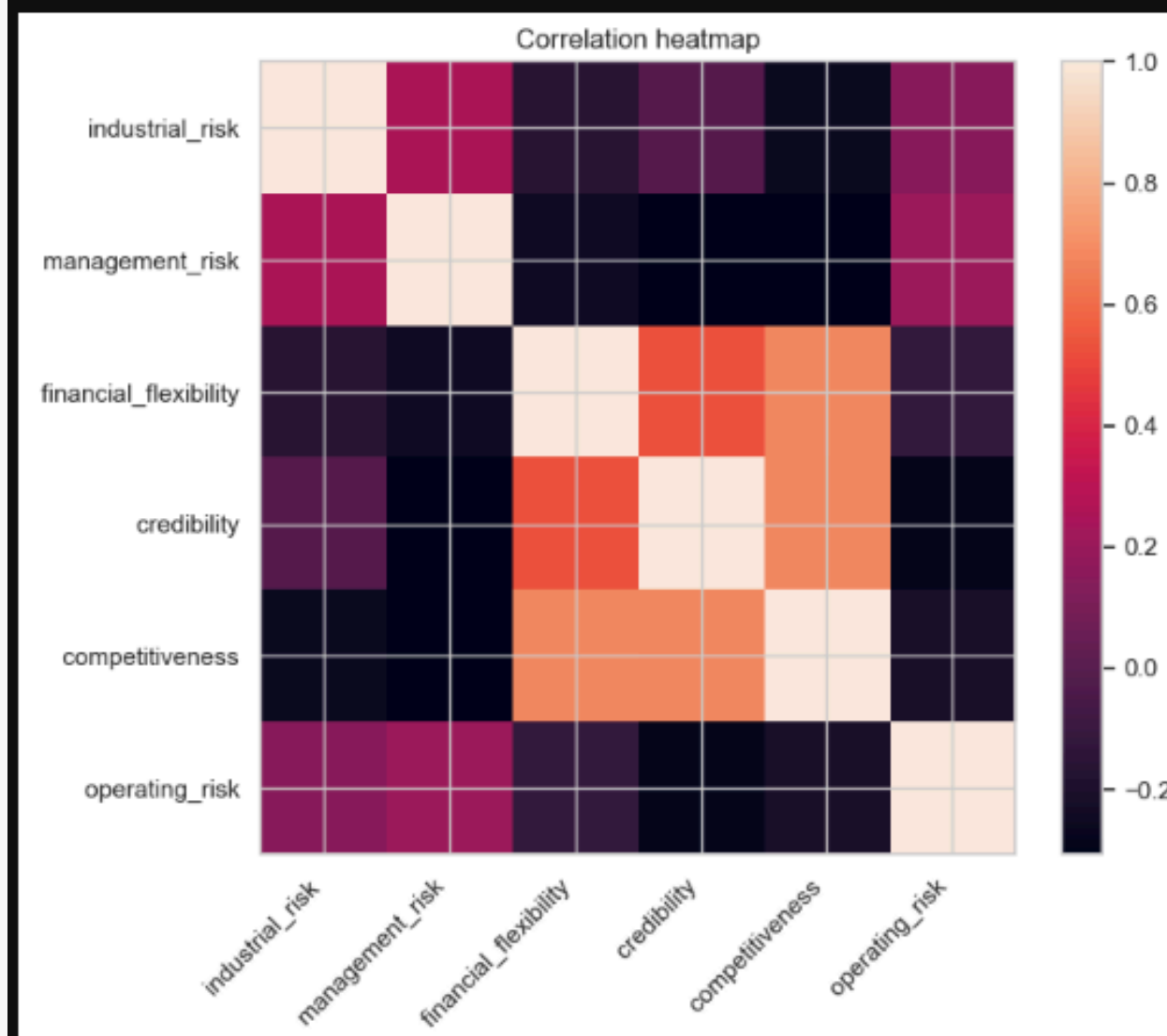
▲ Numeric Diagnostics: Skewness & Kurtosis

	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk
skewness	-0.067011	-0.444701	0.479134	0.112955	0.093906	-0.275547
kurtosis	-1.521223	-1.378700	-1.290455	-1.545520	-1.711286	-1.624863



Correlation (Pearson) - Numeric

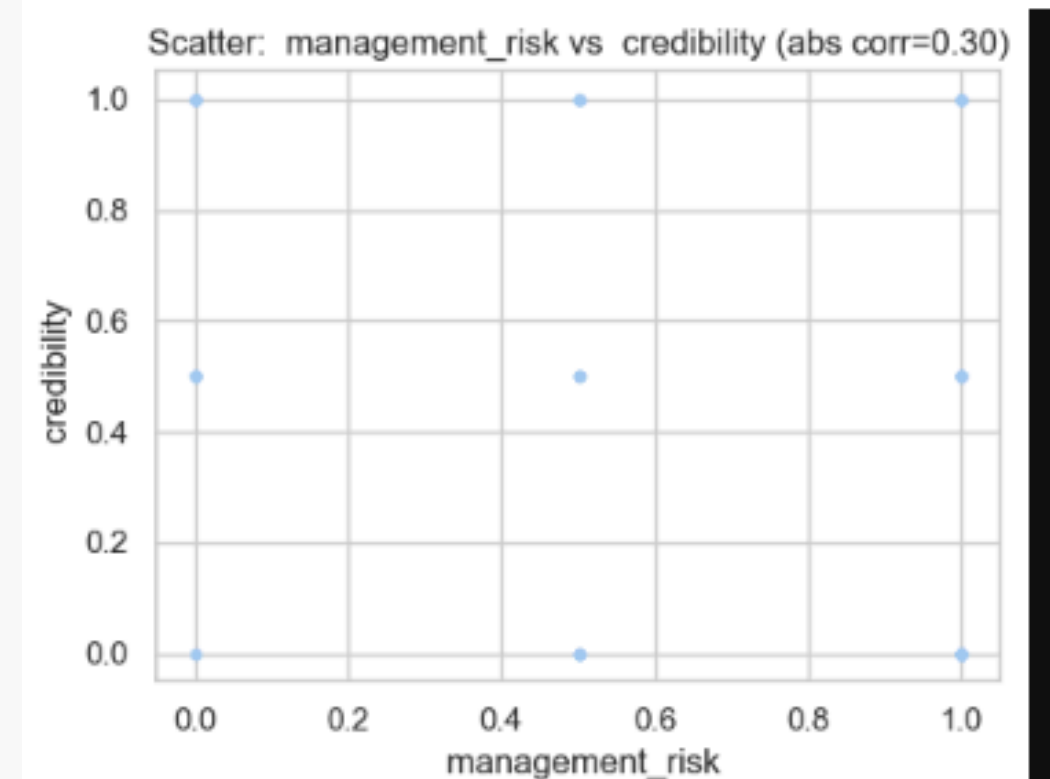
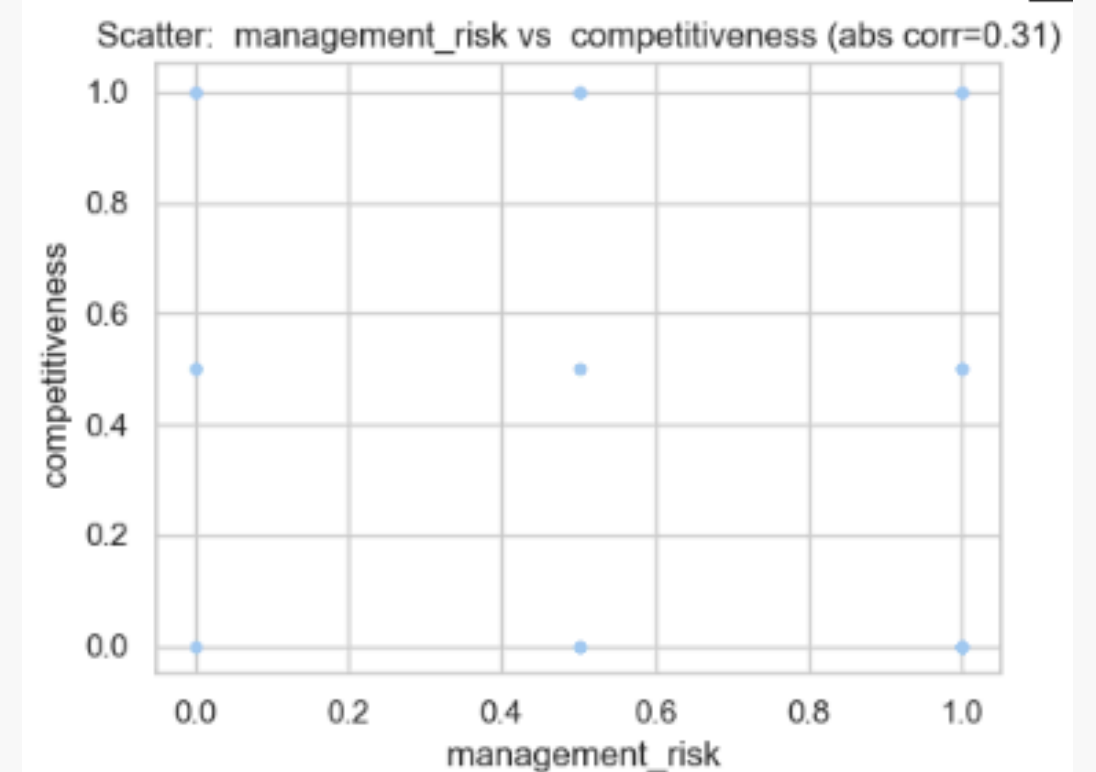
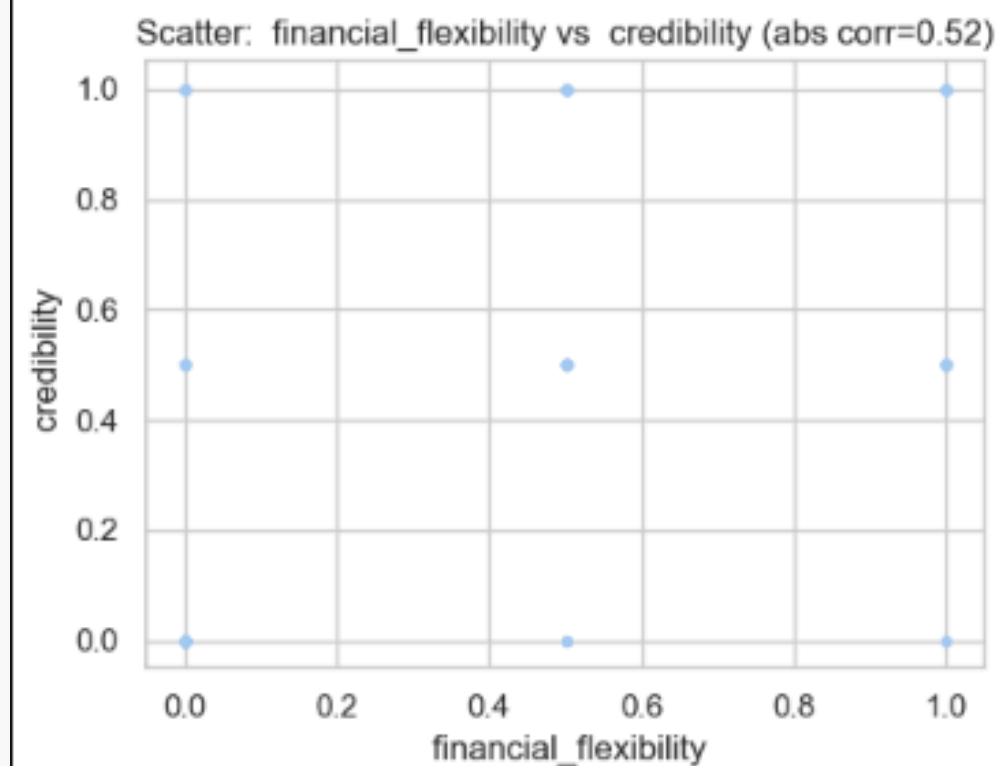
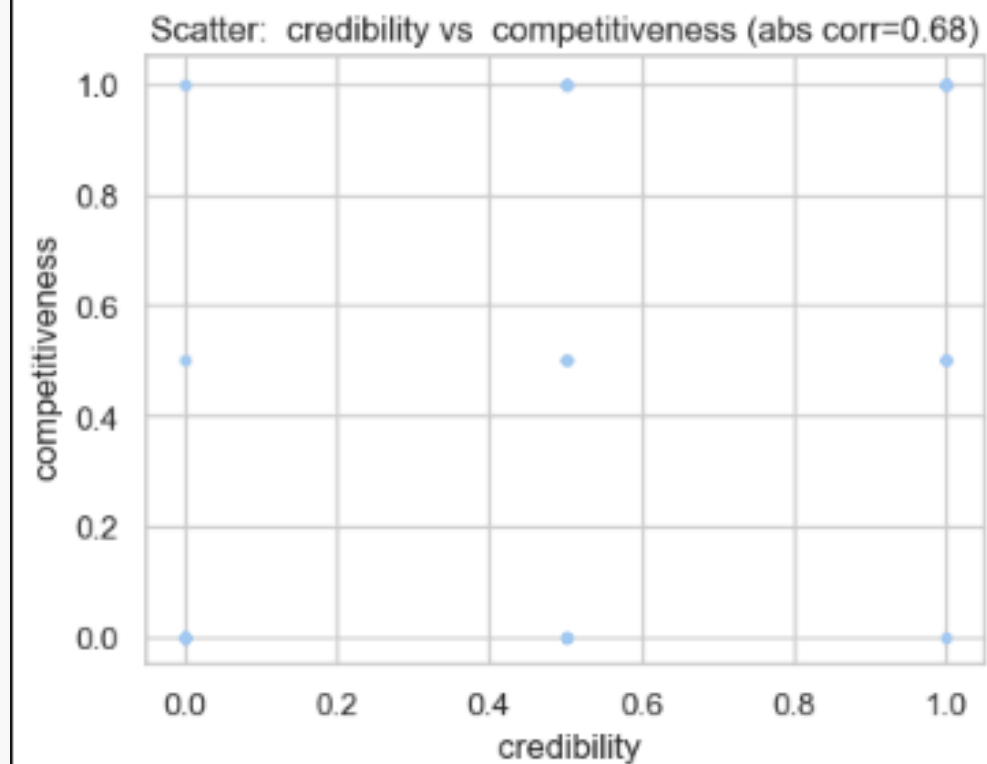
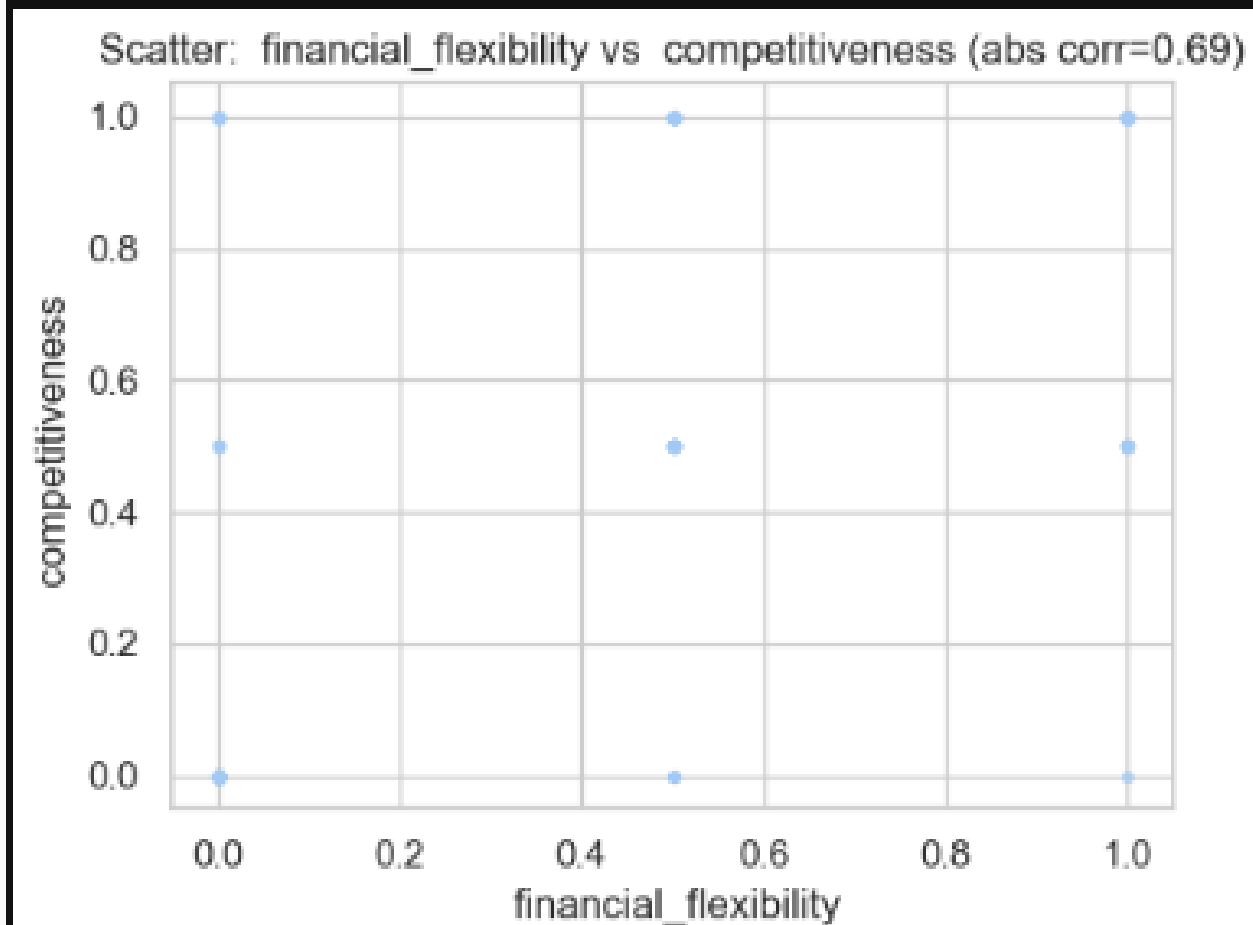
	industrial_risk	management_risk	financial_flexibility	credibility	competitiveness	operating_risk
industrial_risk	1.000000	0.255127	-0.162624	-0.014438	-0.257814	0.144507
management_risk	0.255127	1.000000	-0.254845	-0.303341	-0.306568	0.213874
financial_flexibility	-0.162624	-0.254845	1.000000	0.524951	0.686612	-0.116903
credibility	-0.014438	-0.303341	0.524951	1.000000	0.675689	-0.288458
competitiveness	-0.257814	-0.306568	0.686612	0.675689	1.000000	-0.211383
operating_risk	0.144507	0.213874	-0.116903	-0.288458	-0.211383	1.000000



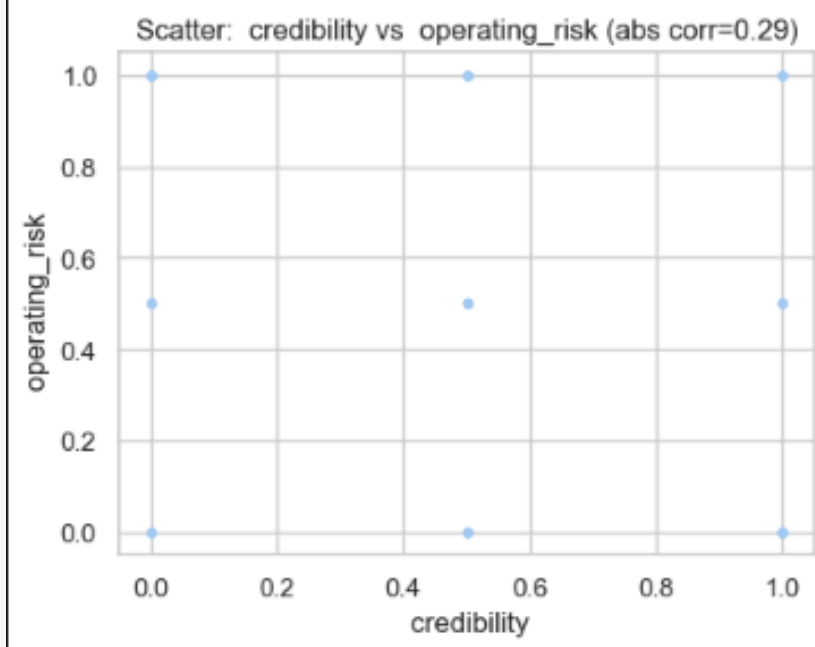
Complete EDA (Inline)

Top 6 correlated numeric pairs (abs)

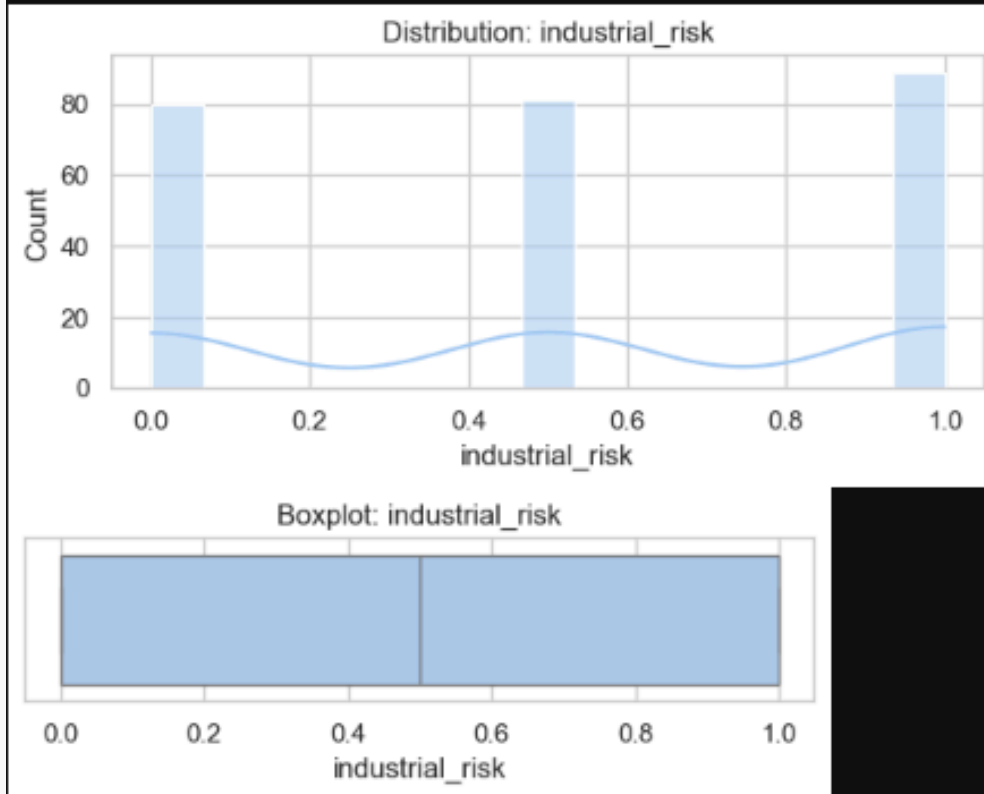
	col1	col2	abs_corr
0	financial_flexibility	competitiveness	0.686612
1	credibility	competitiveness	0.675689
2	financial_flexibility	credibility	0.524951
3	management_risk	competitiveness	0.306568
4	management_risk	credibility	0.303341
5	credibility	operating_risk	0.288458



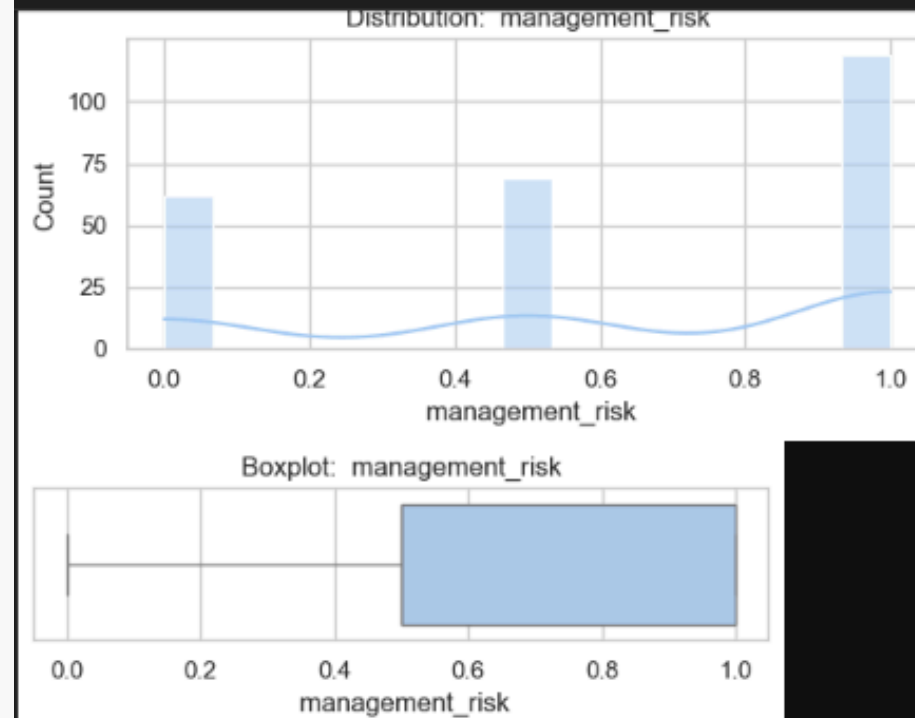
Complete EDA (Inline)



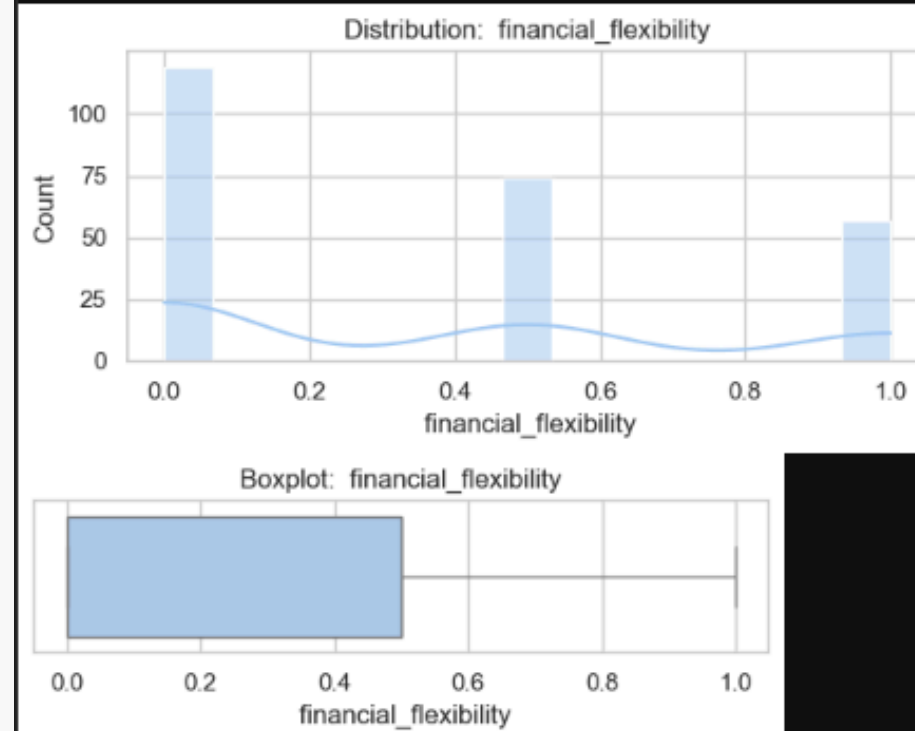
 **Numeric distributions (histograms + boxplots + IQR outliers)**



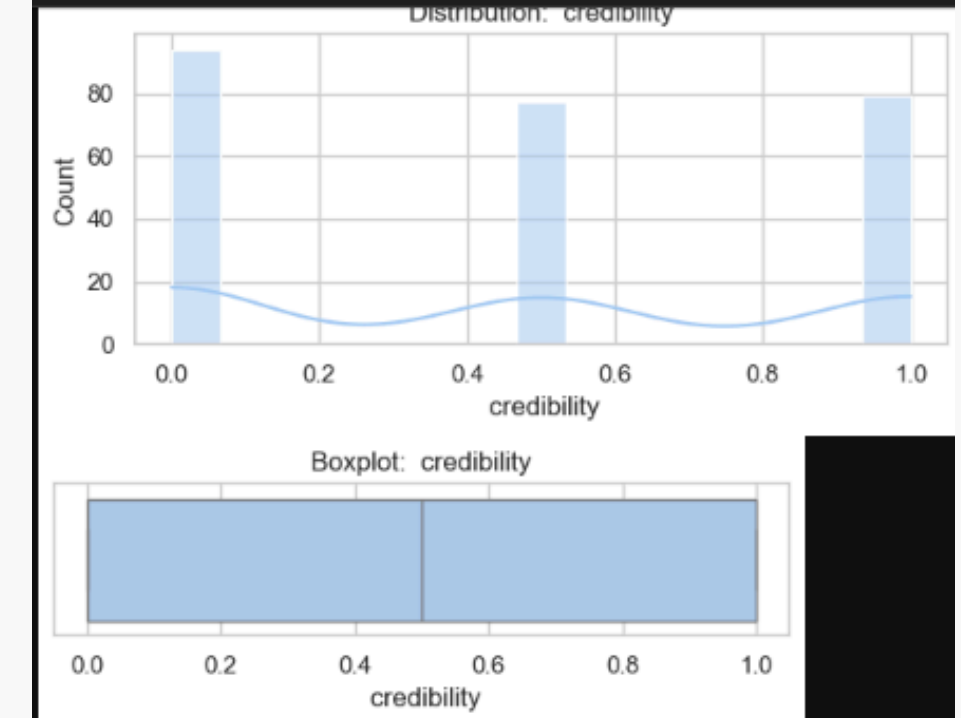
• **industrial_risk**: n=250, IQR-outliers=0



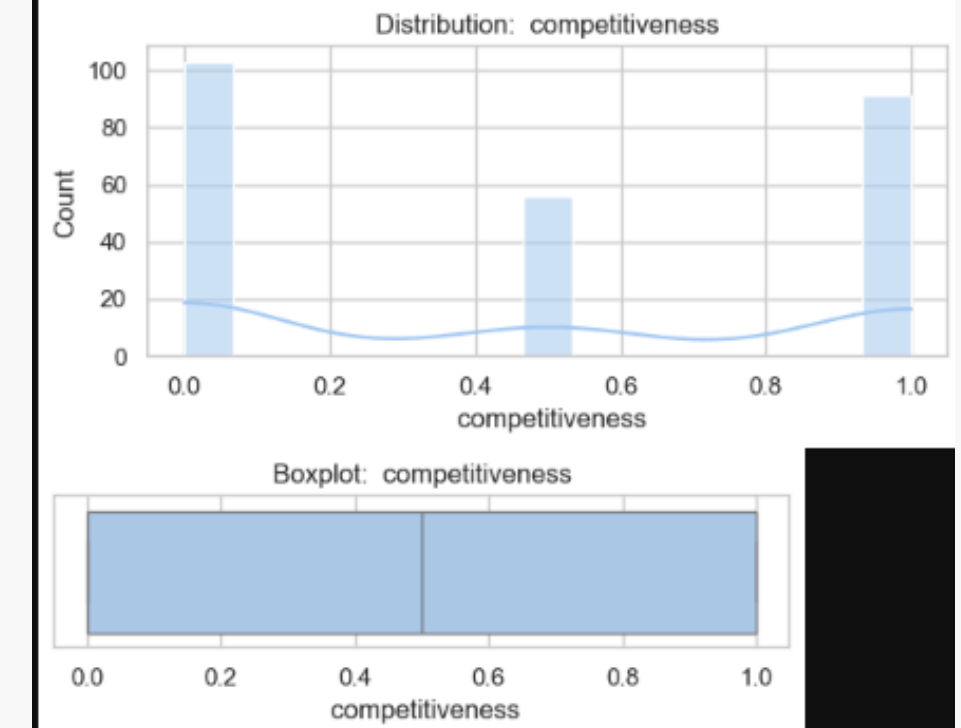
• **** management_risk****: n=250, IQR-outliers=0



• **** financial flexibility****: n=250, IQR-outliers=0



• **** credibility****: n=250, IQR-outliers=0



• **** competitiveness****: n=250, IQR-outliers=0

Feature Engineering (For Boosting Accuracy)

Adding Additional columns in order to get perfect predictions, Clear understanding of dataset, avoid overfitting & Higher accuracy.

Removing unimportant features which are useless, overfitted & diluted for predictions

Here are those *8 steps line by line:

1. Import RandomForest & prepare feature set

Import RandomForestClassifier, separate features (X) and target (y).

2. Keep 7 mandatory columns

Define the essential columns you always need (industrial_risk, management_risk, etc.).

3. Encode categorical variables

Detect non-numeric columns and apply one-hot encoding with pd.get_dummies.

4. Train RandomForest

Fit a RandomForest model on the encoded data to learn feature importances.

5. Rank feature importance

Extract feature importance scores and calculate the mean threshold.



Feature Engineering (For Boosting Accuracy)

6. Select features above threshold

Keep only the features with importance greater than the mean.

7. Add mandatory features back

Ensure the 6 mandatory predictors are always included (even if not “important”).

8. Build final filtered dataset

Merge the selected features with the target column to create filtered_df.

This is how the dataset is reduced to the 56 clean features used for training and scaling.



Logistic Regression (Using 10 fold stratified CV)

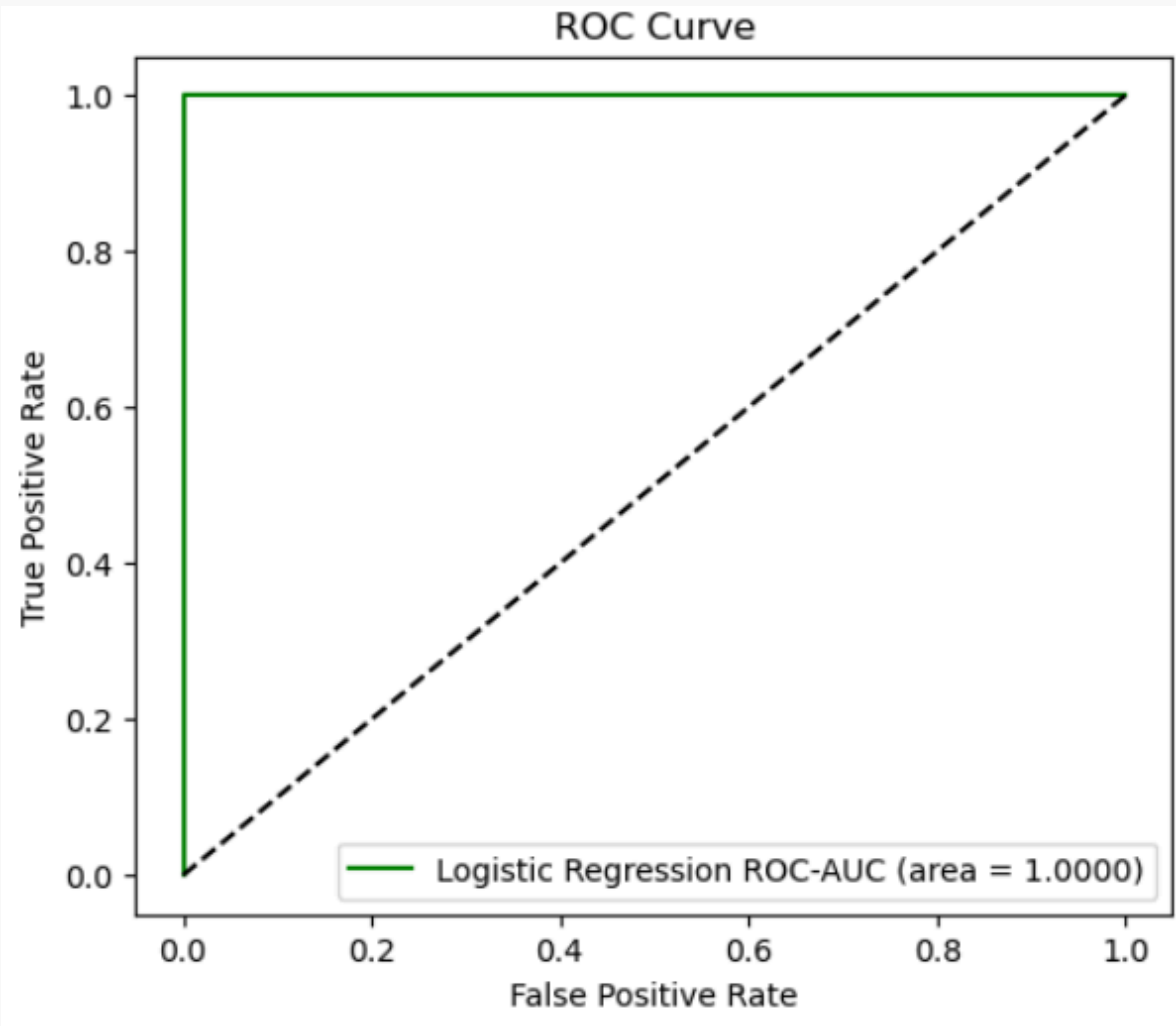
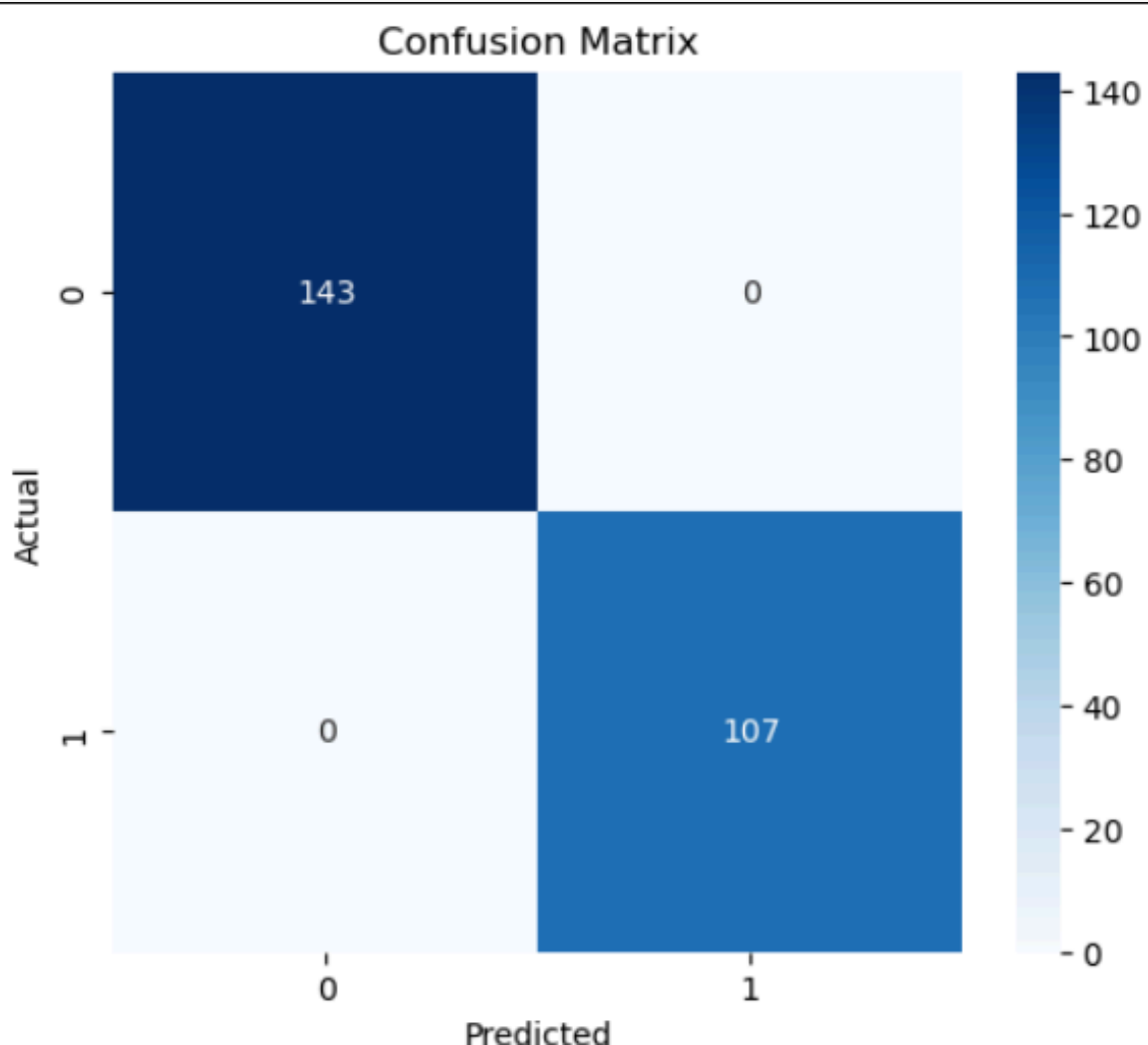
- 

- Class distribution:
 - class
 - 0 143
 - 1 107
- Name: count, dtype: int64
- Fitting 10 folds for each of 20 candidates, totalling 200 fits
- Best hyperparameters: {'C': 0.01, 'class_weight': None, 'penalty': 'l2'}
-
- Mean Accuracy: 1.0000
- Mean F1-Score: 1.0000
- Mean ROC-AUC: 1.0000
-
- Classification Report (All Data):
-
- | | precision | recall | f1-score | support |
|---|-----------|--------|----------|---------|
| 0 | 1.00 | 1.00 | 1.00 | 143 |
| 1 | 1.00 | 1.00 | 1.00 | 107 |
-
- | | | | | |
|--------------|------|------|------|-----|
| accuracy | | | 1.00 | 250 |
| macro avg | 1.00 | 1.00 | 1.00 | 250 |
| weighted avg | 1.00 | 1.00 | 1.00 | 250 |
-

Logistic Regression (Using 10 fold stratified CV)

- Predictions saved for model 5
- Sample Predictions:

	Actual	Predicted	Probability_Class1
•	0	1	0.708022
•	1	1	0.721990
•	2	1	0.681335
•	3	1	0.544603
•	4	1	0.855483



Decision Tree (Using 10 fold stratified CV)

- 

- Class distribution:
- class
 - 0 143
 - 1 107
- Name: count, dtype: int64
- Fitting 10 folds for each of 480 candidates, totalling 4800 fits
- Best hyperparameters: {'criterion': 'gini', 'max_depth': 3, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 2}
- Mean Accuracy: 0.9920
- Mean F1-Score: 0.9905
- Mean ROC-AUC: 0.9921
-
- Classification Report (All Data):
-
- | | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0 | 0.99 | 0.99 | 0.99 | 143 |
| 1 | 0.99 | 0.99 | 0.99 | 107 |
| accuracy | | | 0.99 | 250 |
| macro avg | 0.99 | 0.99 | 0.99 | 250 |
| weighted avg | 0.99 | 0.99 | 0.99 | 250 |
-

Decision Tree (Using 10 fold stratified CV)

- Actual vs Predicted values for all rows:

- Actual Predicted Probability_Class1

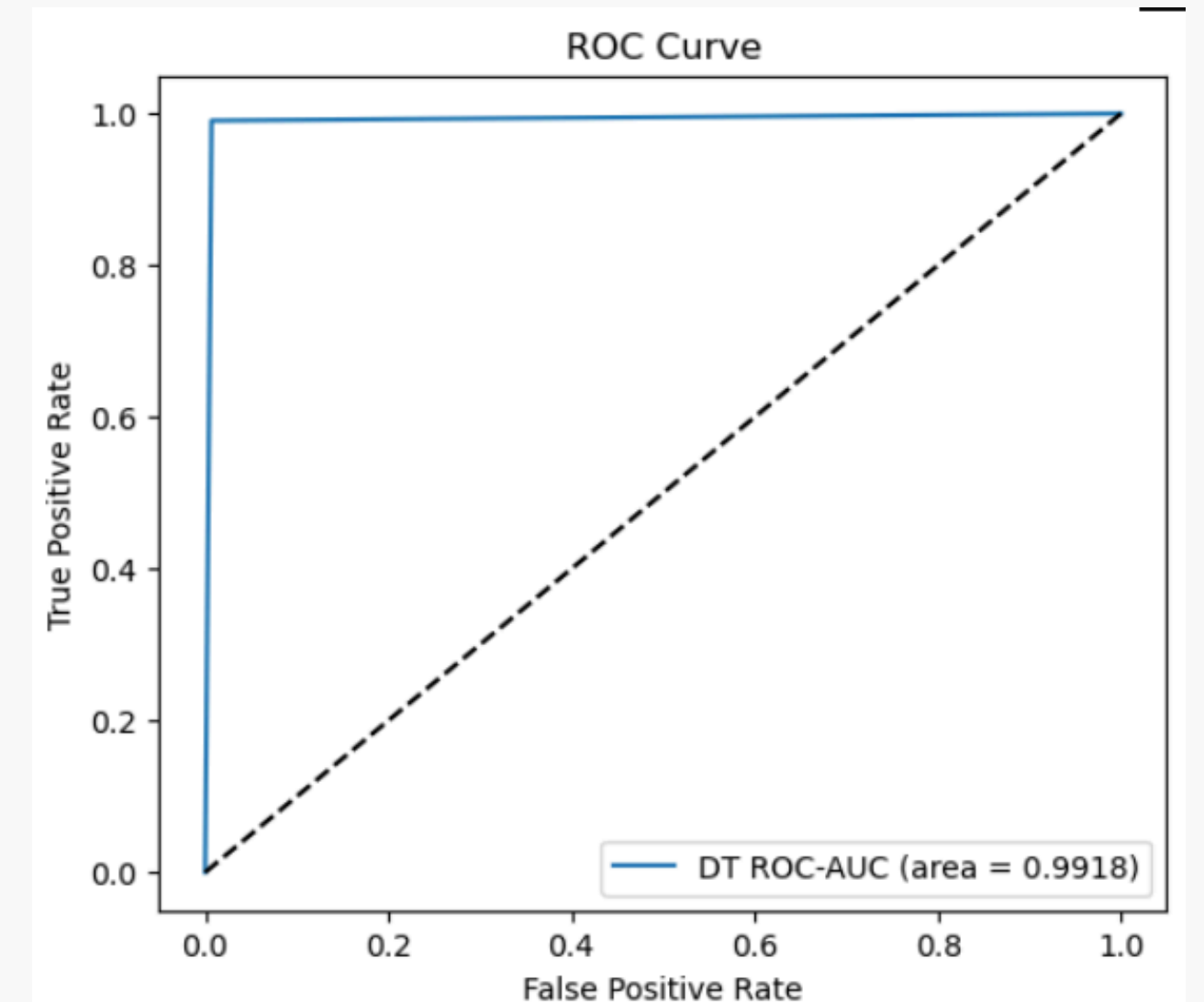
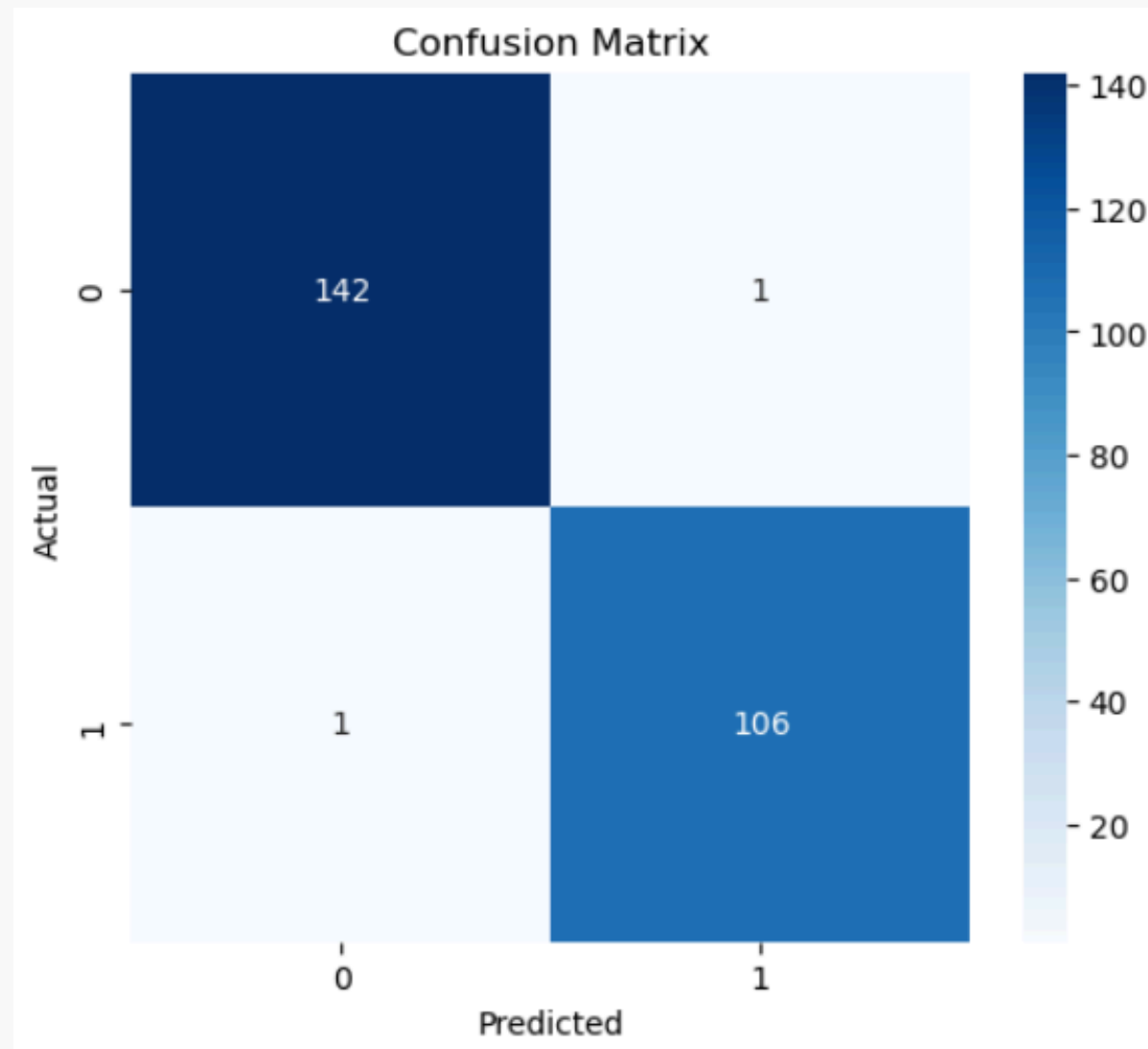
- 0 1 1 1.0

- 1 1 1 1.0

- 2 1 1 1.0

- 3 1 1 1.0

- 4 1 1 1.0



Model Building

We implemented 9 classification models + Ensemble to predict bankruptcy.

Random Forest (Using 10 fold stratified CV)

- Ensemble of multiple decision trees (bagging technique).
- Each tree trained on a random subset of data & features.
- Final prediction = majority vote (classification).
-  Advantages: Handles non-linearity, reduces overfitting.
-  Limitations: Less interpretable compared to a single tree.

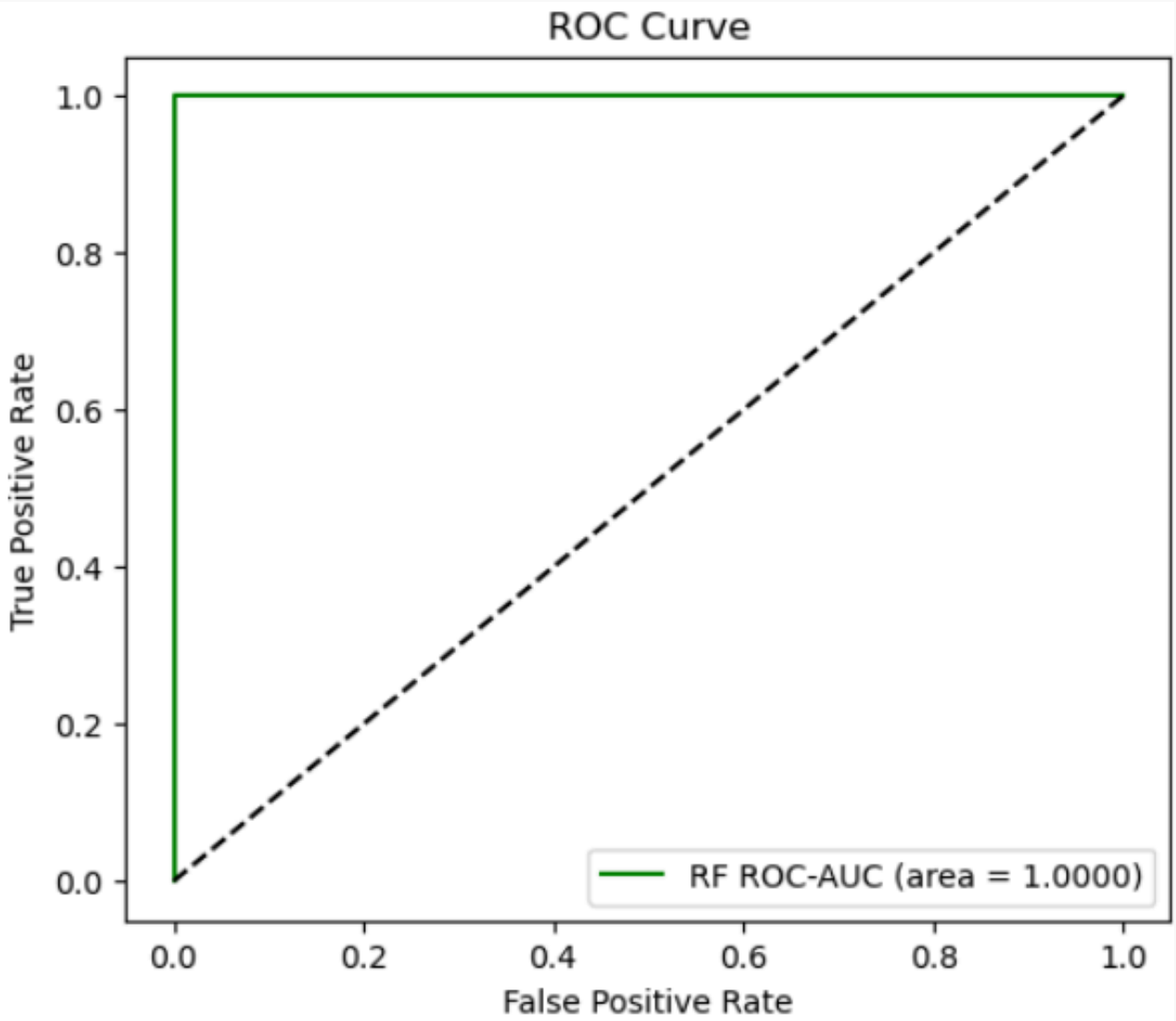
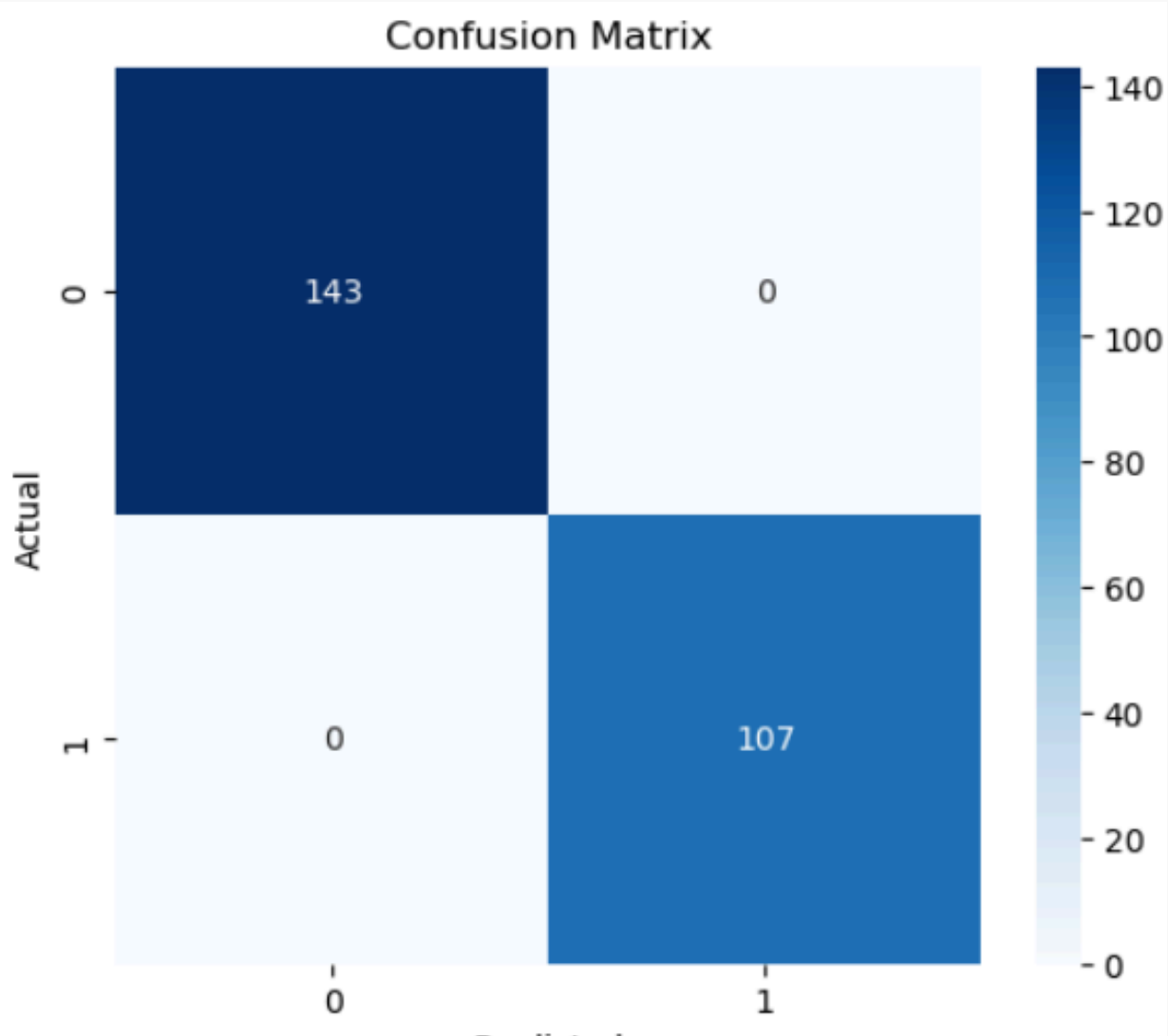
- Class distribution:
- class
- 0 143
- 1 107
- Name: count, dtype: int64
- Fitting 10 folds for each of 1152 candidates, totalling 11520 fits
- Best hyperparameters: {'criterion': 'entropy', 'max_depth': 5, 'max_features': 'log2', 'min_samples_leaf': 1, 'min_samples_split': 20, 'n_estimators': 200}
-
- Mean Accuracy: 1.0000
- Mean F1-Score: 1.0000
- Mean ROC-AUC: 1.0000
-
- Classification Report (All Data):
-
- precision recall f1-score support
-
- 0 1.00 1.00 1.00 143
- 1 1.00 1.00 1.00 107
-
- accuracy 1.00 250
- macro avg 1.00 1.00 1.00 250
- weighted avg 1.00 1.00 1.00 250

Random Forest (Using 10 fold stratified CV)

- Predictions saved for model 2
- Actual vs Predicted values for all rows:

• Actual Predicted Probability_Class1

• 0	1	1	0.995580
• 1	1	1	0.985940
• 2	1	1	0.999929
• 3	1	1	0.999850
• 4	1	1	0.971016



Model Building

We implemented 9 classification models + Ensemble to predict bankruptcy.

SVM (RBF kernel) classification

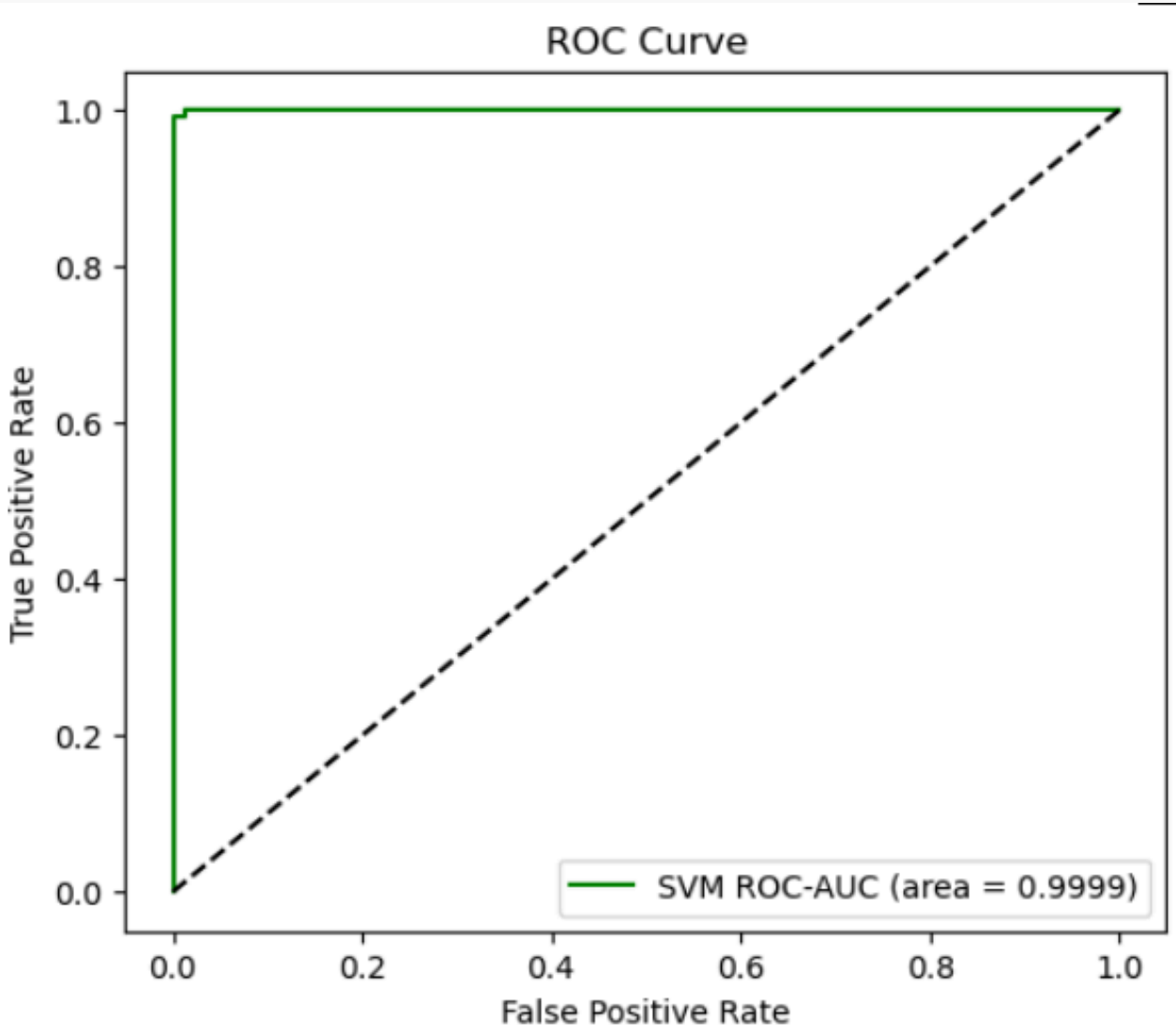
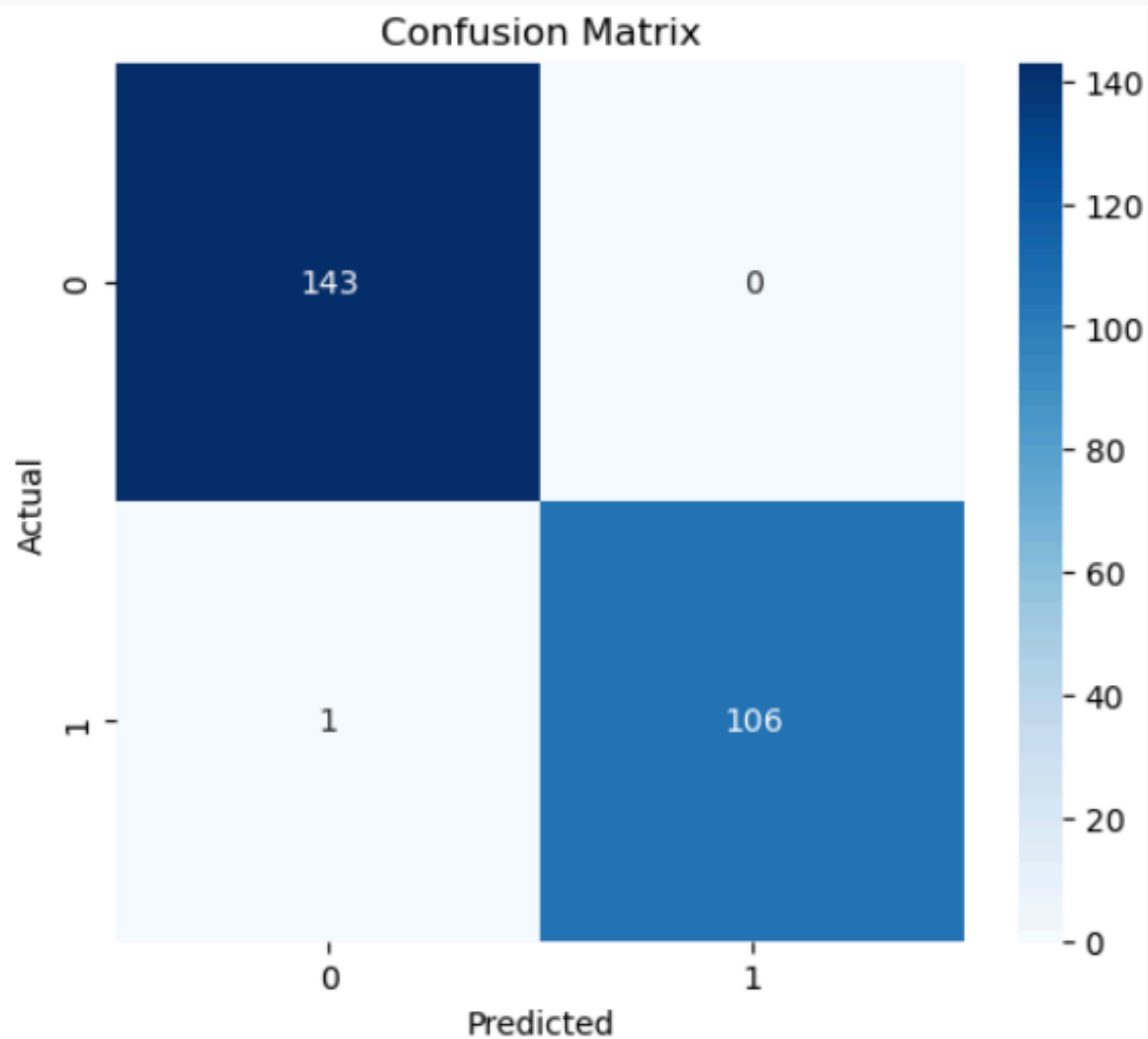
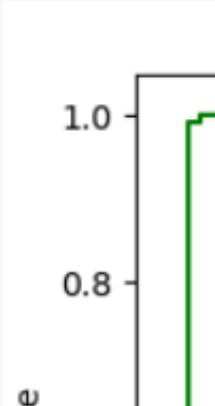
- Creates a hyperplane to separate classes.
- Used RBF kernel to capture non-linear boundaries.
-  Advantages: Works well in high-dimensional space.
-  Limitations: Computationally expensive, sensitive to parameter tuning.

- Class distribution:
- class
- 0 143
- 1 107
- Name: count, dtype: int64
- Fitting 10 folds for each of 30 candidates, totalling 300 fits
- Best hyperparameters: {'C': 1, 'gamma': 'scale'}
-
- Mean Accuracy: 0.9960
- Mean F1-Score: 0.9952
- Mean ROC-AUC: 1.0000
-
- Classification Report (All Data):
-
- precision recall f1-score support
-
- 0 0.99 1.00 1.00 143
- 1 1.00 0.99 1.00 107
-
- accuracy 1.00 250
- macro avg 1.00 1.00 1.00 250
- weighted avg 1.00 1.00 1.00 250

SVM (RBF kernel) classification (Using 10 fold stratified CV)

- Predictions saved for model 6
- Sample Predictions:

	Actual	Predicted	Probability_Class1
• 0	1	1	0.999994
• 1	1	1	0.997130
• 2	1	1	0.979422
• 3	1	1	0.985565
• 4	1	1	0.984048



Model Building

We implemented 9 classification models + Ensemble to predict bankruptcy.

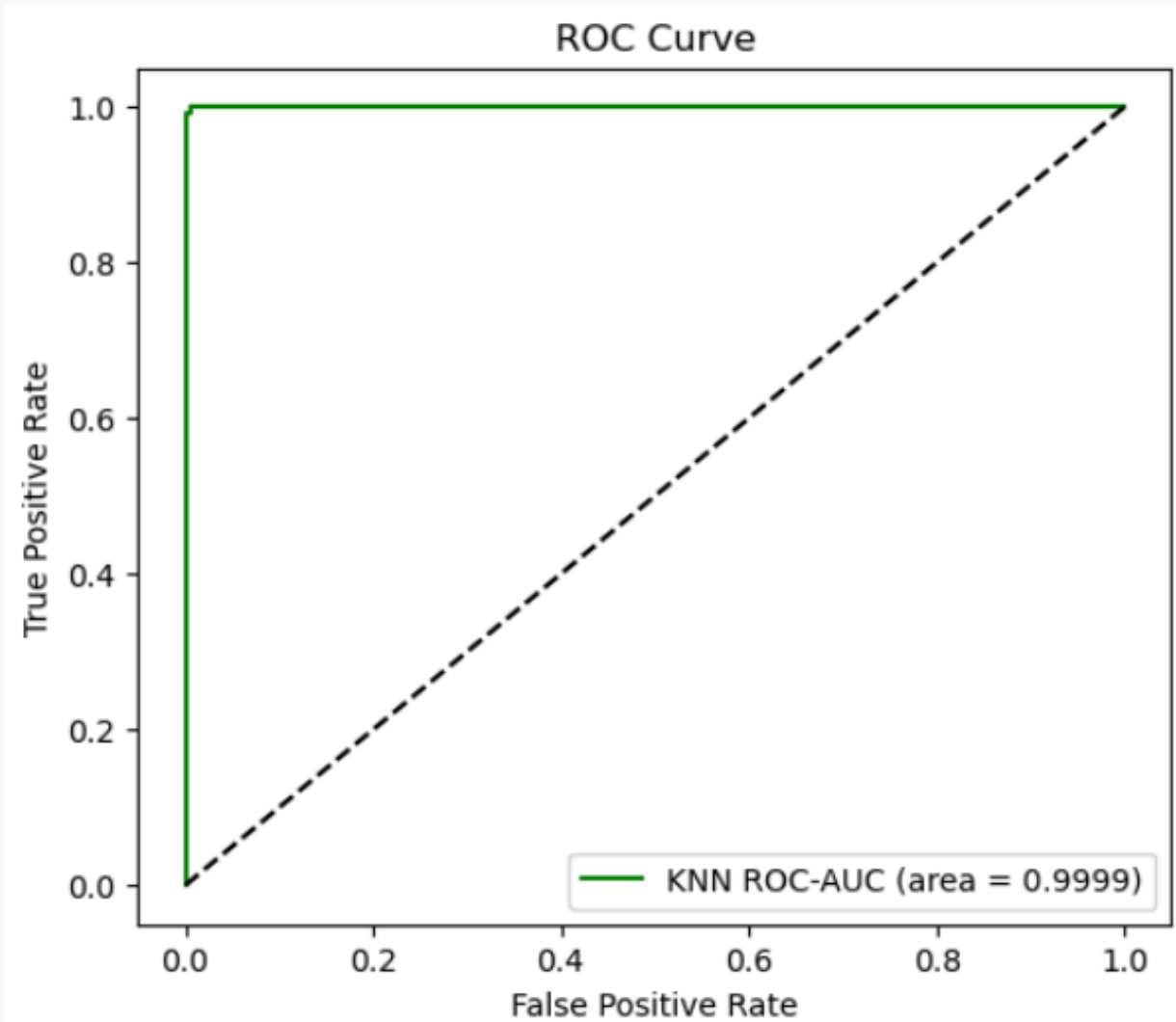
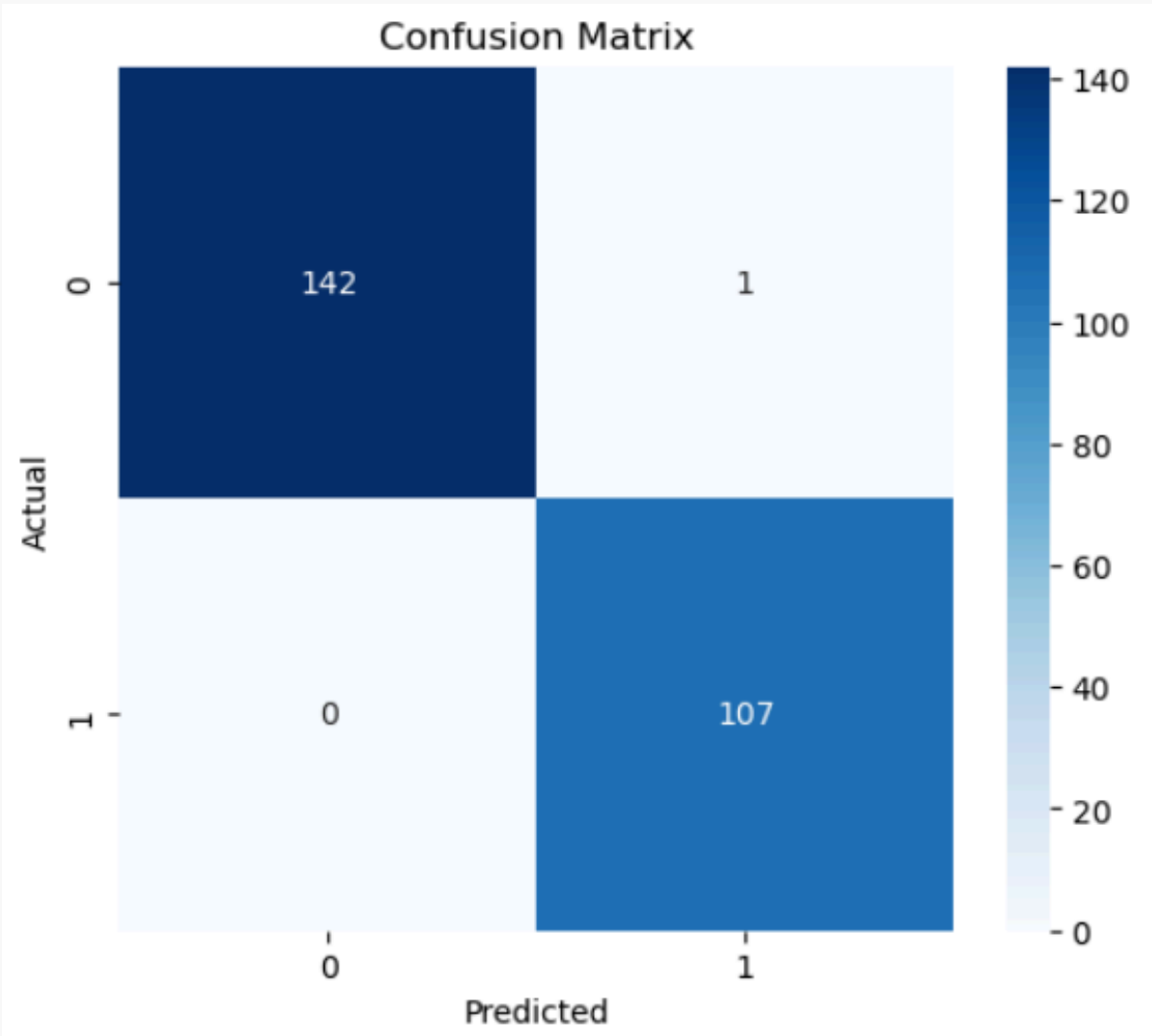
K-Nearest Neighbors (Using 10-fold Stratified CV)

- Classifies based on the majority class among k-nearest neighbors.
- Distance metric (Euclidean) used to measure closeness.
-  Advantages: Simple, no training phase.
-  Limitations: Sensitive to noise, computationally heavy on large datasets.

- Class distribution:
- class
- 0 143
- 1 107
- Name: count, dtype: int64
- Fitting 10 folds for each of 720 candidates, totalling 7200 fits
- Best hyperparameters: {'leaf_size': 10, 'metric': 'chebyshev', 'n_neighbors': 5, 'p': 1, 'weights': 'distance'}
-
- Mean Accuracy: 0.9960
- Mean F1-Score: 0.9957
- Mean ROC-AUC: 1.0000
-
- Classification Report (All Data):
-
- precision recall f1-score support
-
- 0 1.00 0.99 1.00 143
- 1 0.99 1.00 1.00 107
-
- accuracy 1.00 250
- macro avg 1.00 1.00 1.00 250
- weighted avg 1.00 1.00 1.00 250

K-Nearest Neighbors (Using 10-fold Stratified CV)

- Predictions saved for model 7
- Sample Predictions:
-
- Actual Predicted Probability_Class1
- 0 1 1 1.0
- 1 1 1 1.0
- 2 1 1 1.0
- 3 1 1 1.0
- 4 1 1 1.0



Model Building

We implemented 9 classification models + Ensemble to predict bankruptcy.

XGBoost (Extreme Gradient Boosting) (Using 10-fold Stratified CV)

- Boosting algorithm that builds trees sequentially to correct previous errors.
- Optimized for speed and performance.
-  Advantages: High accuracy, handles missing values well.
-  Limitations: More complex tuning required.

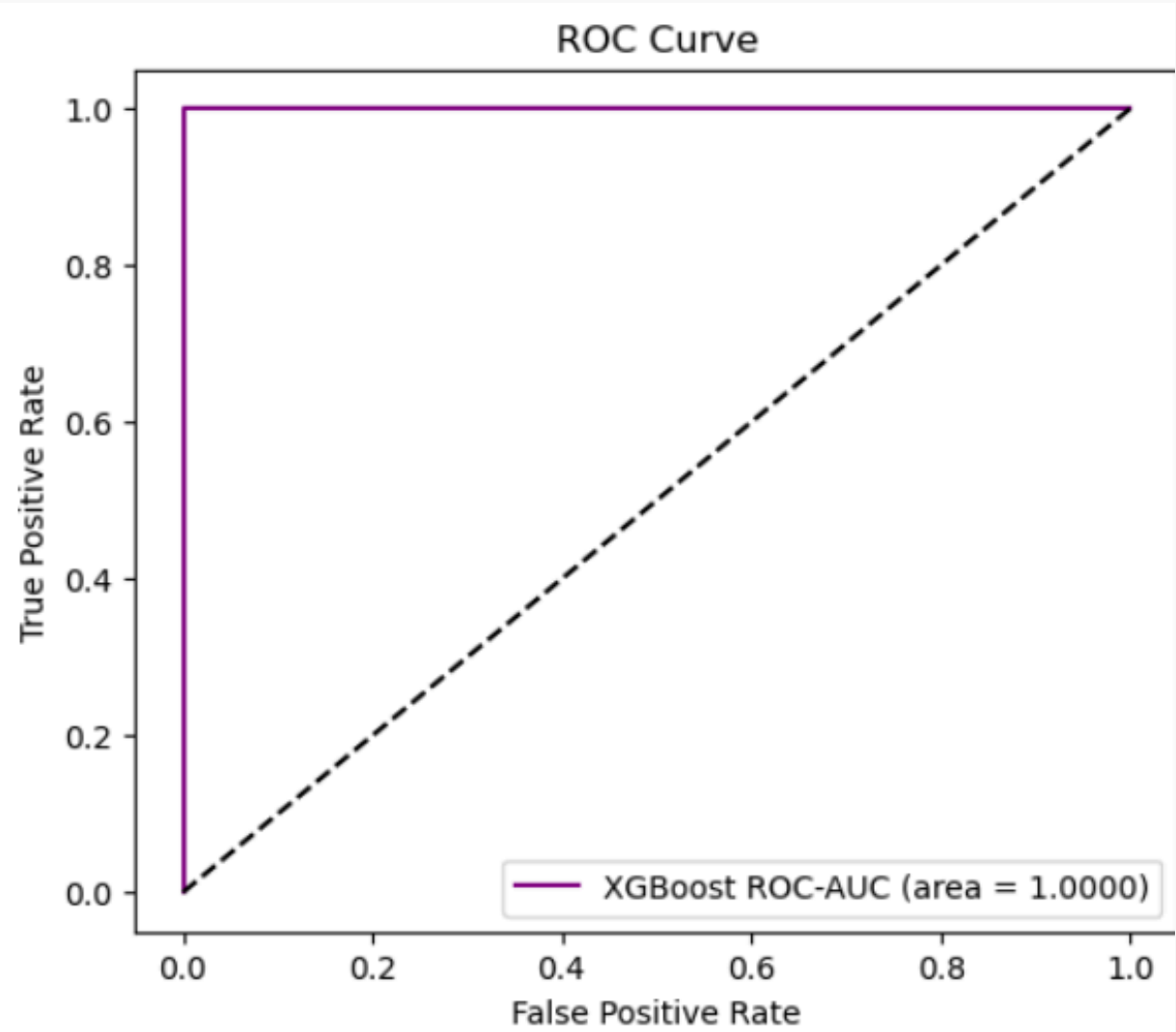
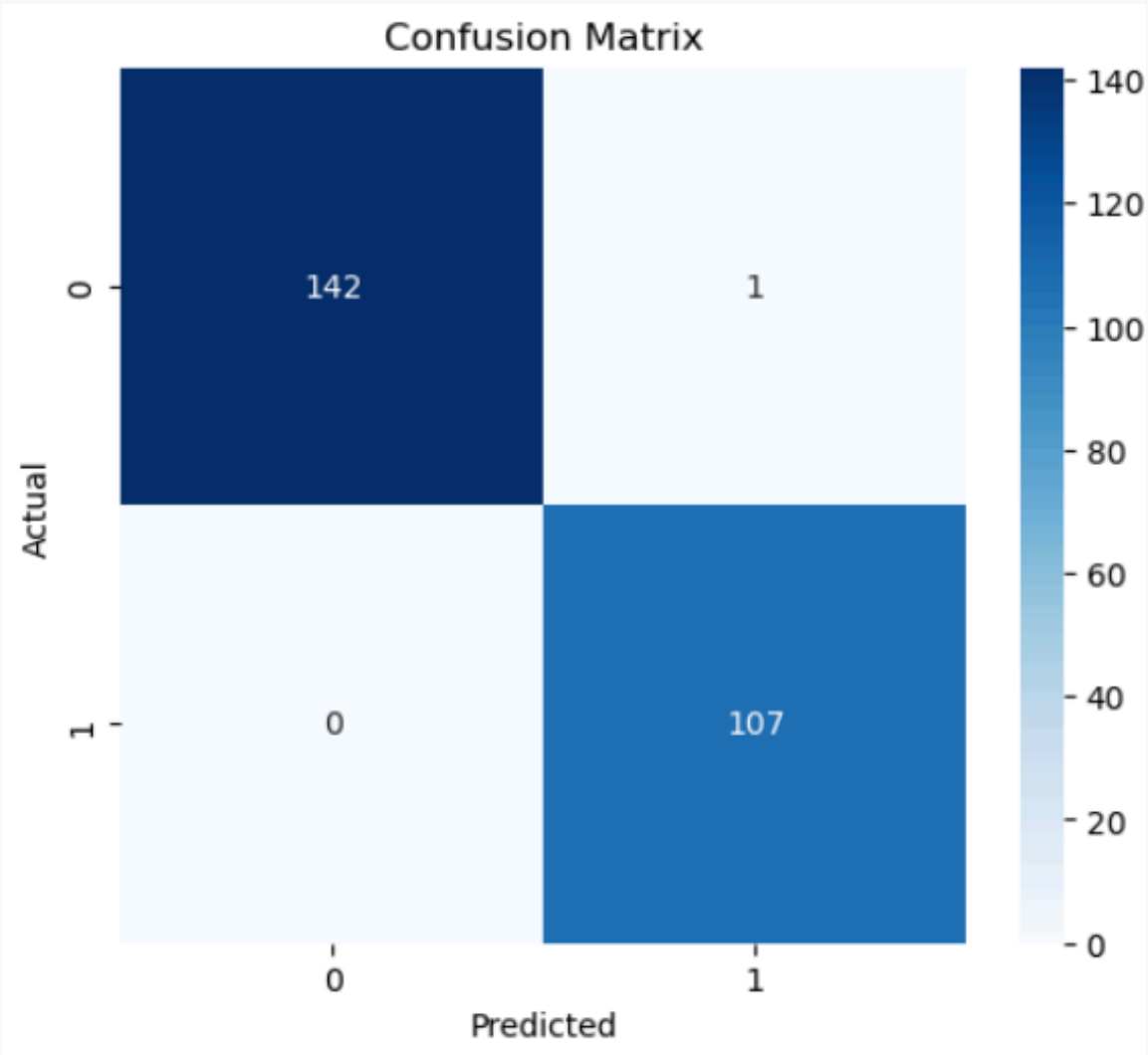
- Class distribution:
- class
- 0 143
- 1 107
- Name: count, dtype: int64
- scale_pos_weight: 1.3364485981308412
- Fitting 10 folds for each of 432 candidates, totalling 4320 fits
- Best hyperparameters: {'colsample_bytree': 0.8, 'learning_rate': 0.01, 'max_depth': 3, 'n_estimators': 300, 'reg_alpha': 0, 'reg_lambda': 1, 'subsample': 0.8}
- Mean Accuracy: 0.9960
- Mean F1-Score: 0.9952
- Mean ROC-AUC: 1.0000
-
- Classification Report (All Data):
-
-
- precision recall f1-score support
-
- 0 1.00 0.99 1.00 143
- 1 0.99 1.00 1.00 107
-
- accuracy 1.00 250
- macro avg 1.00 1.00 1.00 250
- weighted avg 1.00 1.00 1.00 250

XGBoost (Extreme Gradient Boosting) (Using 10-fold Stratified CV)

- Predictions saved for model 8
- Sample Predictions:

• Actual Predicted Probability_Class1

• 0	1	1	0.966506
• 1	1	1	0.966915
• 2	1	1	0.966268
• 3	1	1	0.960596
• 4	1	1	0.966506



Model Building

We implemented 9 classification models + Ensemble to predict bankruptcy.

LightGBM (Using 10-fold Stratified CV)

- Gradient boosting framework using leaf-wise growth strategy.
- Faster than XGBoost on large datasets.
-  Advantages: Very efficient, high accuracy.
-  Limitations: May overfit on small datasets if not tuned properly.

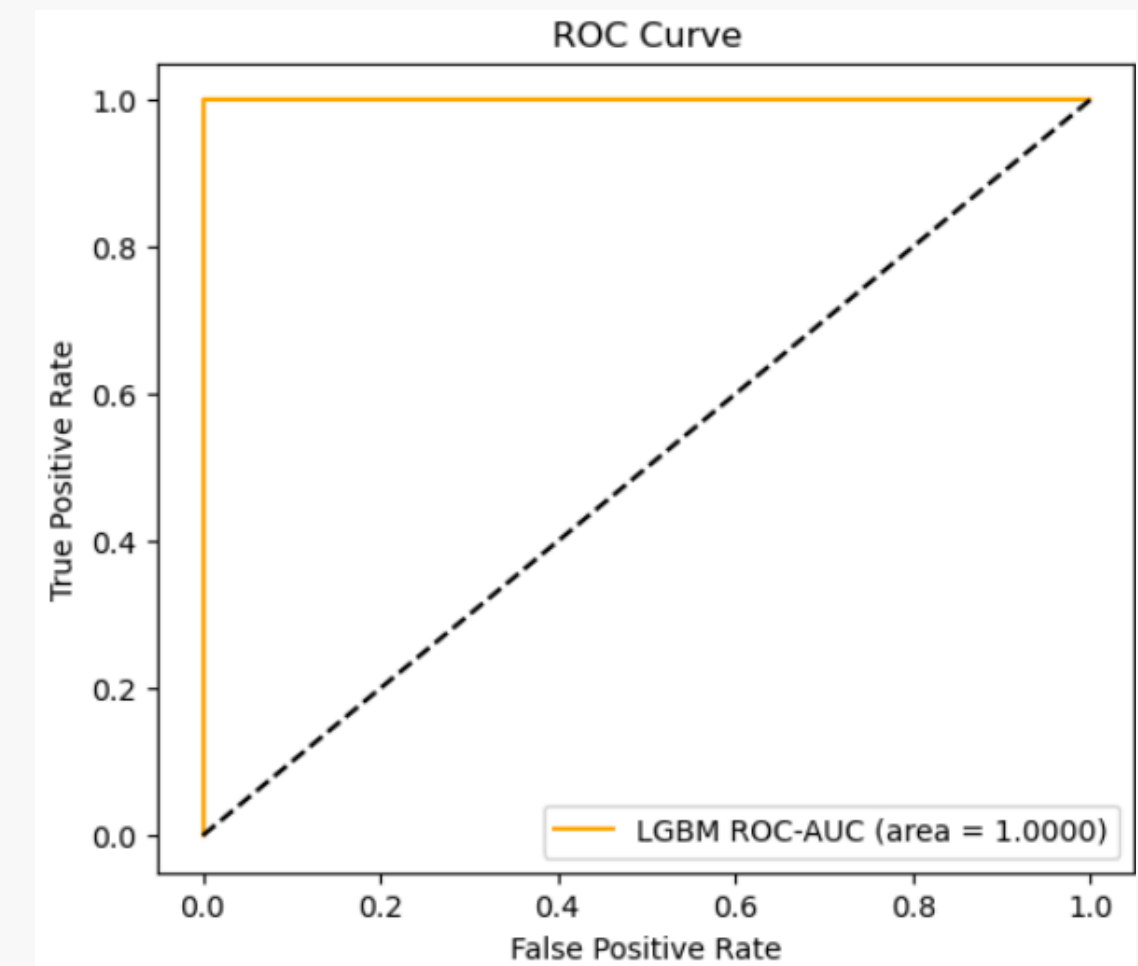
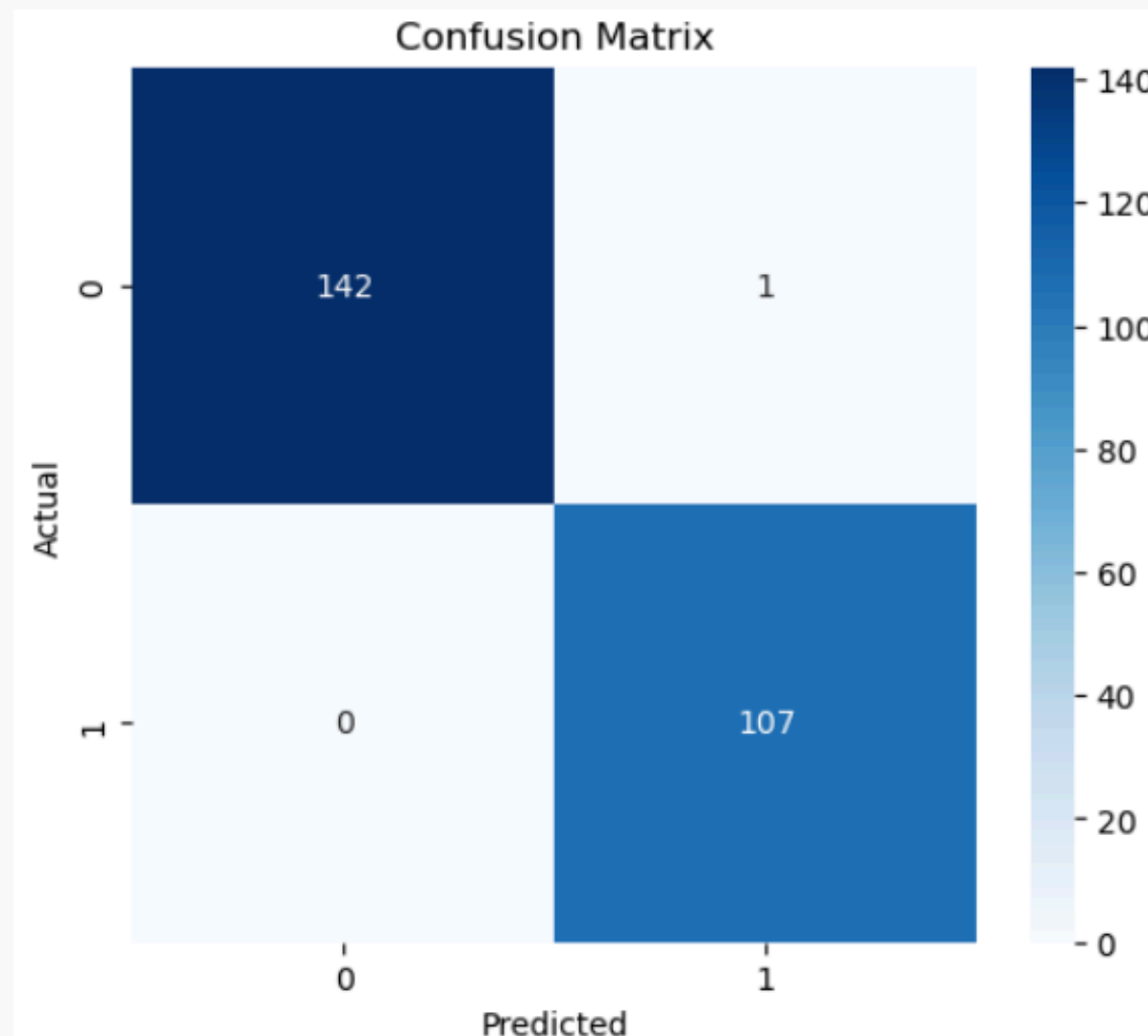
- Class distribution:
- class
- 0 143
- 1 107
- Name: count, dtype: int64
- Fitting 10 folds for each of 1296 candidates, totalling 12960 fits
- Mean Accuracy: 0.9960
- Mean F1-Score: 0.9952
- Mean ROC-AUC: 1.0000
-
- Classification Report (All Data):
-
- precision recall f1-score support
-
- 0 1.00 0.99 1.00 143
- 1 0.99 1.00 1.00 107
-
- accuracy 1.00 250
- macro avg 1.00 1.00 1.00 250
- weighted avg 1.00 1.00 1.00 250
-

LightGBM (Using 10-fold Stratified CV)

- Predictions saved for model 3
- Actual vs Predicted values for all rows:

• Actual Predicted Probability_Class1

- 0 1 1 0.932579
- 1 1 1 0.932647
- 2 1 1 0.932544
- 3 1 1 0.932544
- 4 1 1 0.932330



Model Building

We implemented 9 classification models + Ensemble to predict bankruptcy.

CatBoost (Using 10-fold Stratified CV)

- Gradient boosting library optimized for categorical data.
- Requires less preprocessing (handles categorical values automatically).
-  Advantages: Strong performance with default parameters.
-  Limitations: Training can be slower compared to LightGBM.

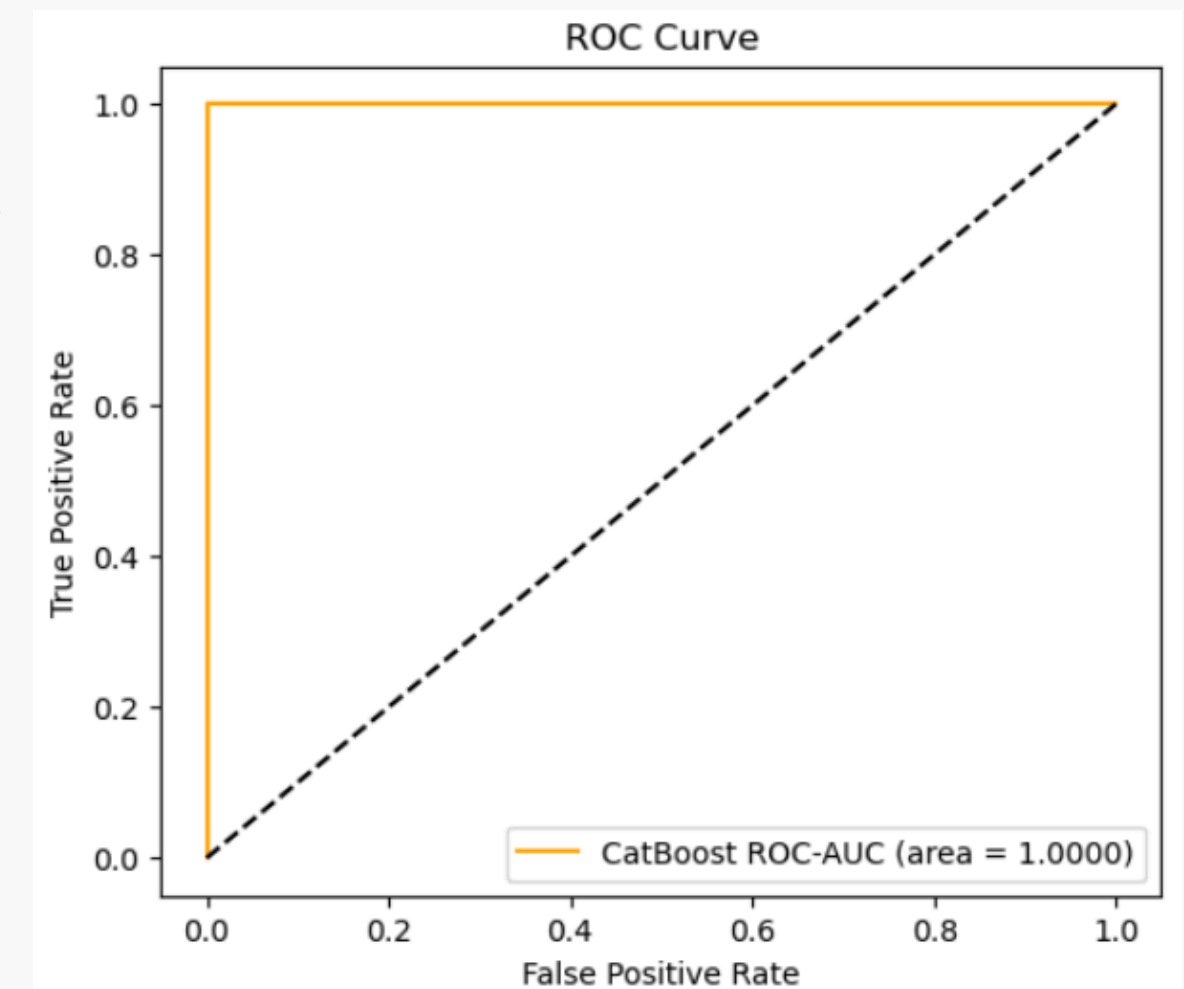
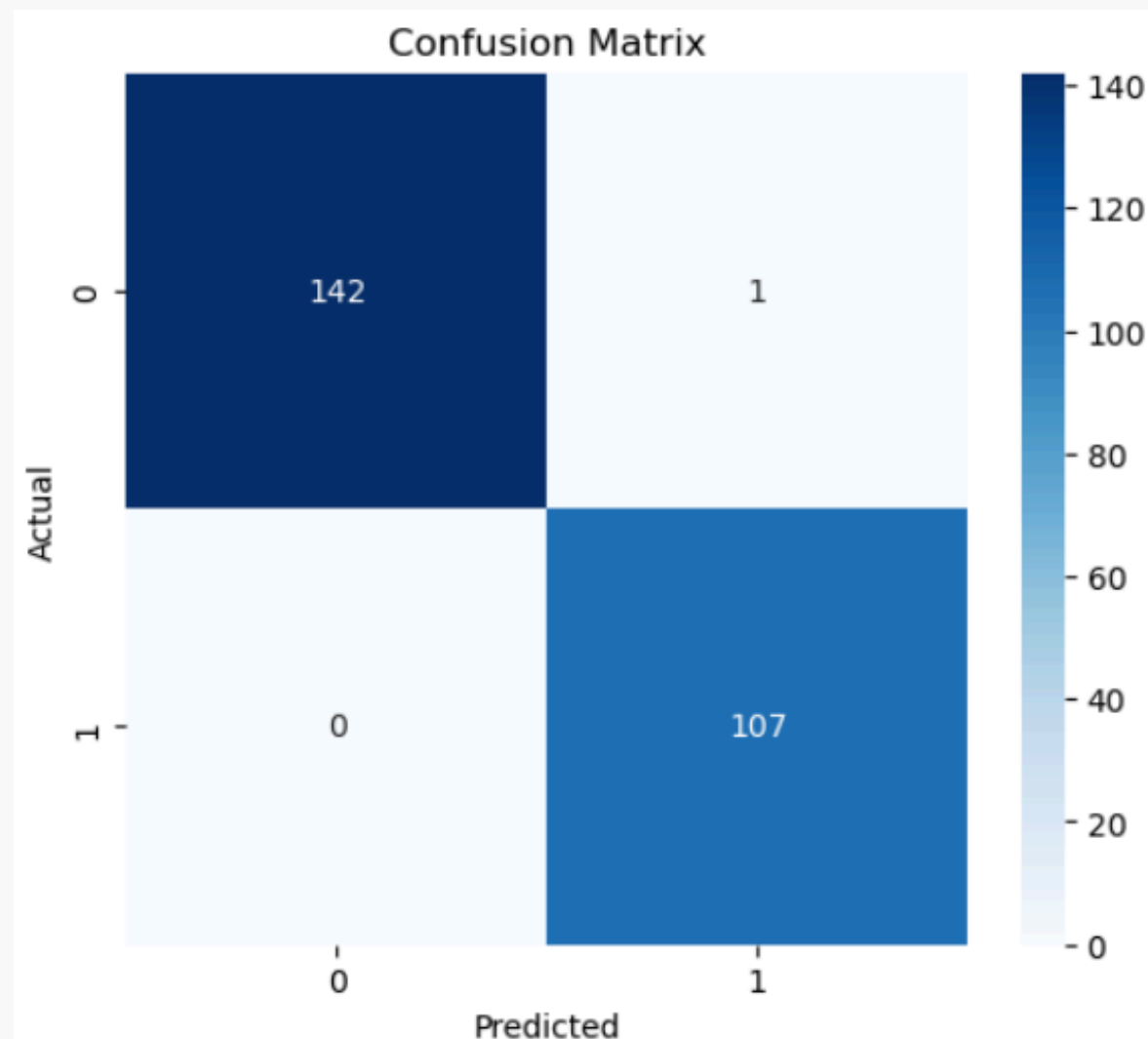
- Class distribution:
- class
- 0 143
- 1 107
- Name: count, dtype: int64
- Fitting 10 folds for each of 144 candidates, totalling 1440 fits
- Best hyperparameters: {'depth': 4, 'iterations': 300, 'l2_leaf_reg': 3, 'learning_rate': 0.01}
-
- Mean Accuracy: 0.9960
- Mean F1-Score: 0.9952
- Mean ROC-AUC: 1.0000
-
- Classification Report (All Data):
-
-
- precision recall f1-score support
-
- 0 1.00 0.99 1.00 143
- 1 0.99 1.00 1.00 107
-
- accuracy 1.00 250
- macro avg 1.00 1.00 1.00 250
- weighted avg 1.00 1.00 1.00 250

CatBoost (Using 10-fold Stratified CV)

- Predictions saved for model 4
- Sample Predictions:

• Actual Predicted Probability_Class1

- 0 1 1 0.997050
- 1 1 1 0.996658
- 2 1 1 0.996147
- 3 1 1 0.994708
- 4 1 1 0.997146



Model Building

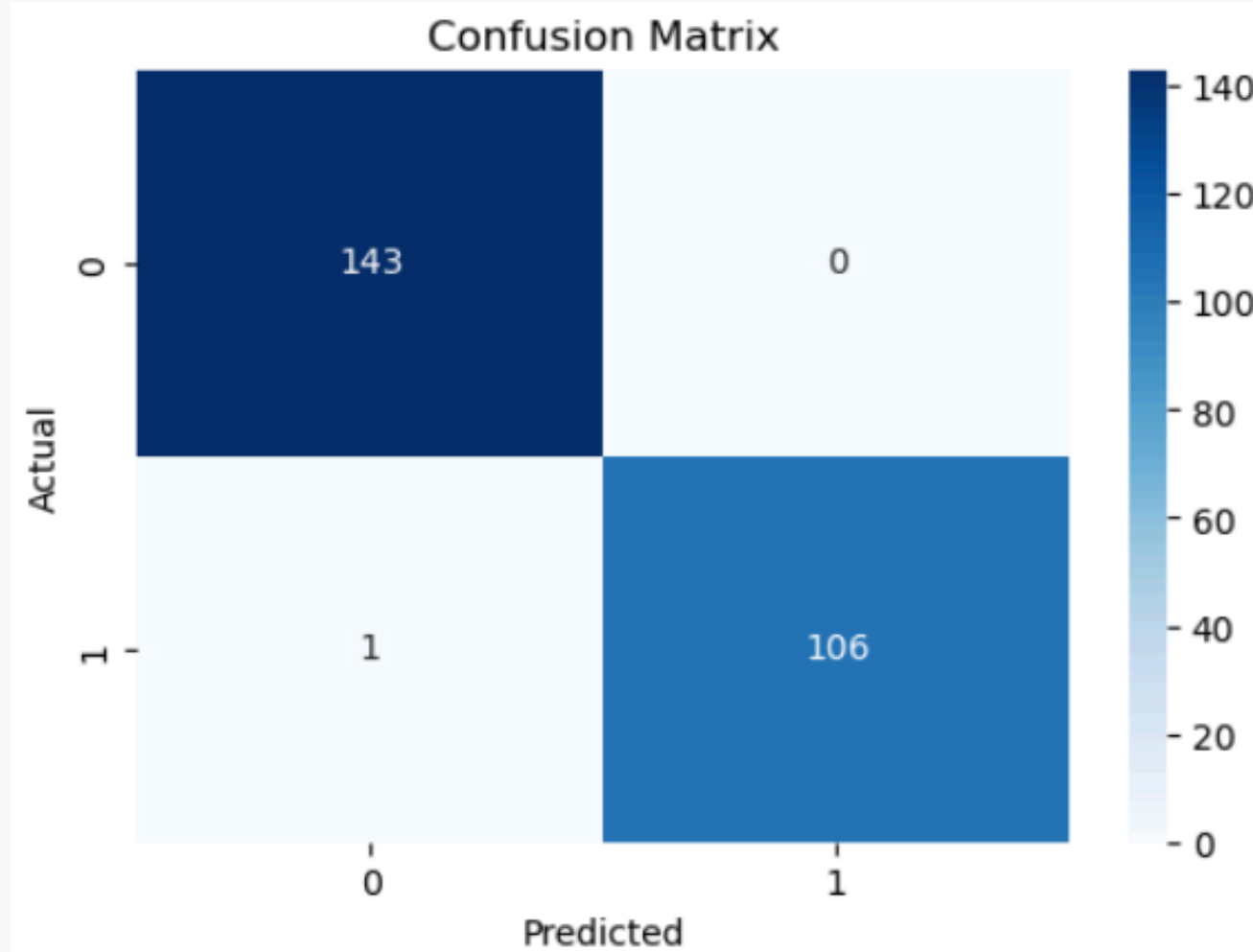
We implemented 9 classification models + Ensemble to predict bankruptcy.

Artificial Neural Network (ANN) (Using 10-fold Stratified CV)

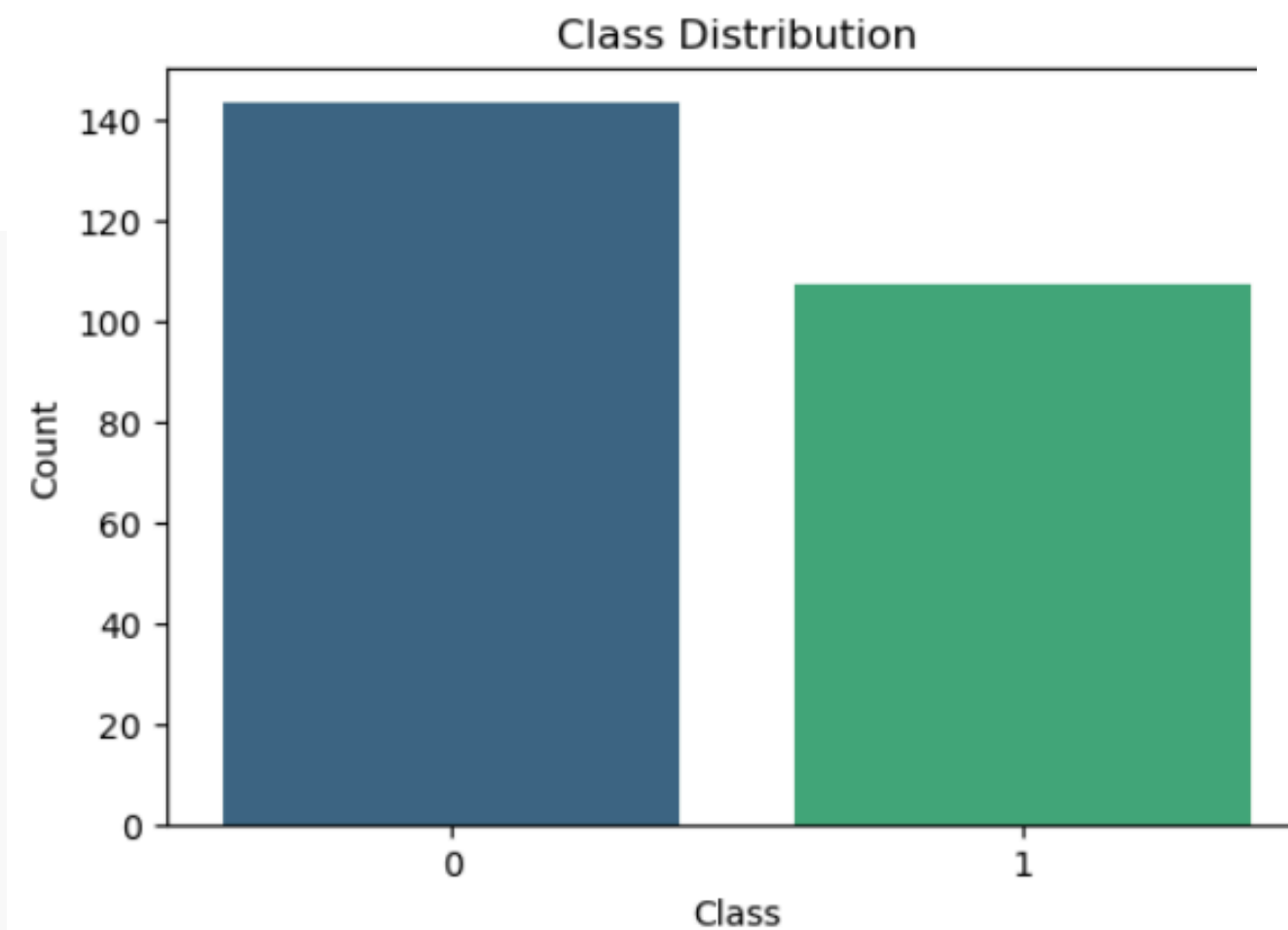
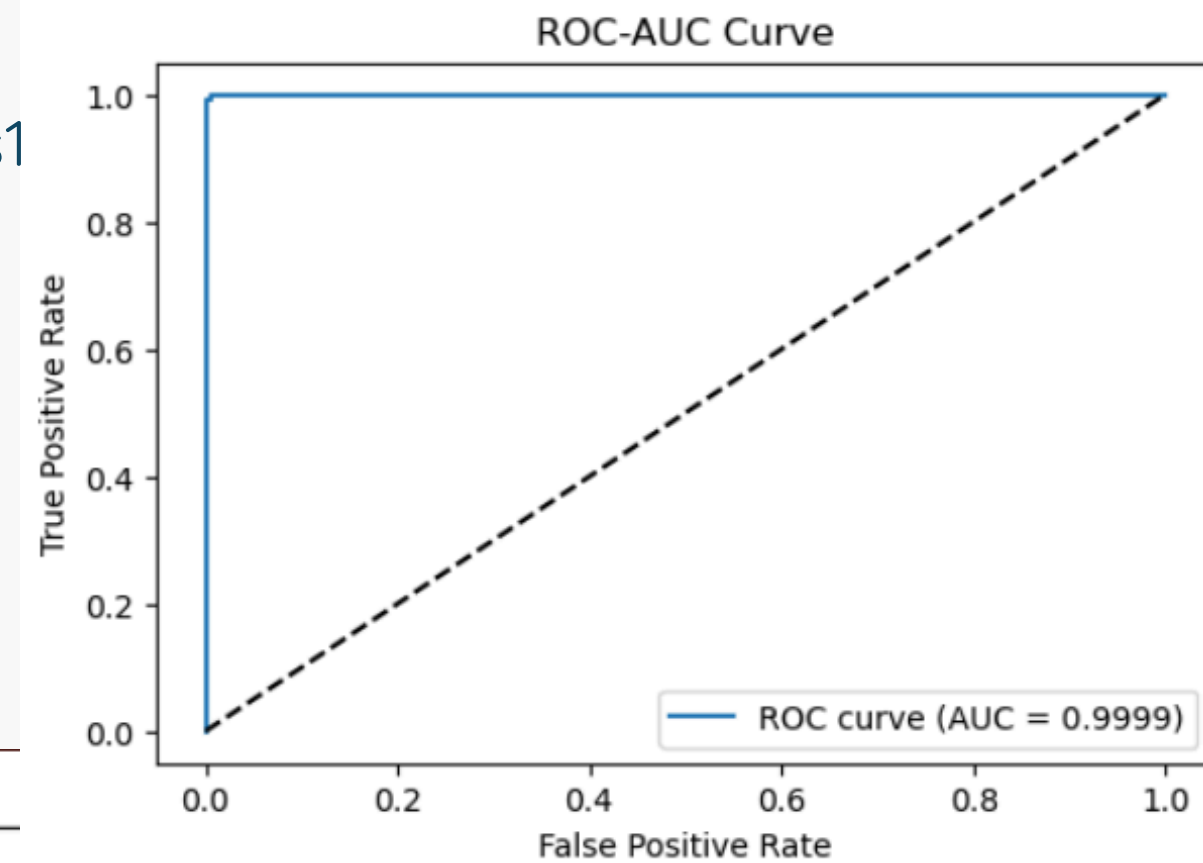
- Inspired by human brain structure with input, hidden, and output layers.
- Captures complex non-linear patterns.
-  Advantages: Very powerful for pattern recognition.
-  Limitations: Requires more data & careful tuning to avoid overfitting.

- --- Overall Metrics ---
- Mean Accuracy: 0.9960
- Mean F1-Score: 0.9952
- Mean ROC-AUC: 1.0000
- Class distribution:
- class
- 0 143
- 1 107
- Name: count, dtype: int64
- Classification Report:
-
- precision recall f1-score support
-
- 0 0.99 1.00 1.00 143
- 1 1.00 0.99 1.00 107
-
- accuracy 1.00 250
- macro avg 1.00 1.00 1.00 250
- weighted avg 1.00 1.00 1.00 250
-

CatBoost (Using 10-fold Stratified CV)



- Predictions saved for model 9
 - Actual Predicted Probability_Class1
- | | | | |
|-----|---|---|-----|
| • 0 | 1 | 1 | 1.0 |
| • 1 | 1 | 1 | 1.0 |
| • 2 | 1 | 1 | 1.0 |
| • 3 | 1 | 1 | 1.0 |
| • 4 | 1 | 1 | 1.0 |



Ensemble all 9 models to get best results (For best predictions, avoiding overfitting & accurate results)

- Combined predictions from all 9 models using Weighted Average Ensemble.
- Weights assigned based on AUC score of each model.
- Ensemble prediction gave higher stability and accuracy than individual models.

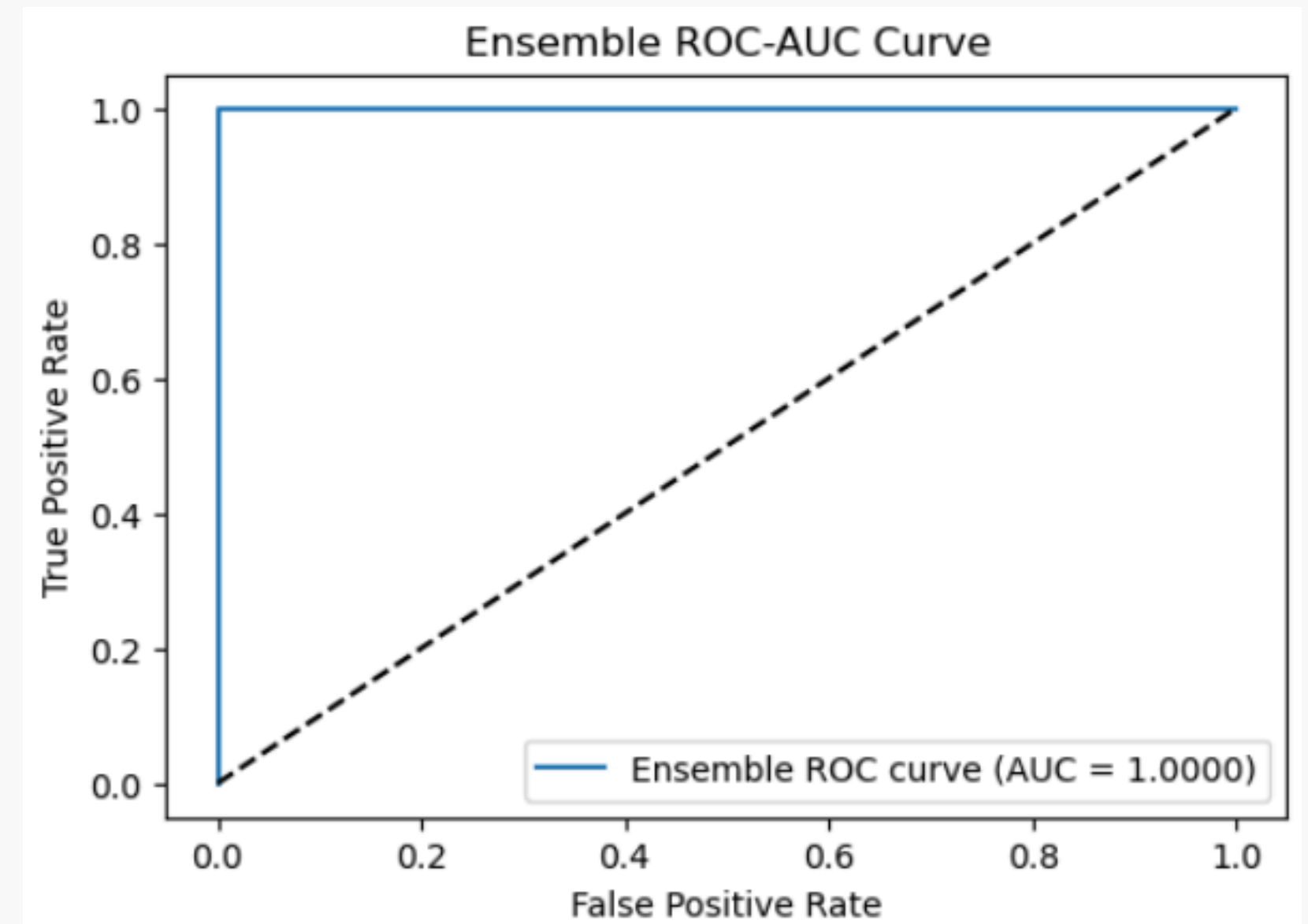
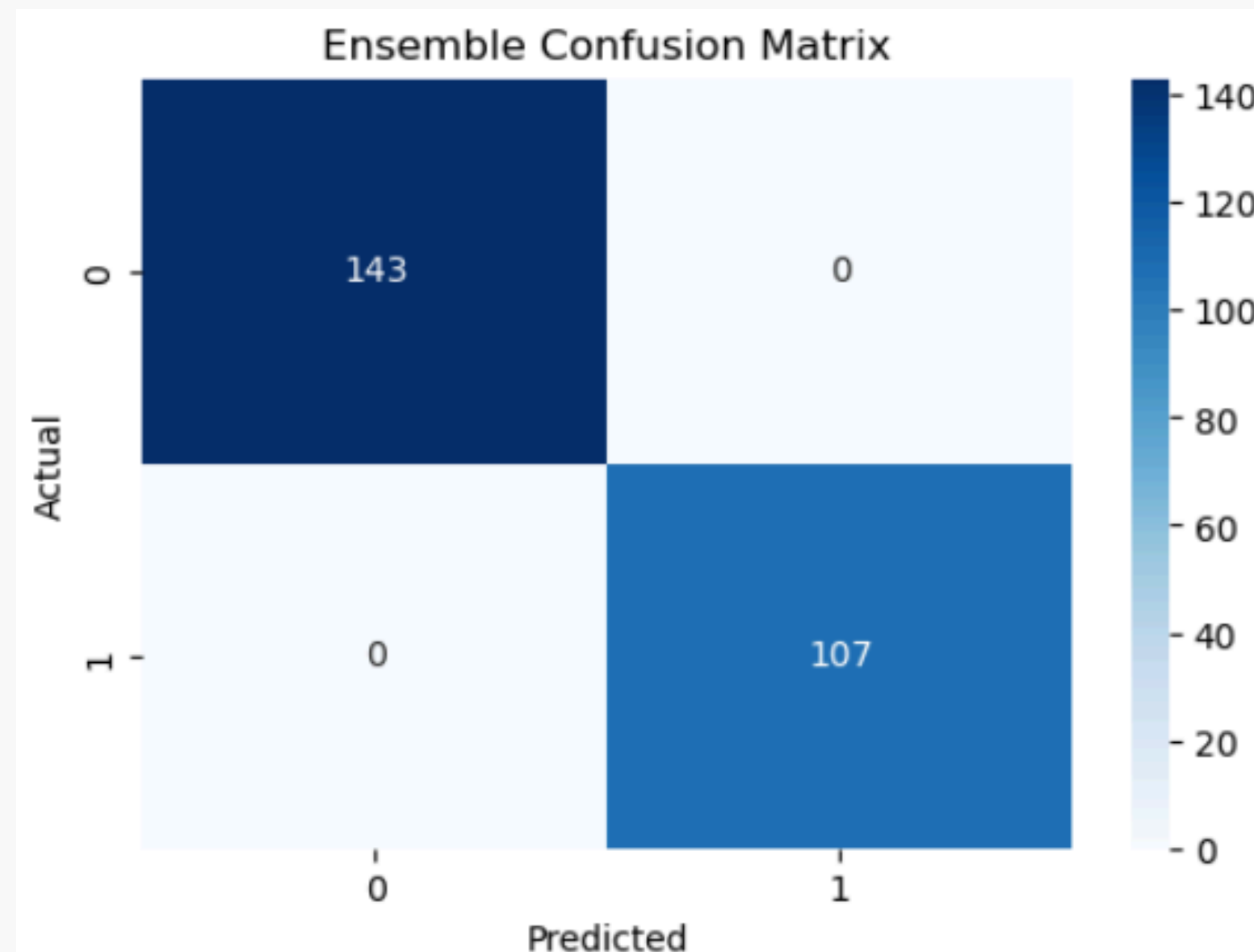
Ensemble Metrics

- Accuracy: 1.0000
- F1-Score: 1.0000
- ROC-AUC: 1.0000

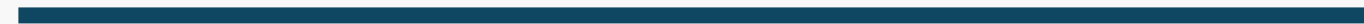
Classification Report:

•	precision	recall	f1-score	support	
•					
•	0	1.00	1.00	1.00	143
•	1	1.00	1.00	1.00	107
•	accuracy			1.00	250
•	macro avg	1.00	1.00	1.00	250
•	weighted avg	1.00	1.00	1.00	250

Ensemble all 9 models to get best results (For best predictions, avoiding overfitting & accurate results)



Streamlit Deployment



Streamlit deployment refers to the process of taking a Python-based data application built with the Streamlit framework and making it accessible to other users via a web browser, allowing for the sharing and productization of machine learning models and data visualizations without requiring front-end web development skills. An intelligent web app that empowers decision-makers to predict bankruptcy risk with near-perfect accuracy using 9 state-of-the-art models, advanced feature engineering, and a weighted ensemble strategy — all wrapped in an intuitive and beautiful Streamlit interface.



Streamlit Deployment

App Controls

Model files will be loaded from working directory. Missing files will be skipped.

You can change the ensemble threshold and preview engineered features.

Ensemble decision threshold: 0.50 (range 0.00 to 1.00)

- ☒ Show individual model probabilities
- ☒ Show engineered features preview
- ☒ Run evaluation on `filtered_df` if available
- ☒ Save ensemble predictions to CSV after predict

Bankruptcy Prediction — Ensemble (Weighted soft-voting)

Enter the 6 base risk factors (0.0 - 1.0). Engineered features are computed automatically.

Industrial Risk	Financial Flexibility	Competitiveness
1.00	1.00	1.00

Management Risk	Credibility	Operating Risk
1.00	1.00	0.00

Quick presets: Custom

Engineered Features Preview

	value
--	-------

Streamlit Deployment

working directory. Missing files will be skipped.

You can change the ensemble threshold and preview engineered features.

Ensemble decision threshold

0.50

0.00 1.00

- ☒ Show individual model probabilities
- ☒ Show engineered features preview
- ☒ Run evaluation on `filtered_df` if available
- ☒ Save ensemble predictions to CSV after predict

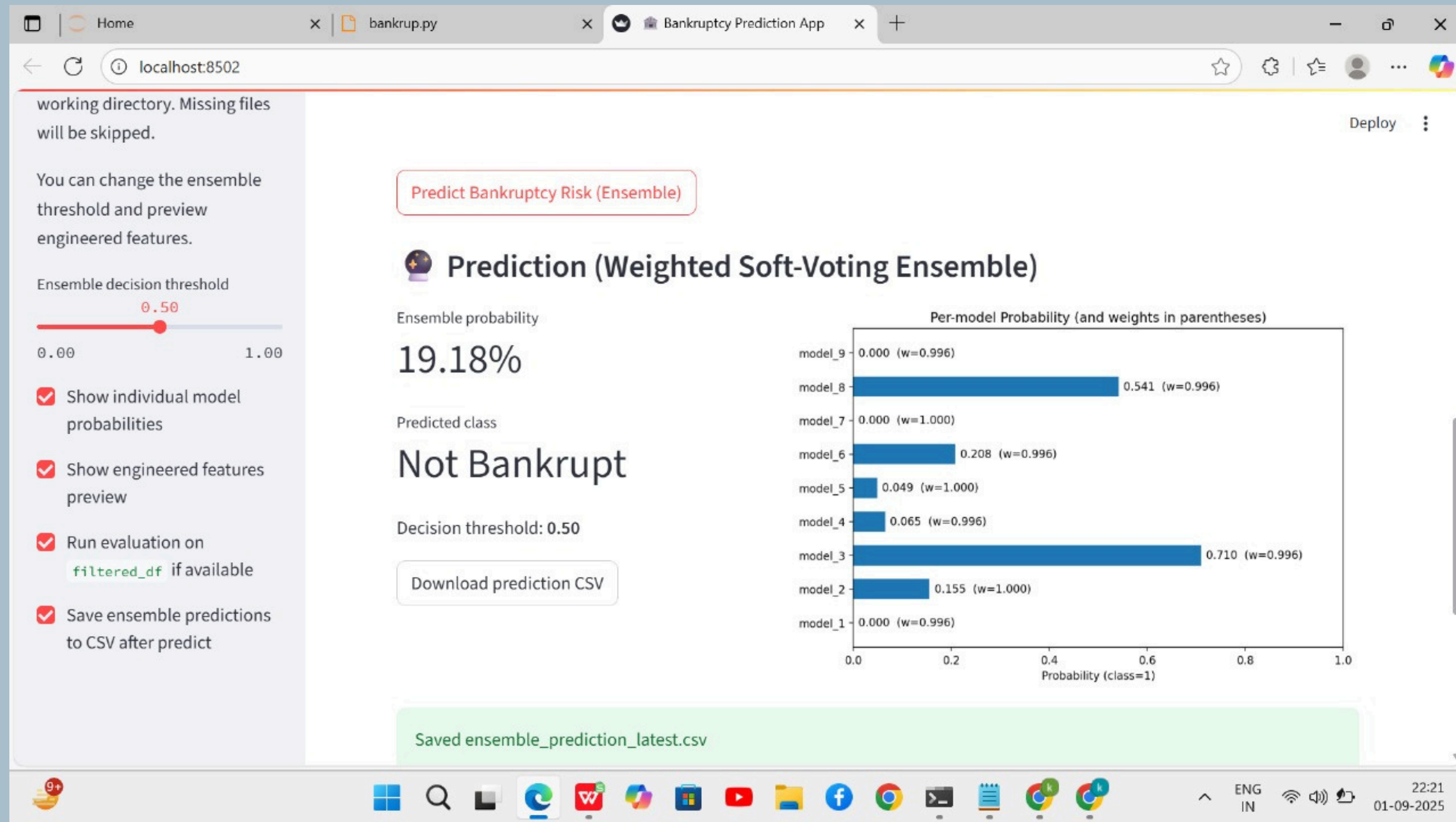
Engineered Features Preview

	value
financial_flexibility^2	1
total_risk	5
flexibility_vs_risk	1
credibility^2	1
management_risk	1
financial_flexibility flexibility_vs_risk	1
credibility competitiveness	1
competitiveness total_risk	5
credibility	1
flex_vs_comp	1

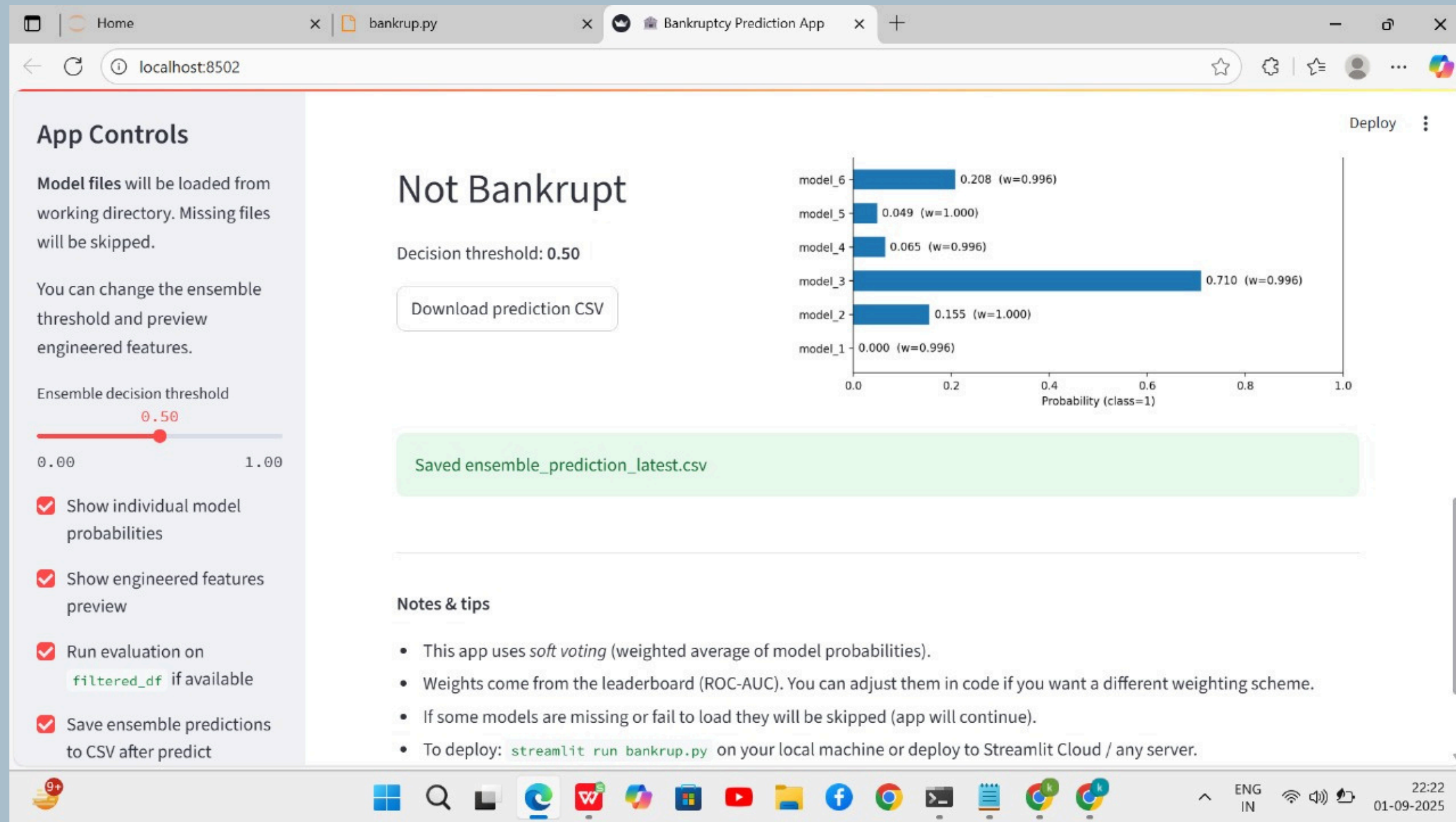
Predict Bankruptcy Risk (Ensemble)

22:20 01-09-2025

Streamlit Deployment



Streamlit Deployment



Streamlit Deployment Result



Result for all the 9 models Ensembled - 100%



Architecture Overview

Frontend (User Inputs)

Users provide 6 simple base inputs.



Industrial Risk



Management Risk



Financial Flexibility



Credibility



Competitiveness



Operating Risk



All values are normalized between 0 and 1 for ease of input.



Architecture Overview

Backend (Core Intelligence)

1. Feature Engineering Engine

Transforms 6 base inputs into *56 engineered features.

Includes:

Aggregates (e.g., total_risk)

Ratios (e.g., flexibility_vs_risk)

Squares (e.g., credibility^2)

Cross-terms (e.g., industrial_risk , management_risk)

Guarantees input alignment with the models' training schema.

2. Model Zoo (9 Experts)

Pre-trained & saved models:

Logistic Regression

Decision Tree / Random Forest / Extra Trees

Gradient Boosting / XGBoost

SVM / KNN

Neural Network (Keras ANN)

Each model brings *unique predictive strength.



Architecture Overview

Backend (Core Intelligence)

3. Weighted Ensemble Brain

- Collects probability outputs from all models.
- Applies performance-based weights* (higher accuracy → higher weight).
- Produces a blended bankruptcy probability.



Architecture Overview

Outputs (User Experience)

1. Feature Preview

Users can preview engineered features before prediction.

2. Prediction Dashboard

- *  Predicted Class: Not Bankrupt
- *  Predicted Class: Bankrupt
- *  Probability (e.g., 53.08%)

Results shown in styled cards with colors, icons, and confidence levels.

3. Model Transparency (optional future step)

Individual model scores & contribution weights.
ROC-AUC curve visualizations.
Confusion matrix diagrams.



Architecture Overview

Design Vision

*Minimalist, Finance-Inspired Theme

- Clean background, accent colors (green = safe, red = risky).

Intuitive Layout

- Inputs on left, outputs on right.
- Collapsible “Advanced Insights” section.

Interactive Charts

- Gauge meter for bankruptcy probability.
- ROC curve & confusion matrix.



Architecture Overview

Key Strengths

- ✓ Combines raw interpretability (base features) with complex intelligence (engineered features).
- ✓ Leverages 9 diverse models for bulletproof accuracy.
- ✓ Built for scalability (can run locally or deploy to the cloud).
- ✓ Designed with usability in mind — even non-technical users can make predictions.



Thank you

